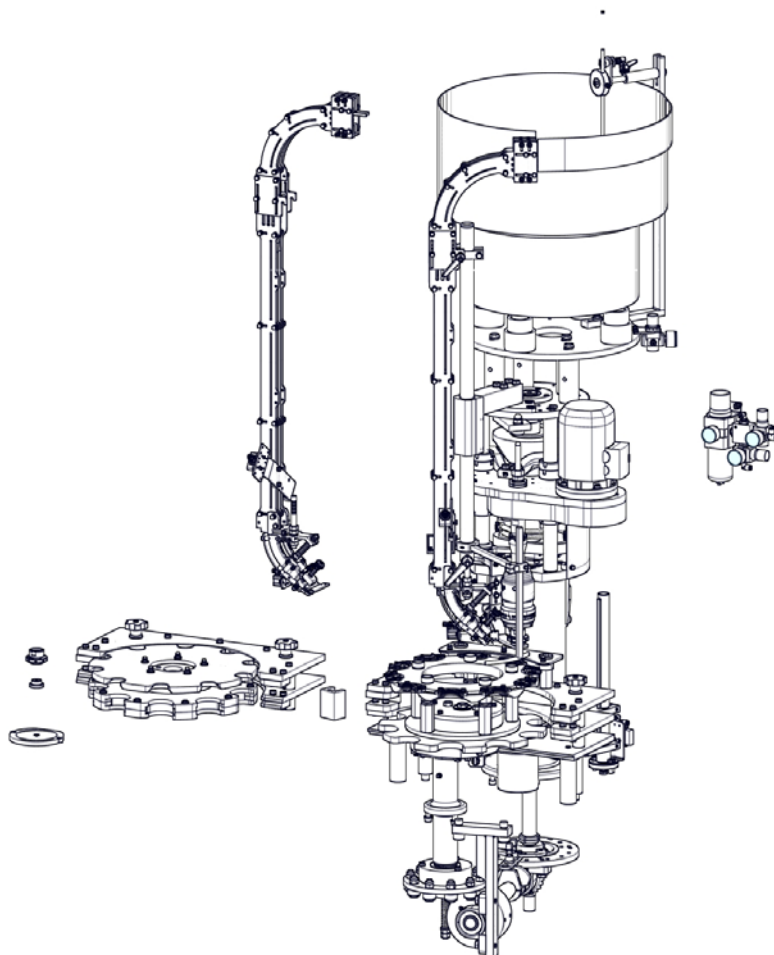




Model
TSI - CLOSYS EAGLE PK

Serial number
A4344

Star-wheel pitch diameter
410

Nr of heads
1

Manufacturing year
2018

USE AND MAINTENANCE

*TRANSLATION OF THE
ORIGINAL INSTRUCTIONS*

Code
ZA4344ABSEN001

Revision
0



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BEFORE START UP THE PARTLY COMPLETED MACHINERY, READ THE TECHNICAL INSTRUCTIONS IN THIS MANUAL AND FOLLOW THE INDICATIONS SHOWN.

KEEP THE USE AND MAINTENANCE MANUAL IN PLACE KNOWN AND ACCESSIBLE TO OPERATORS AND THE SERVICE PERSONNEL.



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1 PURPOSE OF THE MANUAL - EDITION - UPDATING METHOD

The manual is integral to the partly completed machinery TSI - CLOSYS EAGLE PK model, A4344 serial number; it should be kept all the life of partly completed machinery; it must follow the same machinery when it is moved to other premises or if it sold.

It is in Italian and translated in target language whenever required.

In case of doubts about the interpretation of the text, always refer to the original language and contact Customer Service in case of need:

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The text of the manual will refer to the "closing machine" as an abbreviation of the standard name **"Closing machine which uses a closure"** as stated in the UNI EN 415-1 standard (Terminology and classification of packaging machines).

The manual contains information and instruction about all the operating cycle of the closing machine:

- Installation.
- Start up.
- Use.
- Size change.
- Maintenance.
- Dismantling and disposal.

The manual should be kept so that:

- avoid any damage.
- Arrange for its availability to the professional operators working on the closing machine.
- Prevent it from being modified or damaged during the use.
- Make it available in case of need (training, changed ownership of the closing machine, etc.)



Ask for a replacing copy if damaged.

This manual was written with care; if some of the sections needs to be reviewed to add information or instructions, or simply for clarity reasons, AROL S.p.A. will proceed as follows:

- *should the modification involve only a few pages*: the replacement pages will be sent.
- *should the modifications involve more pages of different chapters*: all the chapters involving modification will be sent again.
- *should the modification involve many chapters* : the whole replacement manual will be sent.

Should the manual be supplied in electronic form, the supplied manual will always be the complete and updated one.

The consultation of this manual is made easier by the addition of the initial pages of the general table of contents, for an immediate tracking of the topic concerned.

The chapters are organised a hierarchy structure for a fast tracking of the info required.



The manual shows graphic symbols that are used to distinguish:

- Informations and prescriptions.
- Obligations.
- Warning about residual risks.

The following documents are attached to this manual:

- Pneumatic diagram.
- Closing machine installation diagram.
- Identification tables for size change.
- Instructions manuals for accessories and/or equipment installed on the closing machine and not integrated in this manual.



Before any operation with or on the closing machine read the procedures and warnings herein carefully.



IMAGES OF ANY GENERAL KIND ARE NOT USED IF THE DETECTION OF SPECIFIC STRUCTURE OF CLOSING MACHINE IS ESSENTIAL IN THE CONTEXT TO THE SUBJECT MATTER. The reported instructions and use modes for all described components anyway remain valid.



2 IDENTIFICATION OF THE CLOSING MACHINE

The upper part of the closing machine is provided with an identification plate. (See Figure).

On the plate are reported:

- Closing machine model (**MODEL**)
- Type of closing machine (**TYPE**)
- Pitch diameter (**P.D. in mm**)
- Number of closure elements (**HEADS**)
- Serial number (**SERIAL N.**)
- Manufacturing year (**YEAR**)

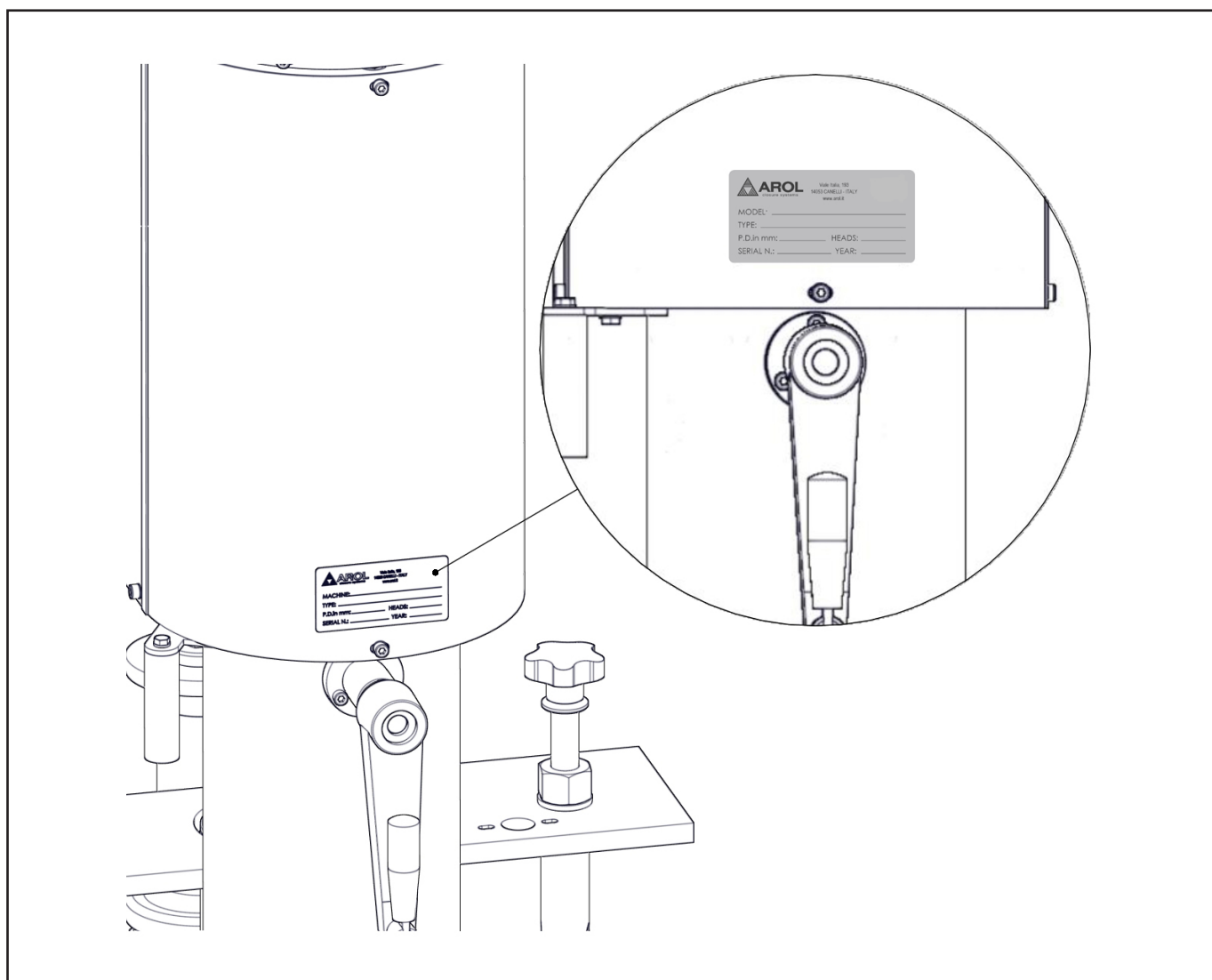


Fig. 1

The data plate is applied using 2 rivets at the opposite corners and should not be removed for whatever reason.

In case of damage, ask for a spare copy to *AROL S.p.A.*.

Clean with neutral detergents paying attention in order not to damage the warning instructions.



3 DECLARATION OF INCORPORATION

The closing machine of this manual is defined as "partly completed machinery" (according to the European Directive 2006/42/EC art. 2 point g), and comes with the *DECLARATION OF INCORPORATION*.

The original copy of the *DECLARATION OF INCORPORATION* is kept at AROL S.p.A..

A compliant copy of the *DECLARATION OF INCORPORATION* is delivered together with the partly completed machinery documents and must be kept till the machine is scrapped.



AROL S.p.A. points out that this statement only and **exclusively** concerns the above-described partly completed machinery, in the conditions and utilisation processes it was placed on the market, thus excluding any added components and/or operations carried out later on without any authorisation by the manufacturer.

In particular, any possible modification or replacement of parts of the closing machine, like as, for example modification to the software of programmable components, made without asking the permission of AROL S.p.A., releases the manufacturer from any responsibility and may cancel the validity of the Declaration of Conformity for the machine where this partly completed machinery is integrated.



4 CLOSING MACHINE DESCRIPTION

The closing machine with pre-threaded cap CLOSYS EAGLE PK has been designed and built for the application of pre-threaded caps (namely screw caps) on bottles/containers.

4.1 MAIN UNITS OF THE CLOSING MACHINE

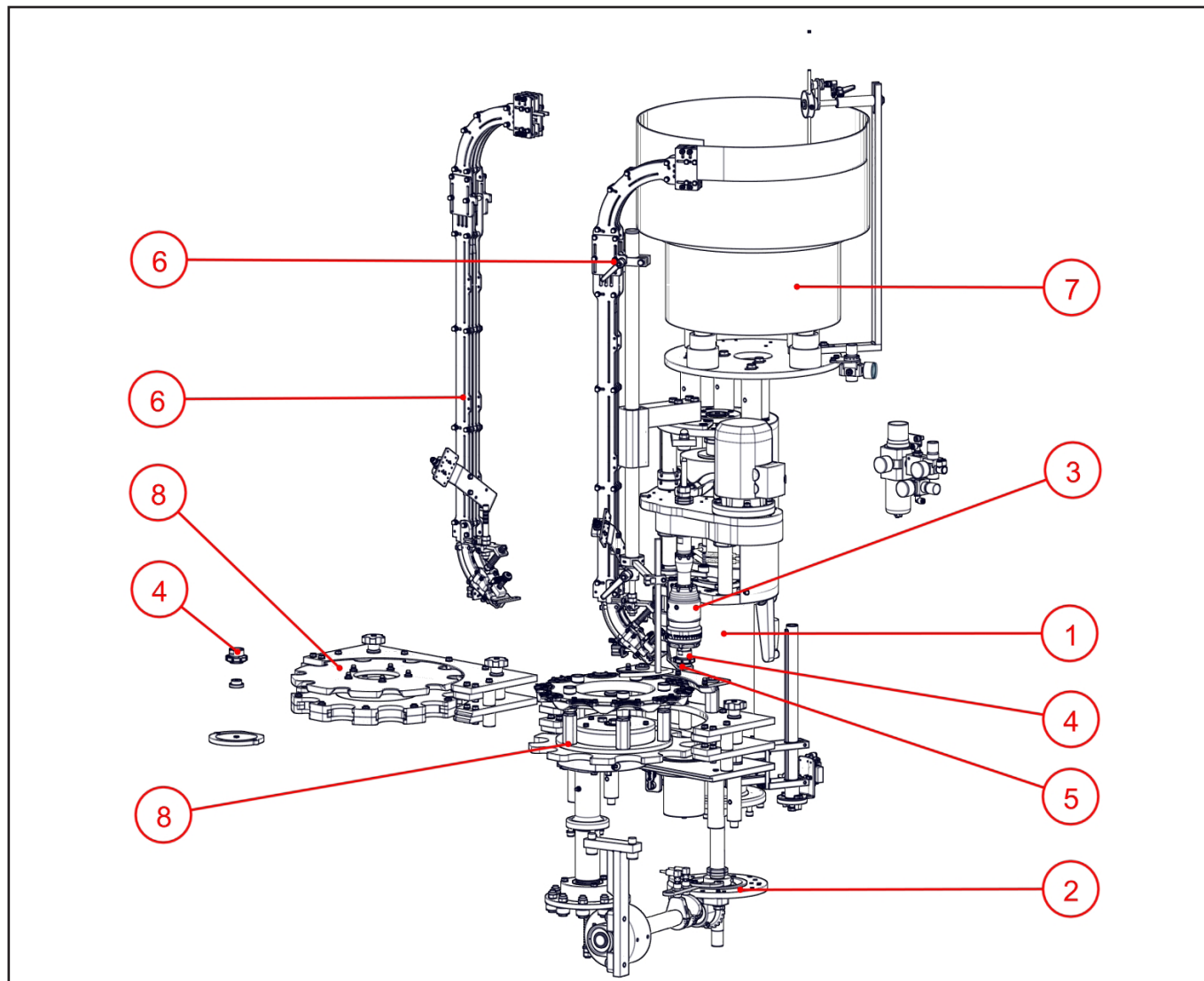


Fig. 2

Legend:

- 1 Upper closure carousel.
- 2 Lower closure carousel.
- 3 Closure head.
- 4 Closure chuck.
- 5 Transport device of closing systems.
- 6 Chute for closure systems.
- 7 Closure system sorter.
- 8 Container conveying equipment.



The components forming the closing machine are located inside the main assemblies:

- Closure head.



Fig. 3

- Closure chucks.

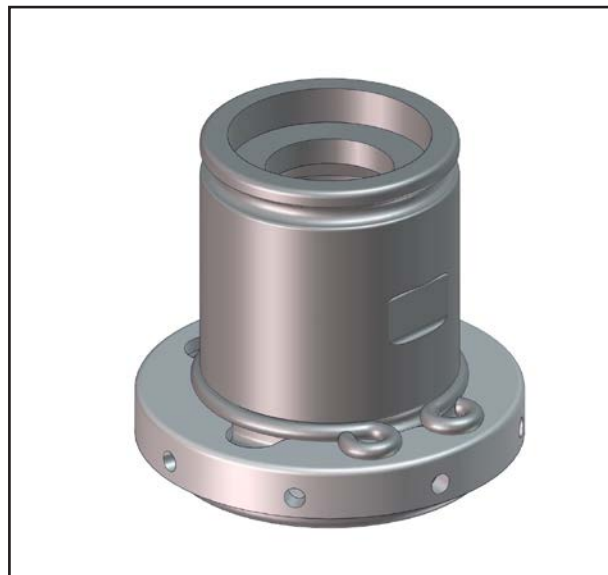


Fig. 4

- Closure carousel height adjustment.

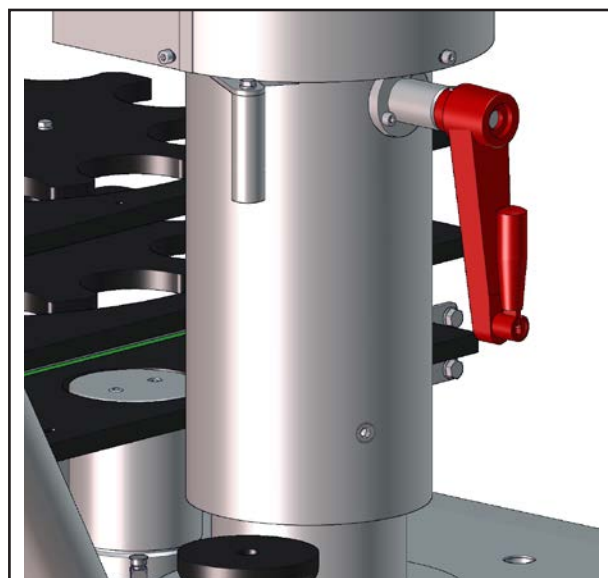


Fig. 5



4.2 DESCRIPTION OF THE PRODUCTION CYCLE

The application and sealing of closure systems on the containers is the following:

- An orientation device selects the caps inside the hopper.



Fig. 6

- The cap placed at about 45°, is taken, by a system calle "pick off", from the container that is coming during the rotation.

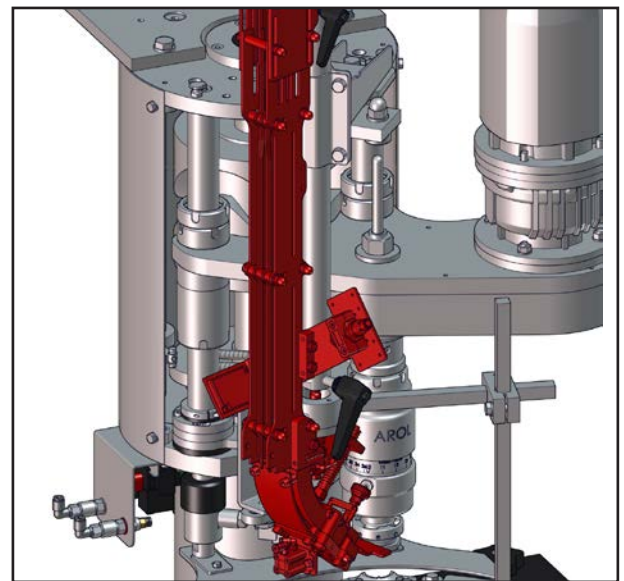


Fig. 7

- A transport device, called "pick & place", moves the caps under the closing head.

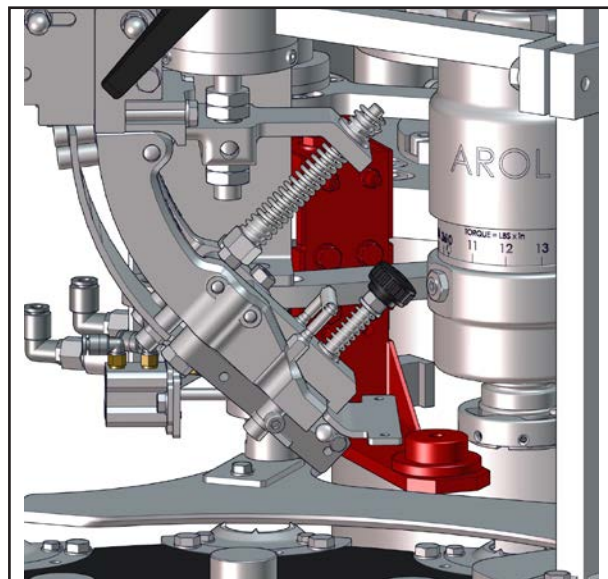


Fig. 8

- The closing head, following the profile of a cam, will pick up the cap from the transport device through a chuck with spheres.

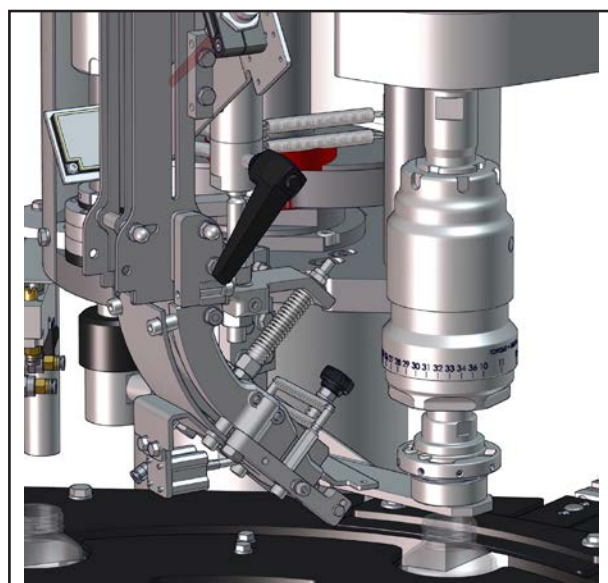


Fig. 9

- The rotation of the star-wheel conveys the bottles until under the closure head, that applies the cap on the container.
- After the closure, the exiting container is released, then there is a reset for the following closure.

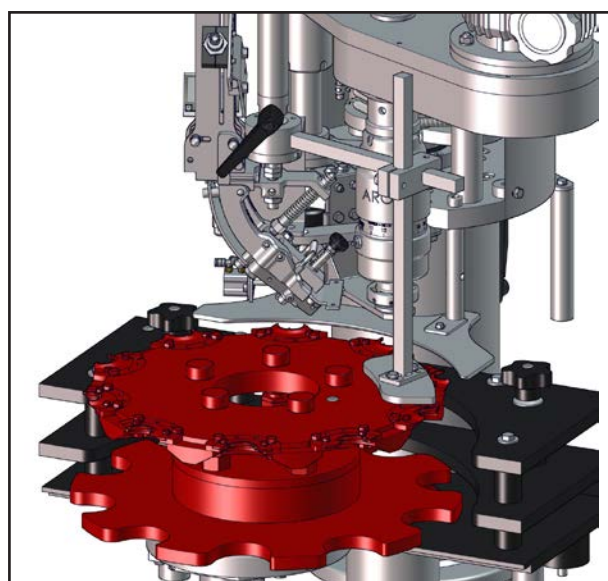


Fig. 10



4.3 CLOSING MACHINE TECHNICAL DATA

TECHNICAL SPECIFICATIONS		
Model	CLOSYS EAGLE PK	
Serial number	A4344	
Pitch diameter	410	
Nr of heads	1	
Direction of rotation	Clockwise	
Manufacturing year	2018	
NOMINAL PRODUCTION		
Pieces/hour	1200	bph
CLOSING MACHINE WEIGHT		
Weight	380	kg.
ELECTRICAL DATA		
POWER SUPPLY VOLTAGE:		
main	480	V ~
	60	Hz
auxiliary	24	V =
Installed power:		
caps sorter motor	0,66	kW
Total installed power:	1,03	kW
PNEUMATIC DATA		
Sterile air capacity	300	l/min (1 bar 20°C)
Sterile air min. pressure	6	bar
Sterile air max. pressure	8	bar



4.4 USE RESTRICTIONS

4.4.1 PRESCRIBED USE AND FORBIDDEN USES

The closing machine has been designed and made to apply closure systems (caps/corks) on preformed rigid containers with production limits indicated in chapter 4.3 CLOSING MACHINE TECHNICAL DATA page 12.

The closing machine must be used to apply closure systems (caps/corks) on preshaped rigid container already produced according to the order, planning and building stages, as described in the tables attached to this manual.

The closing machine receives the closure systems correctly orientated by its orientation system.

Besides the conduction of the closing machine out of what underlined in the chapter 4.3 CLOSING MACHINE TECHNICAL DATA page 12, just as an example and not limited to, the following uses are forbidden and not expected:

- use of closure systems (caps/corks) and/or preshaped rigid containers whose specifications are different to the ones described in the expected use.
- supply to the closure systems (caps/corks) and/or preshaped rigid hand containers or other means.
- revert the rotation direction of motors.
- tamper guards and/or protection devices.
- overlook the protection means using means to rise on the top or climb on parts of the closing machine.
- perform maintenance operations without insulating and locking all the energy sources and dissipating the residual energies.
- do not comply with the specifications of power sources listed in the table page 12.
- Treat products with characteristics other than those communicated at the order, changing the environmental conditions here the closing machine must be fitted (example: temperature of the treated product; with environmental characteristics other than those communicated at the order).

4.4.2 OPERATING MODES

The closing machine cannot work autonomously.

The installation technician must thus install the closing machine under his own responsibility and according to the European Community directives and take care that all requirements and specifications of the laws in force are complied with.



4.4.3 NUMBER OF OPERATORS

The operations described in this manual, referring the cycle of life of the closing machine, have been analysed by AROL S.p.A.. Therefore, the number of phases and operators indicated is the one most suitable to carry out the functions in an optimal way.

4.4.4 PROFESSIONAL ROLES OF USERS

The closing machine User shall check that the operators have the following requirements:

- had read and understood the indications that are in the manual.
- had received job instruction and training appropriate to their tasks;
- have been properly trained about the correct use of the closing machine, about the residual risks, on the closing machine controls and on the PPM.

Only the operators and the person charged with the maintenance with a suitable training can use the closing machine.

The professional terms used are specified here below to understand the manual fully:

TERM	DESCRIPTION
EMPLOYER	Person legally responsible for the use of the closing machine described herein.
OPERATOR	Person charged with the standard closing machine operation.
MECHANICAL MAINTENANCE OPERATORS	Person charged with the routine and extraordinary mechanical maintenance of the closing machine.
ELECTRICAL MAINTENANCE OPERATORS	Person charged with the routine and extraordinary electrical maintenance of the closing machine.

4.4.4.1 OPERATOR

The Operator must have:

- familiarity with the use of the check system (operator interface), with capacity of checking the correlation between the set data and the effect of these on the production;
- knowledge of the technology and specific experience in operating the closing machine acquired through the basic training of this manual;
- base general knowledge and sufficient technical knowledge to read and understand the manual content and the annexes, included the correct interpretation of the drawings.
- Sufficient technical knowledge to carry out intervention specified in the manual, in safe operation condition.
- Knowledge of the accident-preventing norms.
 - General Health and safety, prevention of the accidents on the work place.
 - specification (for the type of product processed by the closing machine, according the laws in force of the country in which the machine is installed).



Besides the conduction of the closing machine from the productive point of view the operator must:

- Know the operation of all the safety devices (for the operator and for the closing machine) so as to be able to check their efficiency and signal any possible fault to the personnel in charge according to the procedures described herein.

When the operator locates a failure that might compromise safety, he must stop the closing machine using the emergency stop controls.



it is forbidden to execute any intervention on the electric and/or pneumatic equipment.

4.4.4.2 MAINTENANCE MAN

The Maintenance Man (mechanical/electrical), besides the above characteristics, must also have a good technical preparation and a suitable working experience in the relevant field. Further, he must have specific and special (mechanical and electrical) technical knowledge needed for the operations described in this manual.

4.4.5 EMPLOYER'S OBLIGATIONS

the employer shall guarantee appropriate training and information of all the professional roles of the closing machine on the following topics:

- Instruction and specific procedures for use and maintenance of the closing machine.
- The user of the closing machine should comply with the safety and health prescriptions on the workplace, in force in the premises where the closing machine is started; all the specific risk conditions should be taken into account.

He must have acquired a good knowledge of the contents of this manual so that he can correctly process the training procedures of the professional roles of the closing machine.

At the end of the training, the employer must make sure that operators and maintenance technicians have understood the above points.

Assess the risks resulting from installing the closing machine in the workplace.

The employer must supply the PP (Personal Protection Means) required for all operations linked to the useful life of the closing machine.



4.4.6 INFORMATION - JOB INSTRUCTIONS - TRAINING

4.4.6.1 INFORMATION - JOB INSTRUCTIONS

The employer shall inform and train the personnel charged to use the closing machine and make sure that:

- On the work place there are suitable information concerning dangers, risks and prevention and protection solutions including the use of P.P.M., if provided.
- the workers the task of using of work equipment receive a suitable training accordingly.
- The workers the task of using of equipment requiring special knowledge and responsibility receive a suitable and specific training, so that they can use such equipment safely and suitably, also as far as risks caused to other people are concerned.

4.4.6.2 TRAINING

Training serves to explain the correct use of the closing machine, in compliance with the obligations prescribed by national laws transposing 2009/104/EC directive concerning minimum safety and health requirements for the use of working tools by the operators while working.

4.4.7 ARRANGEMENTS BY THE CUSTOMER

Excluding special contract conditions, the user must arrange the following:

- The closing machine must be electrically supplied by the electric board of the machine to which the partly completed machinery is to be installed. The protection against indirect contact must be envisaged during installation, according to point 6.3 of EN 60204-1.
- Compressed air mains connection.
- Generic tools for the closing machine maintenance and consumables.
- Lifting means suitable for the closing machine handling.



4.4.8 SIGNS FOR WARNING, OBLIGATION, PROHIBITION AND DANGER

SYMBOLS CONTAINED IN THE MANUAL

The graphic symbols inside the manual are described and shown here below.

GRAPHIC SYMBOL	DESCRIPTION
	It indicates a generic warning. Pay utmost care to the text blocks marked by this symbol.
	Indicates the danger of hot surfaces. Pay utmost care to the text blocks marked by this symbol.
	It refers to a warning about key function or useful information. Pay utmost care to the text blocks marked by this symbol.
	Notes, usage tips, or additional information
	Part of the manual dedicated to the OPERATOR.
	Part of the manual dedicated to the MECHANICAL MAINTENANCE OPERATOR.
	Part of the manual dedicated to the ELECTRICAL MAINTENANCE OPERATOR.
	Obligation to insulate all energy sources. The symbol placed close to this description means that all the energy sources must be insulated, dissipating the residual energies by the user, because this involves the risk of an accident.
	Mandatory use of <i>PROTECTIVE GLOVES</i> . The presence of the symbol close to the descriptions informs that the operator must wear safety gloves due to the risk of injury.
	INSPECTION AND OPERATION Visual inspection.
	INSPECTION AND OPERATION Inspection hearing.
	INSPECTION AND OPERATION Intervention with tools
	INSPECTION AND OPERATION Manual intervention.
	INSPECTION AND OPERATION Intervention of lubrication.



4.5 ENVIRONMENTAL LIMITS

4.5.1 ENVIRONMENTAL OPERATING RISKS

USE ENVIRONMENTAL CONDITIONS

The closing machine must work in premises protected by weather agents, with a temperature ranging between 5°C and 40°C, a maximum relative humidity of 80% and height above sea level not over 2000 m.

LIGHTING

The closing machine can work with the standard lighting of the workplace (about 250 lux); local lighting of areas concerned with maintenance is not required very often and must be performed with low voltage lights.

EXPLOSIVE ATMOSPHERE

It is strictly forbidden to operate the closing machine in an area with potentially explosive atmosphere, even if only momentarily.

4.5.2 RESIDUES AND ENVIRONMENTAL CONTAMINATION

No toxic substance is released if the closing machine is used and maintained according to the instructions herein.

The disposal of spent lubricants must respect the norms in force for environment protection in the country of use.



5 SECURITY AND ACCIDENT PREVENTION

5.1 GENERAL INFORMATION

The employer will must provide for the training of the staff about dangers caused by accidents, about equipments arranged for the operator safety, about dangers of noise emission and about the general accident rules provided by international directives and by legislation of the country of designation of the closing machine.

Anyway the behaviour of staff, maintenance operator, cleaning, control, etc. will must scrupulously respect the accident norms of the country of designation of the closing machine.



Before starting the work, the operator must be thoroughly familiar with the location and operation of all controls and characteristics of the closing machine must also have fully read this manual.



The closing machine must be used only from operators that had received a training and understand totally the instructions of this manual.



It is necessary to respect totally instructions, cautions, general accident rules that are in this manual.



Before any maintenance and/or size change operation, the closing machine must be safety conditions:

- **insulate the closing machine from all energy sources dissipating residual energies.**
- **divide and lock all the cutting off devices;**
- **Prevent access to the closing machine to unskilled staff, not charged with maintenance operations, using suitable signs.**



5.2 RISKS - PROTECTIONS - CAUTIONS AND PRECAUTIONS

5.2.1 SAFETY DEVICES OF THE CLOSING MACHINE

To guarantee health and safety of the people exposed, the closing machine is provided with this safeties:

FIXED GUARDS

All the fixed guards installed on the closing machine are connected to the structure by means of fastening devices which do not allow manual removal, since they necessarily require the use of specific tools.

It is forbidden to use the closing machine after having disassembled the fixed guards; operation can resume after the reassembly of guards.

The housing of the fixed guards in vertical position in the structure of the closing machine has been designed so that the guards, once the fastening devices have been removed, do not remain in closed position, falling by gravity.



To access the area protected by the guard, the closing machine must be insulated from all power sources. (See sub-chapter 5.4 SAFETY PROCEDURES FOR THE MAINTENANCE page 22).

5.2.1.1 POSITION OF THE SAFETIES

FIXED GUARDS

- Head protection guard.

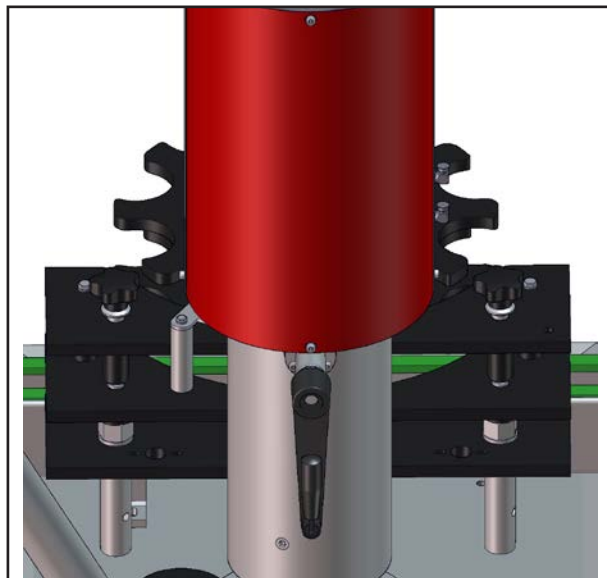


Fig. 11



5.3 SAFETY PROCEDURES FOR THE OPERATION



Wrenches and tools used for the maintenance or the access to closing machine hazardous sections must not be left at the disposal of operators not suitably trained for the size change or the maintenance.

All check, adjustment, maintenance and lubrication steps must be carried out by staff previously trained and instructed.

All check, adjustment and maintenance steps of the electric system must be carried out by staff previously trained and instructed.

People on drugs, alcohol or drowsiness-causing medicines are not allowed to use or maintain the machine.

All check, adjustment, maintenance and lubrication steps must be carried out with the closing machine stopped and separated and split electric power supply.



The cap/cork feeding system has been designed and manufactured to be loaded by an automatic elevator with ground loading.

If the cork/cap loading is manual, it should be performed leaving the closing machine at a standstill, with separated and split electric power supply.



Before accessing the top of the machine, separate and split the energy sources and dissipate the residual energies.

Access at any height must occur using certified means according to the norm UNI EN ISO 14122.



5.4 SAFETY PROCEDURES FOR THE MAINTENANCE



BEFORE CARRYING OUT ANY MAINTENANCE OPERATIONS, the closing machine must be in safety conditions to avoid untimely screwing and/or electrocution.



It is therefore compulsory to insulate all energy sources dissipating residual energies.

The insulation for power sources must be charged to qualified technicians and be performed as follows:



Wrenches and tools used for the maintenance or the access to closing machine hazardous sections must not be left at the disposal of operators not suitably trained for the size change or the maintenance.

After the maintenance works and before starting the closing machine again it is always necessary to:

- check that the possibly replaced parts and/or used tools have been removed from the closing machine.
- check that all guards, protections and safety devices possibly removed during the intervention have been re-installed correctly and are operating.



Replace the electrical safety components within their life cycle, approximately 20 years.



The standard operating conditions will be restored only after the intervention and the checks.



5.4.1 GENERAL SAFETY WARNINGS

The safety for the closing machine and the operators also depends on a regular maintenance according to the manufacturer's indications.

The routine and extraordinary maintenance interventions must be exclusively charged to skilled staff, that is to technicians qualified according to the specifications in this manual.

If outsourced operators intervene, they must ensure the same performance of the inhouse qualified technicians and must work in safe operating conditions.

The access to high levels is allowed only after separating and disconnecting any residual energy and using means certified according to UNI EN ISO 14122 standard.

The qualified technicians must:

- respects the limits of their tasks (mechanical, electrical, etc.).
- Follow the procedures and warnings in the manual, within their tasks.
- Follow the times and intervals indicated in the manual for scheduled maintenance interventions.



The following general warnings must be noted:

- Electrocution hazard due to direct contact with the closing machine power supply terminal boards and in the electrical system connector blocks (the hazard is signalled by plates with the suitable triangle with yellow bottom).
- Be extremely careful during the interventions on the electrical system, if the system must be left live. Such interventions must be carried out by staff skilled in this type of maintenance only, trained and correctly equipped for the steps to be made. The intervention with live system must always be authorized in advance by the employer who must oversee its operations by assessing the specific risk situations.
- Carry out maintenance operations on pneumatic systems and components only after cutting off and separating the electric power and pneumatic sources and checking that the residual pressures have been relieved.
- It is forbidden to carry out maintenance and lubrication interventions on moving components.
- While working on heated parts, the operator must always wait until they cool to a temperature below 55°C to prevent serious burns.

5.5 RESIDUAL RISKS

The residual risks are shown in Annex VII B of the European Directive 2006/42/EC about "Technical documentation for partly completed machinery".



6 HOW TO ASK FOR INTERVENTIONS

AROL S.p.A. puts its own Technical Service at the disposal of its customers to solve any problem regarding the machine use and maintenance.

The interventions must be required after carefully evaluating the faults and their reasons, and stating:

- The details on the occurred faults.
- The performed checks.
- The performed adjustments and resulting effects.
- Any further information deemed useful.

7 HOW TO ORDER SPARE PARTS

All orders for spare parts must be sent to AROL S.p.A..

The installation of not original spare parts will invalidate the warranty.

To order spare parts, fill in the suitable form in the “*SPARE PART CATALOGUE*” for the machine.



The replacement of safety devices must be performed with the same components. The 2006/42/EC directive provides that the components supplied by the machine genuine manufacturer do not need the Declaration of Conformity. Vice versa for all components purchased on the market, the Declaration of Conformity is required, along with the control that the replacement component has the same features and performance of the replaced one. Unsuitable components might cause the decay of the safety performance expected by AROL S.p.A.. It is therefore recommended to contact AROL S.p.A..



8 INSTALLATION AND CONNECTIONS

8.1 CAPS CHUTE INSTALLATION



To install the caps chute on the machine it is necessary:

- Loosen the return handles located on the support chute.

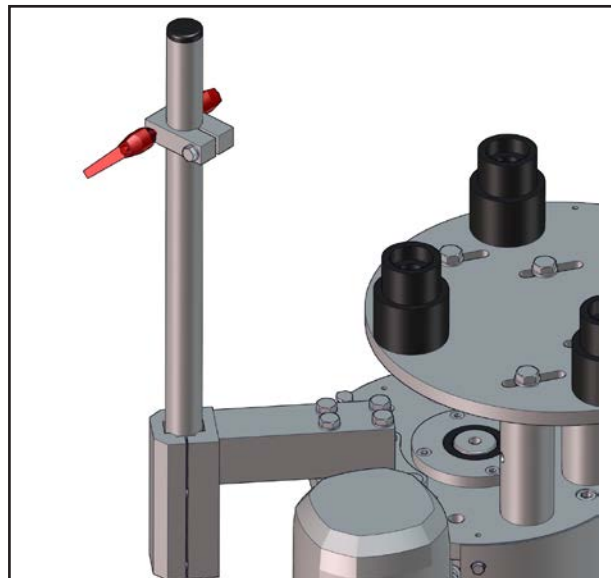


Fig. 12

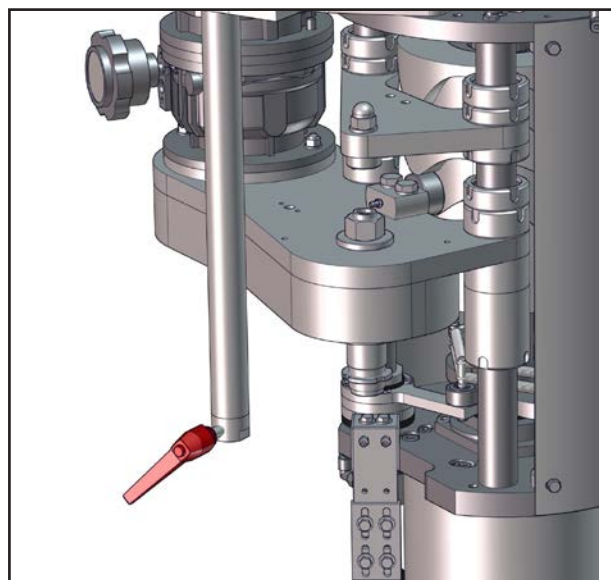


Fig. 13



- Insert the head height adjustment pin into the hole of the chute support and lift the complete caps chute.

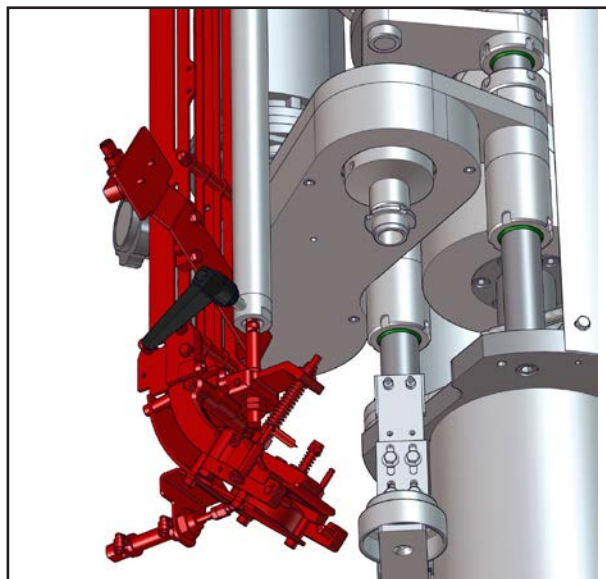


Fig. 14

- Place the caps chute so that the fixing brackets are positioned on the pins of the return handles.

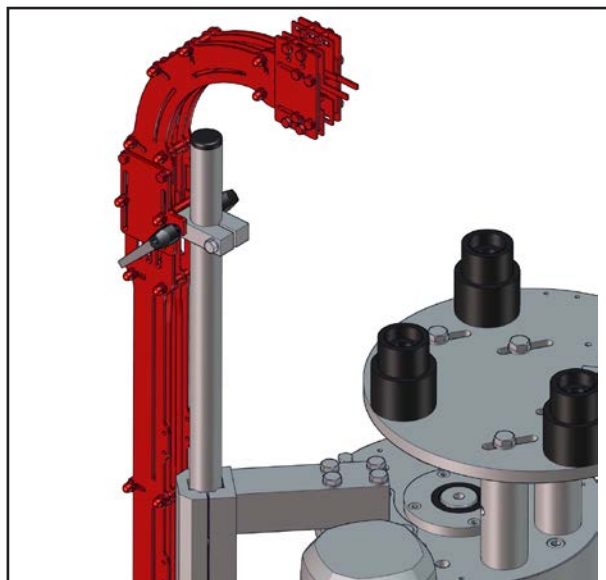


Fig. 15

- Lock the return handles located on the caps chute support.

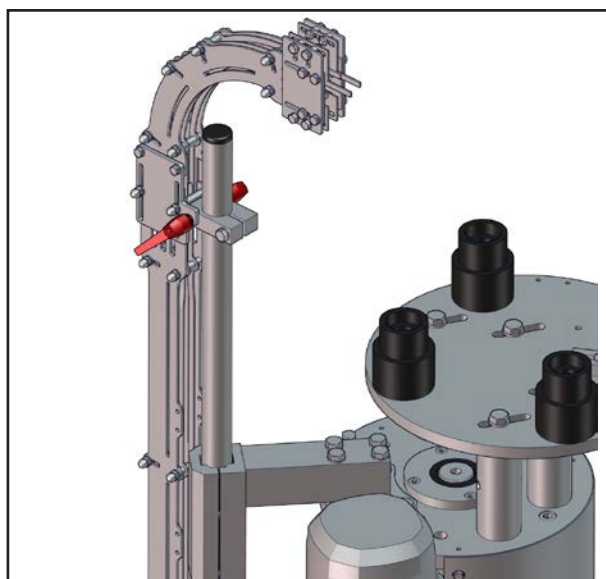


Fig. 16



- Lock the return handles of the caps distribution head.

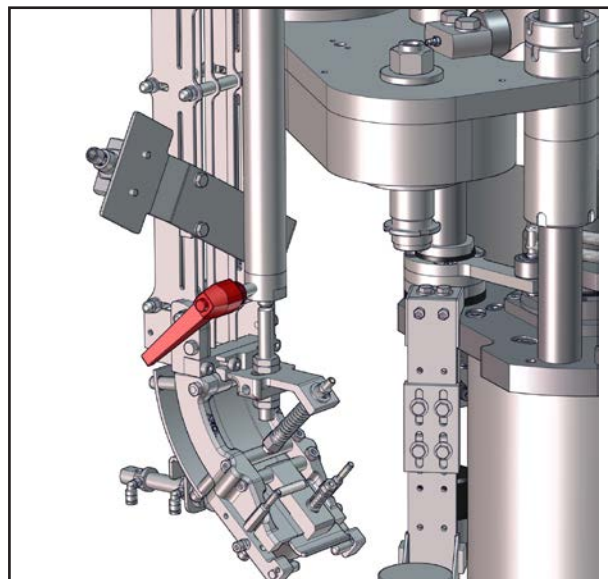


Fig. 17

- Connect the electrical connection of the caps presence photocell (one or more).

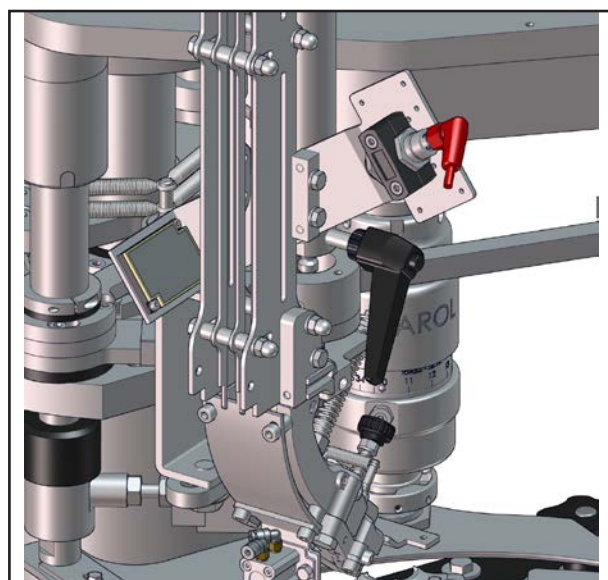


Fig. 18

- Connect the pneumatic connections of the caps stop cylinder to their fittings.

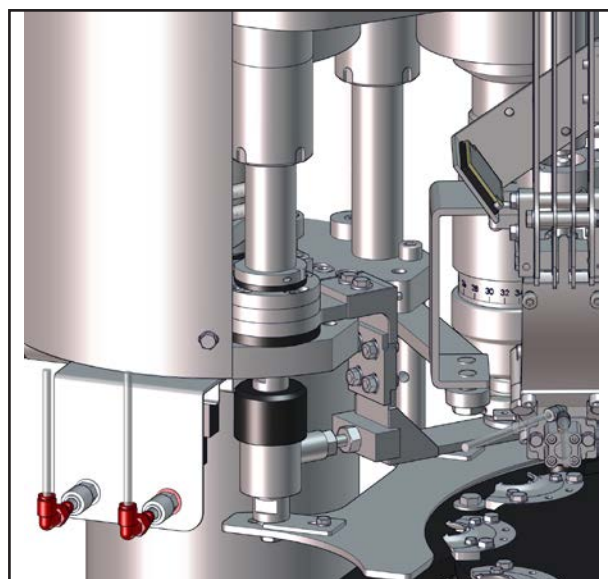


Fig. 19



8.2 INSTALLATION OF THE VIBRATING CAPS SORTER



For the vibrating caps sorter installation follow the instruction below:

- unscrew and remove the central screw that fix the tank to the vibrating body.

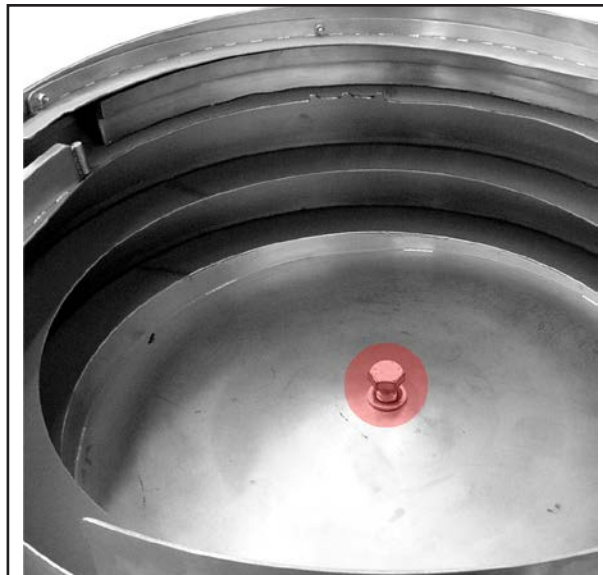


Fig. 20

- At the place of the central bolt, previously removed, mount the supplied eyebolt.



Fig. 21



- Fasten a belt or a chain with hooks to the eyebolt.
- Lift the caps sorter with adequate lifting means.
- Place the caps sorter in such way that the supporting feet are inserted into the centering cups, in such way that the exit mouth and the caps chute coincide.

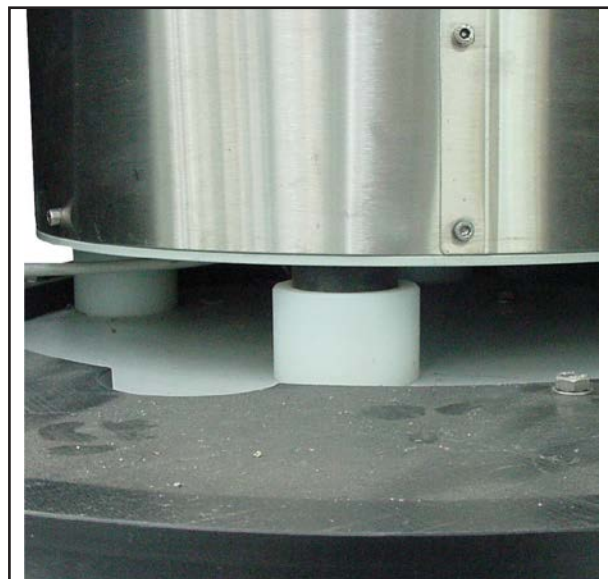


Fig. 22

- Unhook the belt or the chain with hooks from the eyebolt.
- Remove the eyebolt.
- Fasten the tank to the vibrating body with the previously removed central screw.



Fig. 23

- Place the caps level device on the caps sorter proceeding as described below:
 - Insert the level blade in the tank and the levelling rod fixing tie rod.

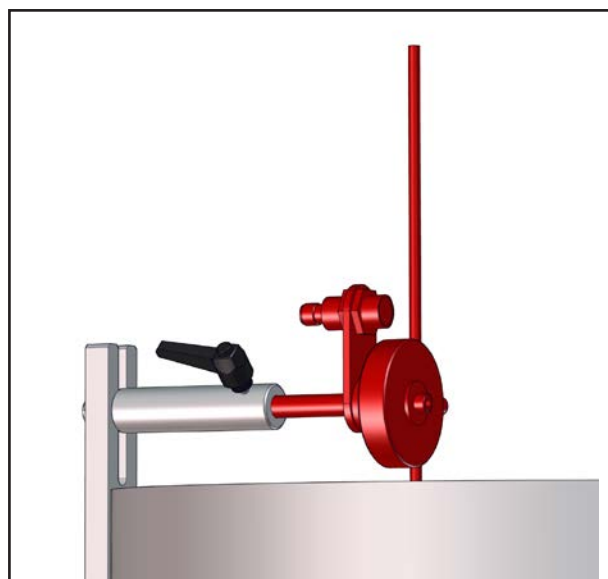


Fig. 24



- Fix the assembly locking the return handle against the levelling rod.

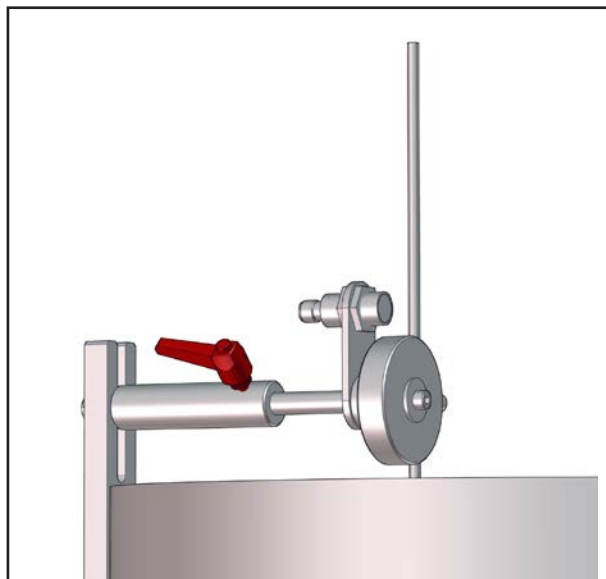


Fig. 25



8.3 PNEUMATIC CONNECTION



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



The air used in the pneumatic installation must be filtered, so without impurities, and dried.

For pneumatic data see the table page 12.

Connect with the mains as follows:

- connect the coupling of the pneumatic assembly to the main distribution network.

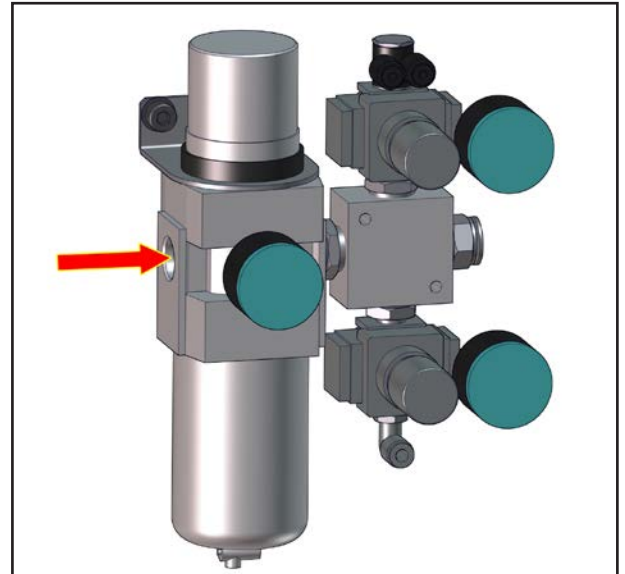


Fig. 26

- Connect the flow rate pipe coming from the main pneumatic unit to the caps orientation unit coupling

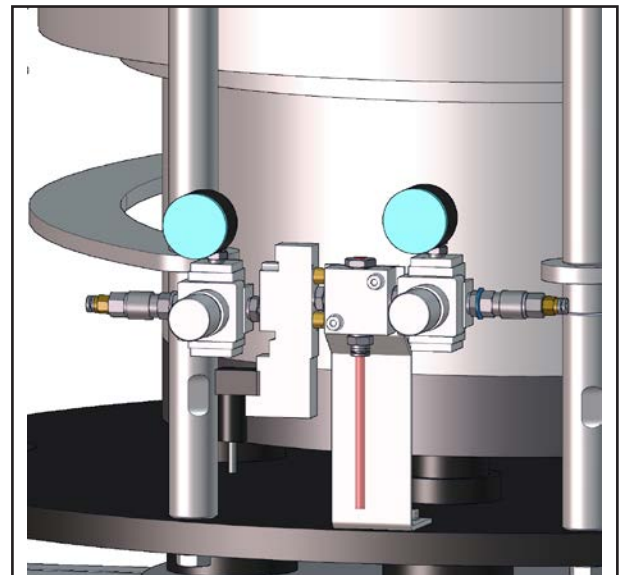


Fig. 27



9 CLOSING MACHINE USE

9.1 SETTING UP, CHECKS AND TEST FOR THE FIRST START UP



The setting up operations of the closing machine for the first start up must be carried out by qualified technicians.

The closing machine has been tested by AROL S.p.A. premises before the shipping; all the calibration and required adjustments have been carried out during the test.

9.1.1 CHECKS BEFORE THE START UP



In order to prevent errors and/or accidents, before the start up of the closing machine it is necessary to carry out some checks:

- Check the sequence of the motor supply phases.
- Check that there are no leaks from the pneumatic circuit.



9.2 PRELIMINARY CHECKS BEFORE STARTING



Before starting the closing machine up, check the following:

- Check that all other safety devices installed on the closing machine are operating properly. (See also sub-chapter 5.2.1 SAFETY DEVICES OF THE CLOSING MACHINE page 20).
- Check all closing machine components are lubricated (See Chapter 11.6 LUBRICATION page 126).
- Make sure the specific installed equipment are suitable for the size to be processed.
- Check the correspondence between container and cork to be treated.



Each set of equipment is marked as indicated in the relevant tables enclosed in this manual. Each mark corresponds to a size to be processed.

- Check the height of the closure carousel according to the container height.
- Check the caps sorter is sufficiently full.
- Check that the working pressure is correct (the pressure is indicated on the pressure gauges).



9.3 PREPARATION FOR STARTING

9.3.1 CLOSING MACHINE HEIGHT ADJUSTMENT



For the adjustment and check, operate as described here below:

- Unlock the fastening screw of the sliding column.

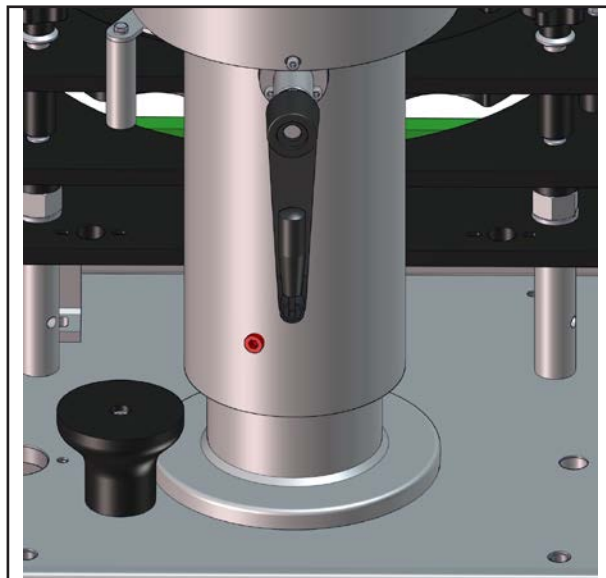


Fig. 28

- Rotate the crank clockwise to lift the closing machine till passing the height of the bottle to be processed

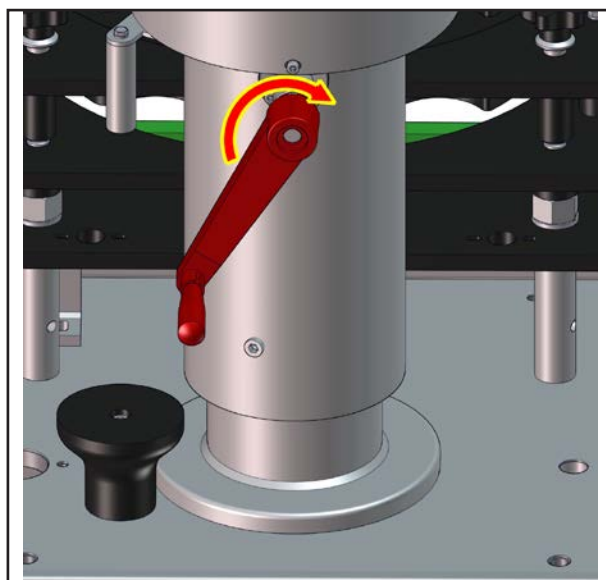


Fig. 29



- Lay a bottle of the size to be treated on the reference plate.

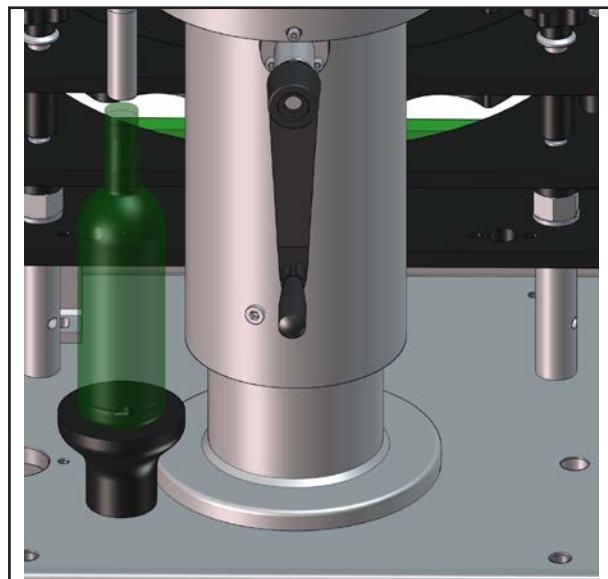


Fig. 30

- Rotate the crank anticlockwise to lower the closing machine till the reference spacer lays on the bottle.

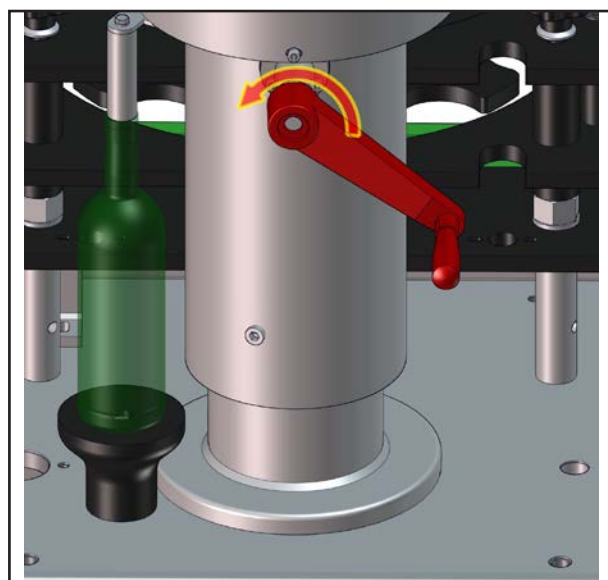


Fig. 31

- Remove the bottle used for the adjustment from the reference plate.
- Lock the fastening screw of the sliding column.

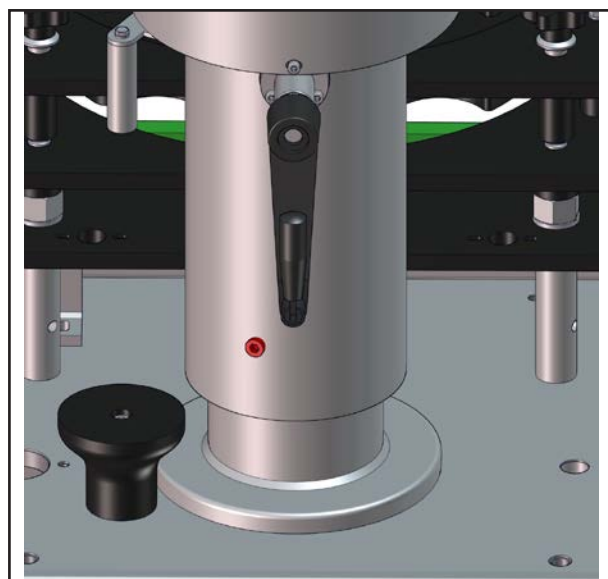


Fig. 32



9.3.2 CLOSURE HEAD ROTATION SPEED ADJUSTMENT



If the head rotation speed of the closing head is properly set, once the screwing operation is completed, the closure chuck stops on the cap few tenths of a second before the bottle abandons the closing position.

If the closure chuck stops rotating for a few more tenths of a second, it means that the rotation speed is too high. If this is the case, reduce it.

If the closure chuck does not stop rotating at all, the capsule may not be screwed enough, therefore the rotation speed of the head must be increased.



Make some closure tests before starting the production.



9.3.3 VIBRATING CAPS SORTER LOADING



On the caps sorter is installed a caps level sensor having the task to signal to the automatic elevator when it is necessary to put a new quantity of caps.

For a correct operating, if the caps sorter is loaded with different system from the automatic elevator, it is necessary to supply a quantity of caps that arrives at least 5 mm under the lower plate of the device.



Fig. 33

9.4 EMPTYING OF THE CAPS DISTRIBUTION SYSTEM

9.4.1 CAPS SORTER EMPTYING



To discharge the caps sorter it is necessary to rotate it till totally empty.



9.4.2 CAPS CHUTE DISCHARGE



Before working on the closing machine, isolate and segregate the sources of energy and dissipate the remaining energy.

To discharge the caps chute, operate as follows:

- Disconnect the pneumatic fittings of the cap stop cylinder from their connections.

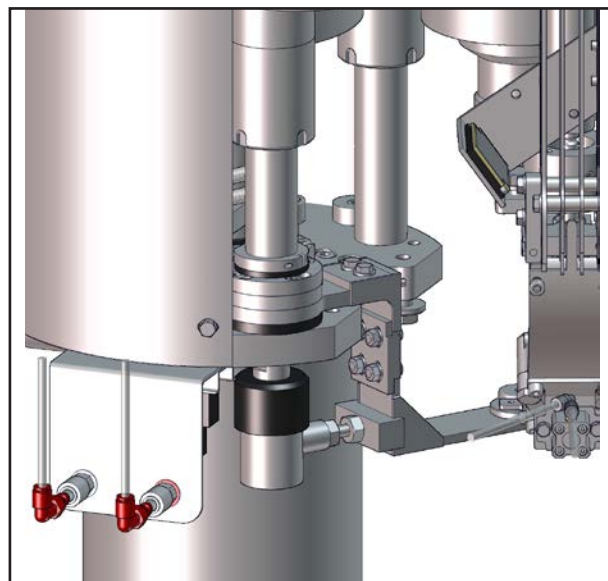


Fig. 34

- Retract the stem of the cap stop cylinder.

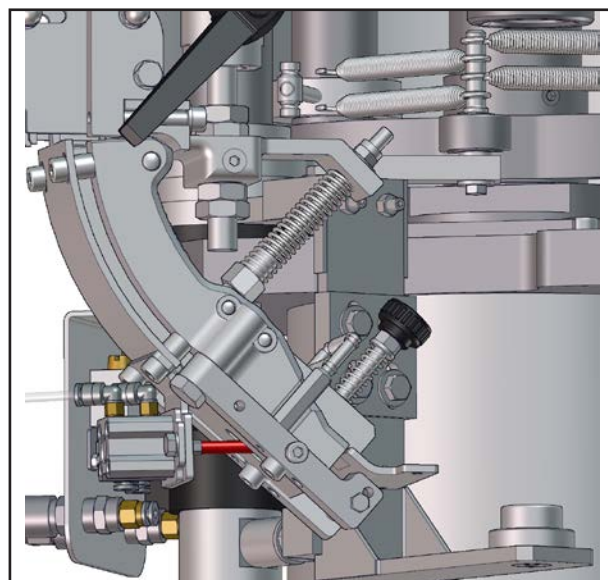


Fig. 35



- Insert a collection container under the cap distribution head.
- Remove the cap taking head springs.

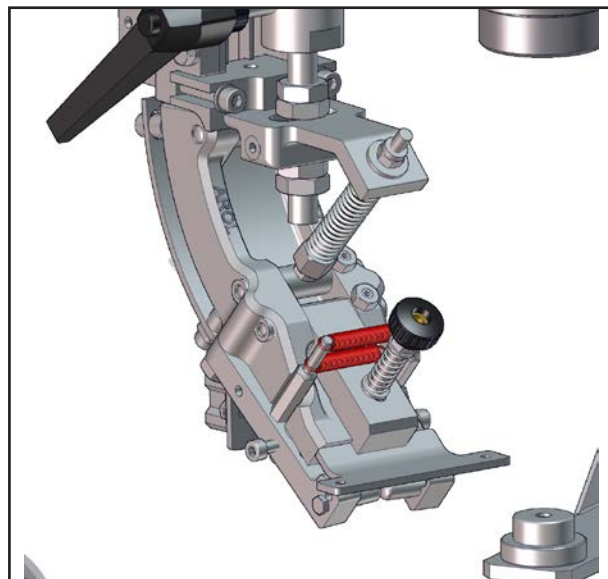


Fig. 36

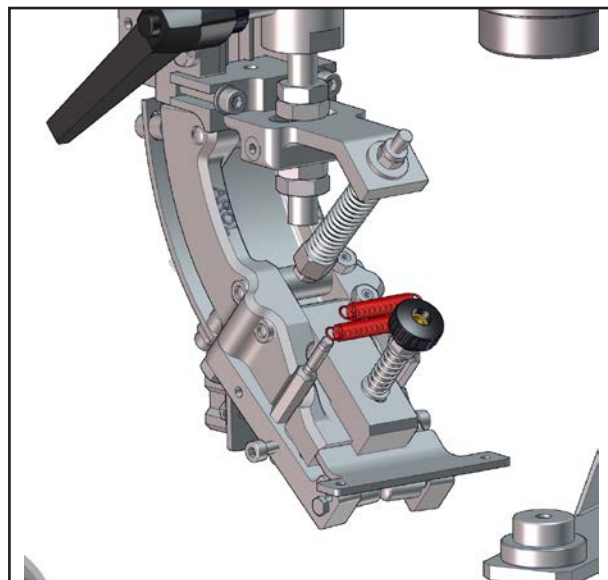


Fig. 37

- Rotate externally the caps stopping hammers so as to allow the discharge of the caps.
- If necessary, empty the cap sorter.
- Empty the cap chute.
- Restore the position of cap stop hammers.
- Mount the cap taking head springs on the spring hooking pins.
- Connect the pneumatic connections of the caps stop cylinder to their fittings.

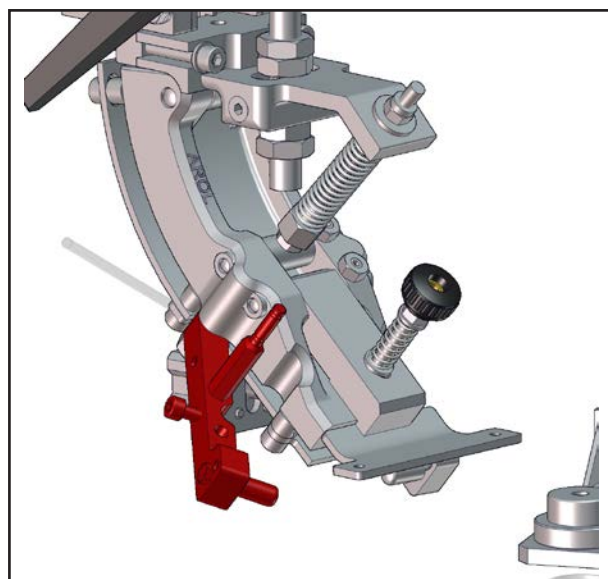


Fig. 38



9.5 NON-CONTINUOUS USE OF THE CLOSING MACHINE



- If the closing machine is not used in a continuous way, before the period in which the machine is not used, it is necessary to:
 - carry out a complete cleaning as described in paragraph 9.7.2 **EXTERNAL WASHING OF THE CLOSING MACHINE** page 44.
 - Lubricate all the moving parts as described in paragraph 11.6.5.2 **LUBRICATION POINT LIST AND SCHEMES** page 131.
 - Let the closing machine operate without corks or bottles for about 10/15 minutes in such a way as to allow for the distribution of lubricant on the moving parts.



If the closing machine has not been used for more than 1 month, before using it again it is necessary to carry out a careful cleaning followed by a further lubrication as previously described.



9.6 BY-PASSING FROM THE WORKING CYCLE



Should it be necessary to have the closing machine by-passed from the working cycle, using the conveyor belt for the container passage, it is necessary:

- Remove the central star-wheel. (See sub-chapter 10.1.1 CENTRAL STAR-WHEEL page 50).
- Remove the conveyor. (See sub-chapter 10.1.2 CONVEYOR page 54).
- Unscrew the pins for gear box using a 13 mm hexagonal wrench and remove the pins.

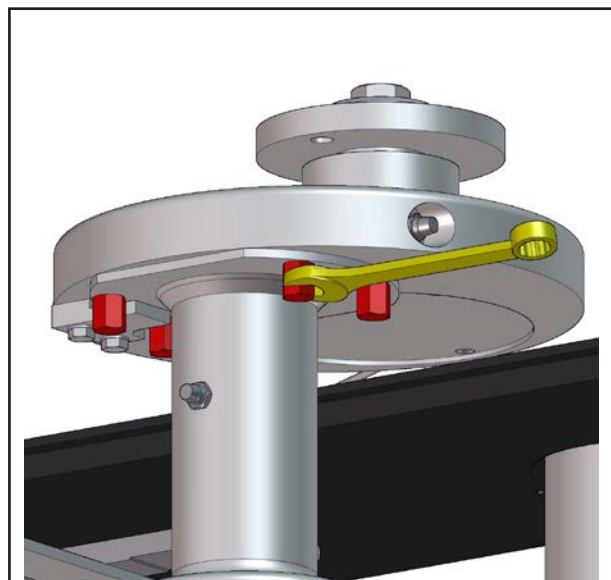


Fig. 39

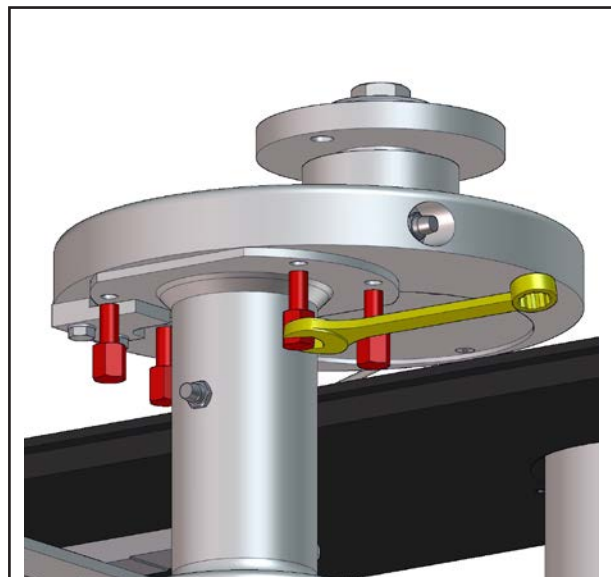


Fig. 40



- Rotate the gear box unit by 180°.

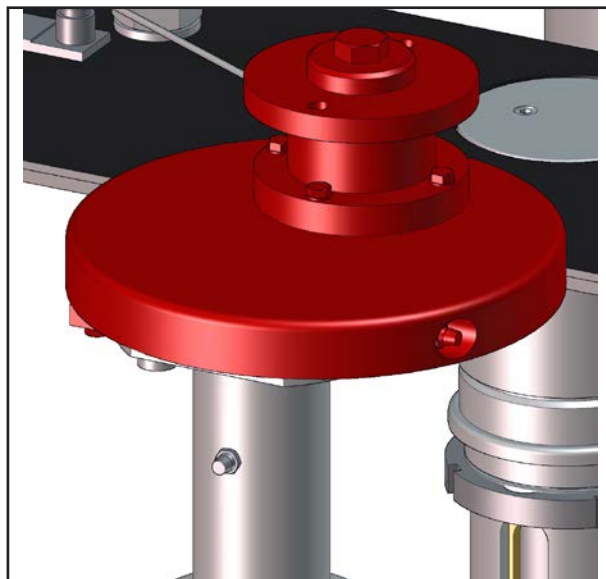


Fig. 41

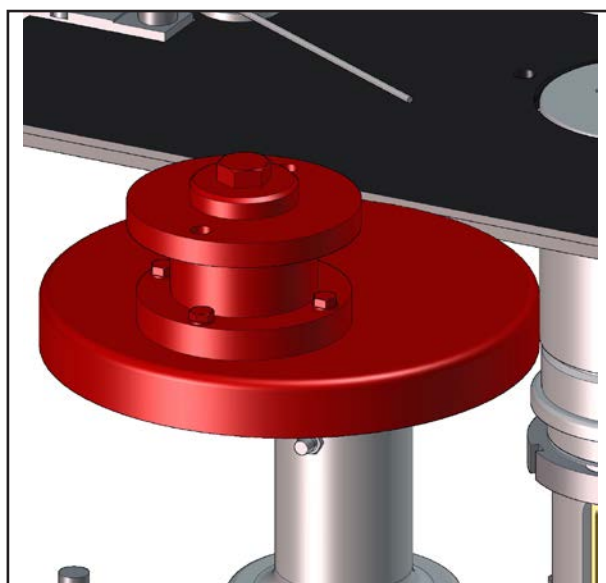


Fig. 42

- Screw 2 pins per gear box using a 13 mm hexagonal wrench in such way to fasten the gear box unit to the star-wheel shaft support.

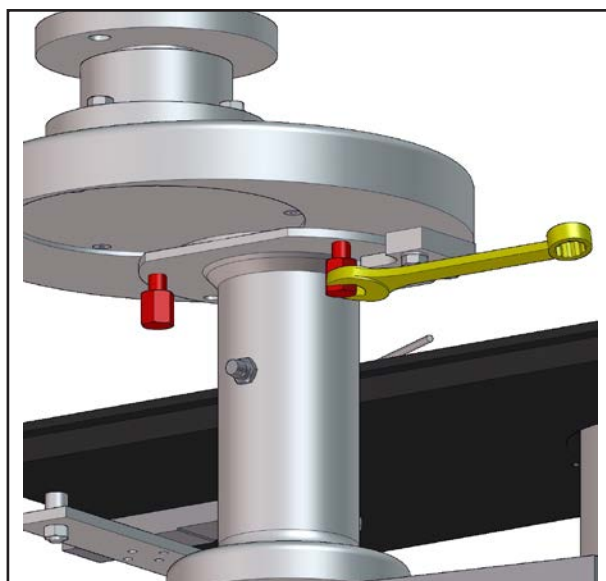


Fig. 43



9.7 CLOSING MACHINE CLEANING



The determination of the frequency of cleaning operations during the closing machine continuous use is to be checked by the user, depending on the environment in which it is being operated.

The following subchapter describes the cleaning intervals of the closing machine and its components, when considering their use in a clean industrial production area.

9.7.1 CLOSING MACHINE CLEANING INTERVALS

The following table shows the cleaning intervals of the closing machine, expressed in hours.

Cleaning	Work shift (hours)		
	1 shift	2 shifts	3 shifts
Closing machine external washing. See sub-chapter 9.7.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 44.	8	16	40
External washing of the closure head. See sub-chapter 9.7.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 44.	8	16	40
External washing of the closure chuck. See sub-chapter 9.7.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 44.	8	16	40
Cork/cap sorter cleaning. See sub-chapter 9.7.2 EXTERNAL WASHING OF THE CLOSING MACHINE page 44.	120	120	120



9.7.2 EXTERNAL WASHING OF THE CLOSING MACHINE



Cleaning and washing are allowed with cold closing machine only.

The description below concerns: cleaning areas, cleaning systems and specifications, products allowed for cleaning the closing machine and its components.

Legend:

- 1 Closure system direction area
- 2 Upper area of the closure carousel
- 3 Closure area
- 4 Bottle-guide and conveying area



The washing jet should be directed from top to bottom. Do not direct air jet towards area "3".

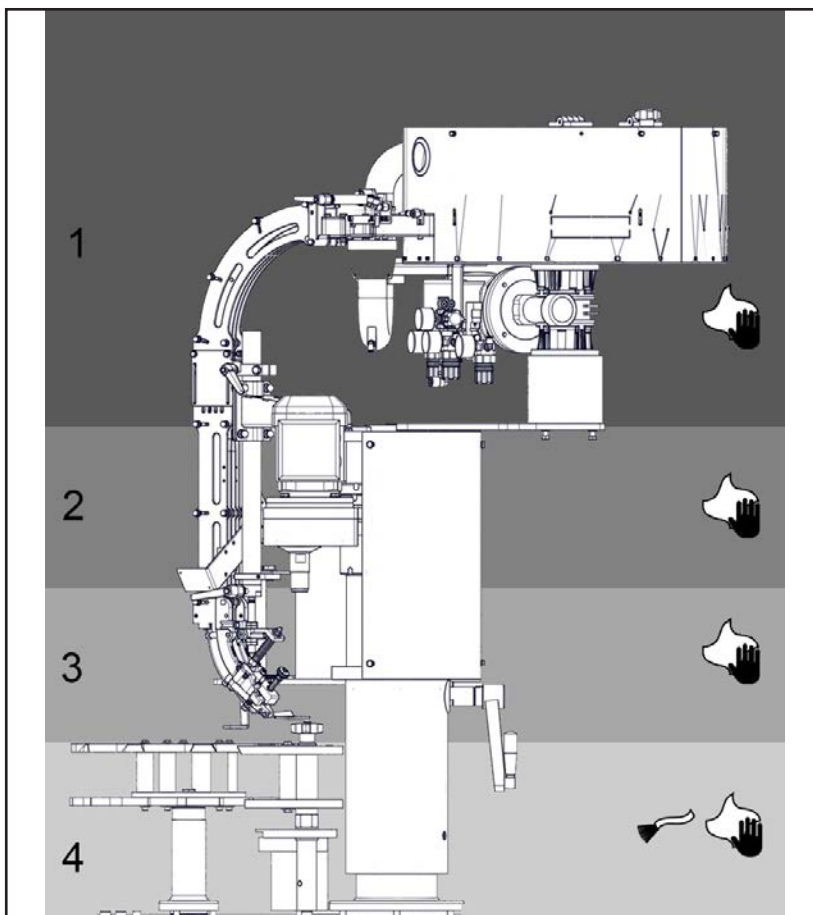


Fig. 44

CLEANING SYMBOL		MAX PRESSURE OF THE CLEANING [bar]	DISTANCE FROM THE SURFACE [mm]	WASHING SYSTEM TILTING [°]
	washing hose	≤ 2,5	-	
	cloth.	-	-	-



PRODUCT ZONE 1 - 2 - 3 - 4	MAX. TEMP. (°C)	pH
Water	35	
Neutral detergents	35	5 ÷ 9

It is absolutely forbidden to clean the closing machine with:

- **water jets under pressure.**
- **aggressive detergents.**
- **steam.**



Before carrying out cleaning, you should refer to the product data sheets relating to detergents.



Before starting production, the closing machine should be suitably rinsed, to remove all the detergents used, then dried up.



IMPORTANT : AROL S.p.A. supplies instructions for washing in order to clean the parts are remove coarse grease stains, powder and scale. On the other hand, it does not guarantee the hygiene of the surfaces, which must be checked by the user depending on the type of treated products, pollutants and disinfection, sanitisation or sterilisation procedures.



Detergents residuals or residuals of liquids used for the cleaning of the closing machine must be disposed according the laws in force of the country in which it is used.



10 SIZE CHANGE



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



The closing machine has been designed to handle containers and caps / corks as indicated in the relevant tables enclosed in this manual.



Every container and/or cap must be processed exclusively with the equipment there indicated.

Each series of equipment is marked according to the size to be processed:

- the container guide equipment are marked with coloured inserts.
- The cap guide equipment, in case of multiple sizes, are marked with an identification initials.

In order to run properly and without obstructions the format change operations, it is necessary to raise the closing turntable at its maximum height.



10.1 CONTAINERS BODY GUIDES EQUIPMENTS SIZE CHANGE



Having to change the size of the container should be treated:

- lift the closure carousel so as to allow the removal of bottles body guide equipment.
- Remove the central star-wheel. (See sub-chapter 10.1.1 CENTRAL STAR-WHEEL page 50).

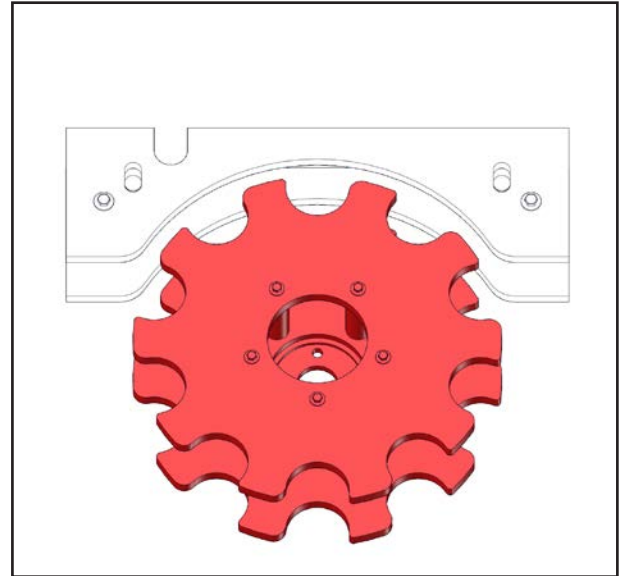


Fig. 45

- Remove the conveyor. (See sub-chapter 10.1.2 CONVEYOR page 54).
- Replace the container guide equipment with those suitable size to be handled in the reverse order to that described above.



To identify the correspondence between container guide unit and the same container, see the TOOLING COLOUR TABLE attached to this manual.

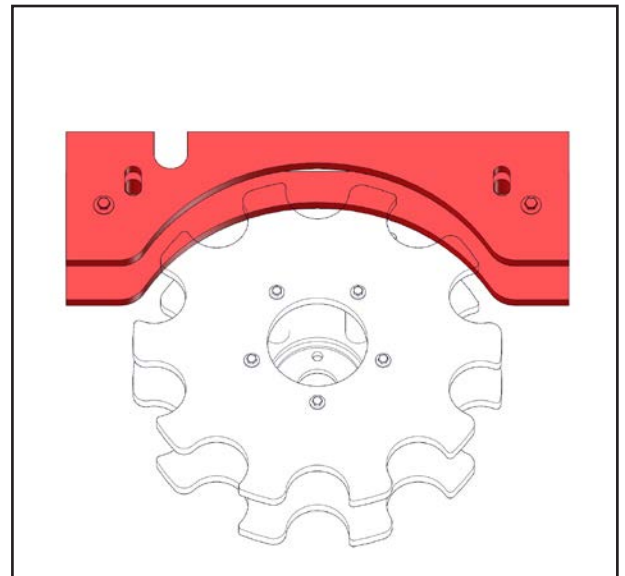


Fig. 46



10.1.1 EQUIPMENT CHANGE FOR CONTAINERS FROM PET TO GLASS AND VICE VERSA



- lift the closure carousel so as to allow the removal of bottles body guide equipment.
- Replace the central star-wheel. (See sub-chapter 10.1.1 CENTRAL STAR-WHEEL page 50).

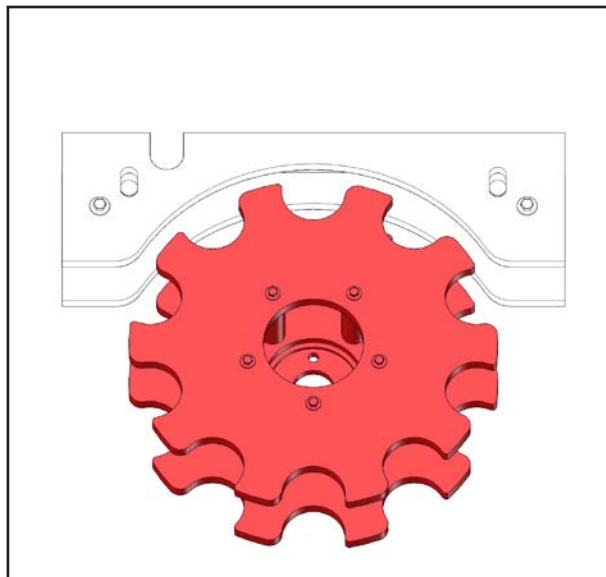


Fig. 47

- Replace the conveyor. (See sub-chapter 10.1.2 CONVEYOR page 54).

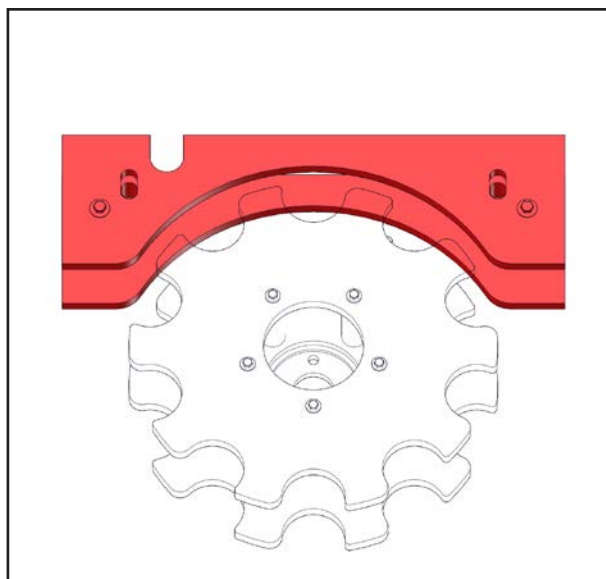


Fig. 48



- Enable or disable the functioning of the containers anti-rotation device. (See sub-chapter 10.1.3 CONTAINER ANTI-ROTATION DEVICE page 56).

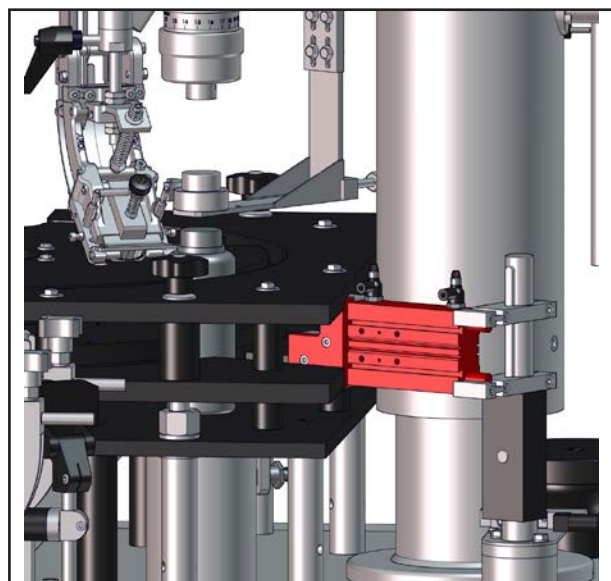


Fig. 49

- Replace the bottles supporting plate. (See sub-chapter 10.1.4 REPLACEMENT OF THE BOTTLE SUPPORTING PLATE page 58).



To identify the correspondence between container guide unit and the same container, see the TOOLING COLOUR TABLE attached to this manual.

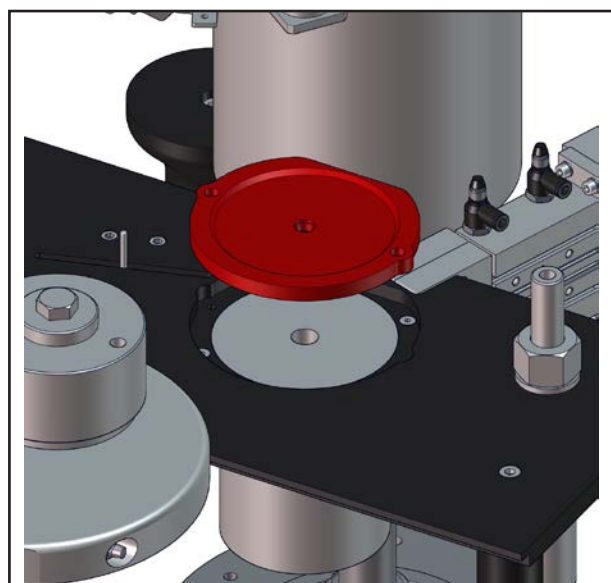


Fig. 50

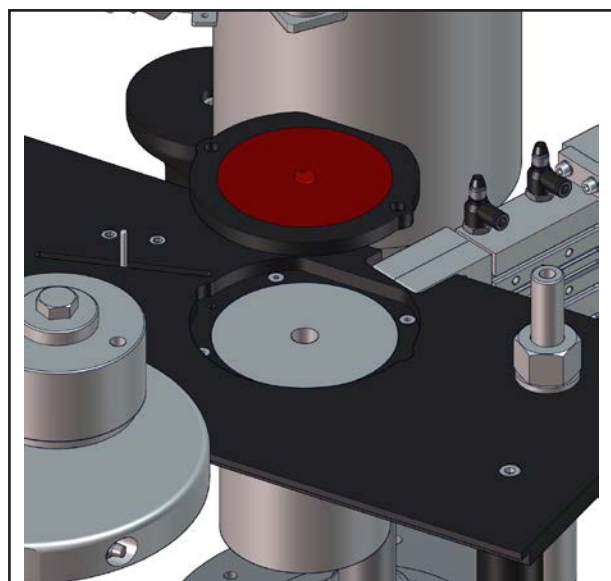


Fig. 51



10.1.1 CENTRAL STAR-WHEEL



10.1.1.1 REMOVAL

EQUIPMENT FOR PET CONTAINERS

To remove proceed in the following way:

- Unscrew and remove the fixing bolts.

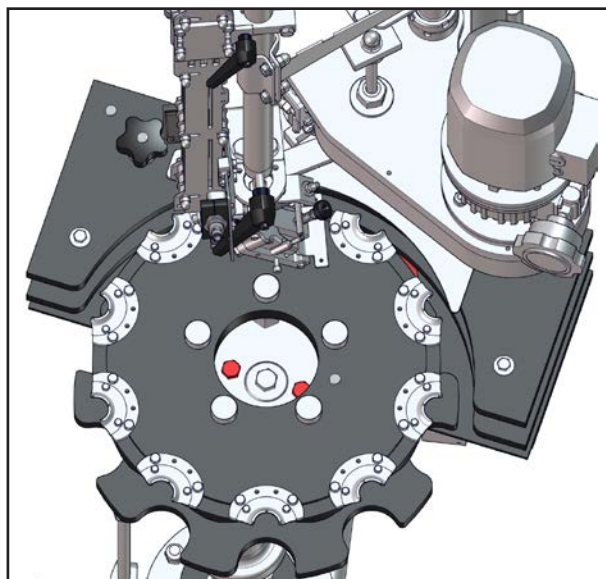


Fig. 52

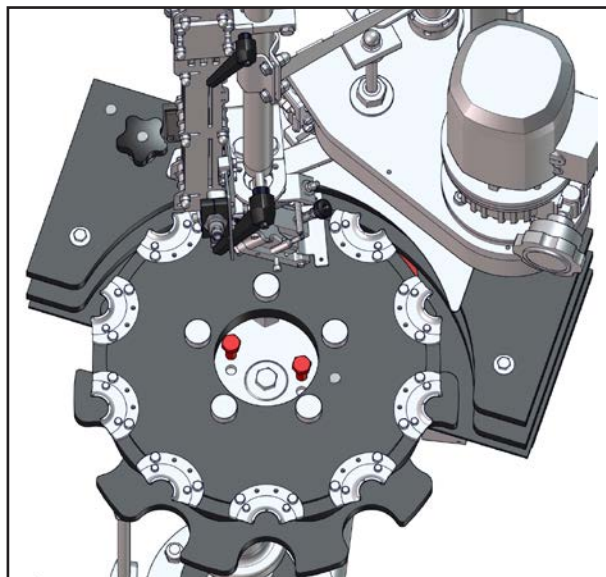


Fig. 53

- Remove the central star-wheel from the support hub.

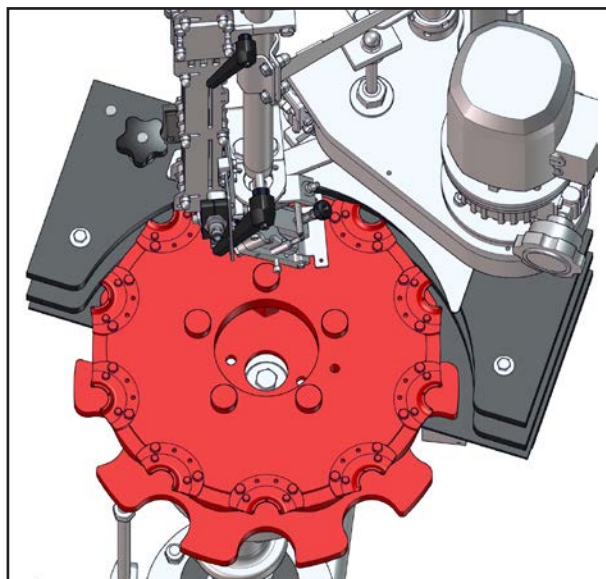


Fig. 54

EQUIPMENT FOR GLASS CONTAINERS

To remove proceed in the following way:

- Unscrew and remove the fixing bolts.

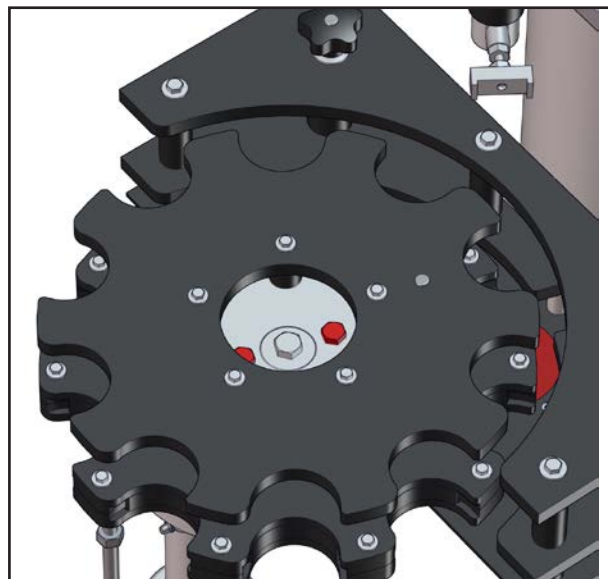


Fig. 55

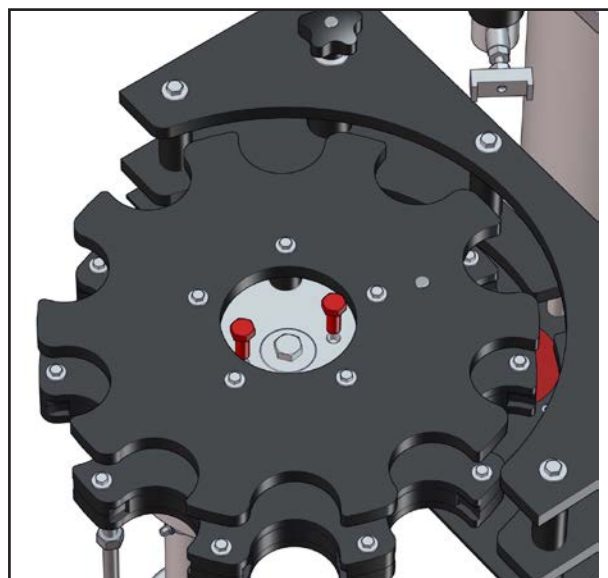


Fig. 56

- Remove the central star-wheel from the support hub.

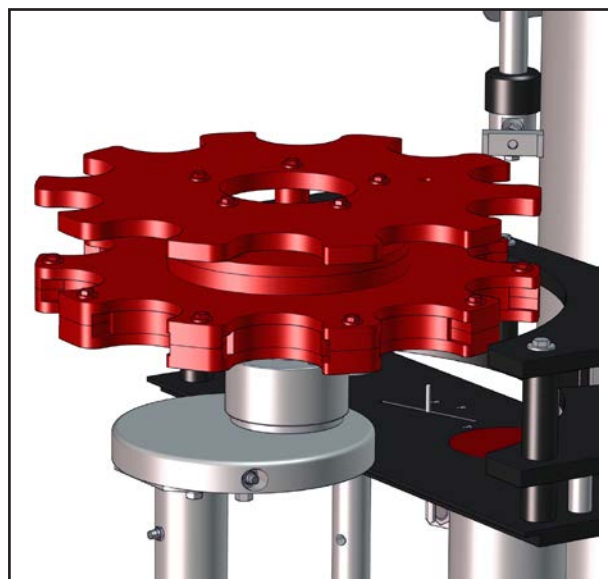


Fig. 57



10.1.1.2 INSTALLATION

EQUIPMENT FOR PET CONTAINERS

For installation proceed as follows:

- Position the central star-wheel on the support hub.

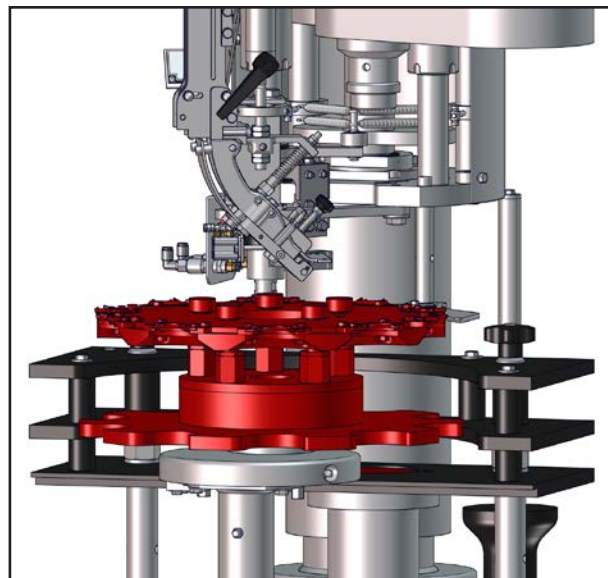


Fig. 58

- Lock the central star-wheel through the fixing bolts.

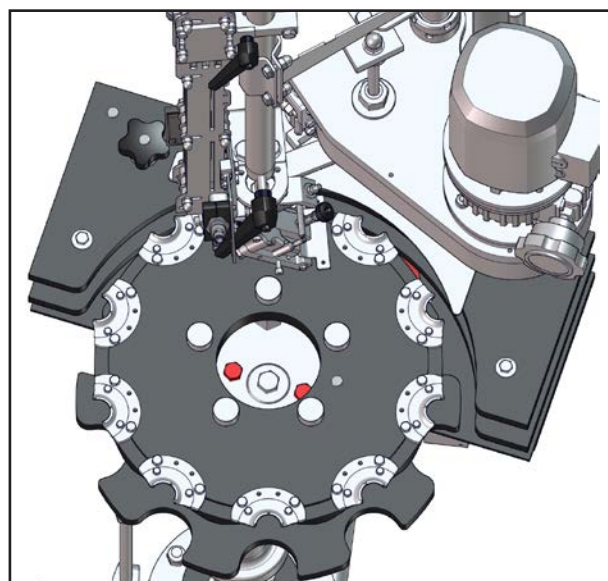


Fig. 59



EQUIPMENT FOR GLASS CONTAINERS

For installation proceed as follows:

- Position the central star-wheel on the support hub.

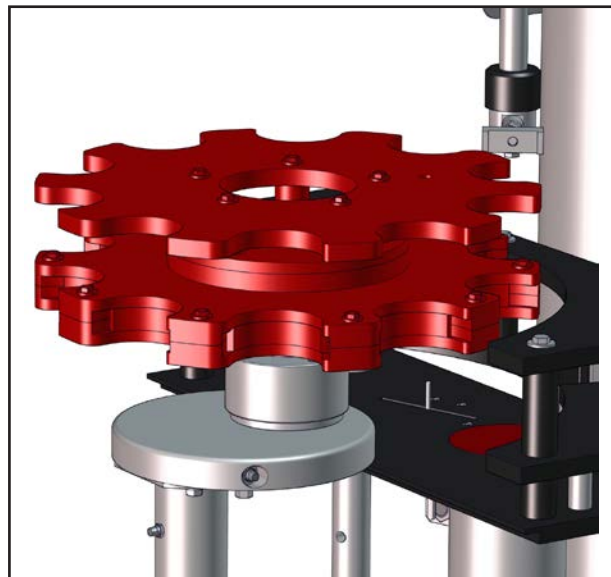


Fig. 60

- Lock the central star-wheel through the fixing bolts.

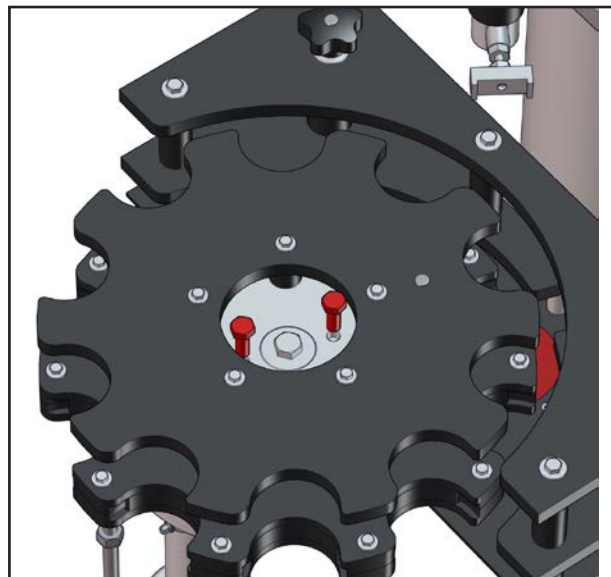


Fig. 61



10.1.2 CONVEYOR



10.1.2.1 REMOVAL

To remove proceed in the following way:

- Untighten conveyor fastening knobs.

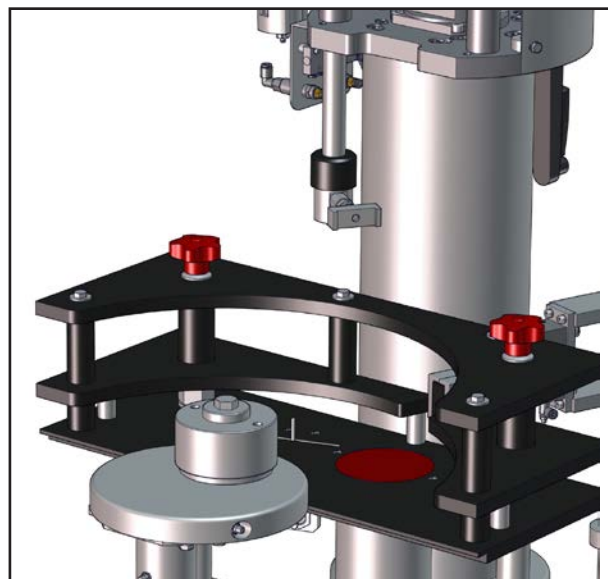


Fig. 62

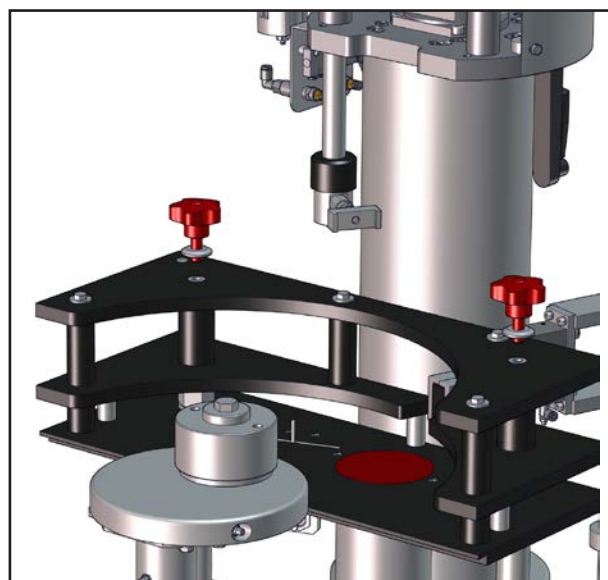


Fig. 63

- Lift and extract the conveyor

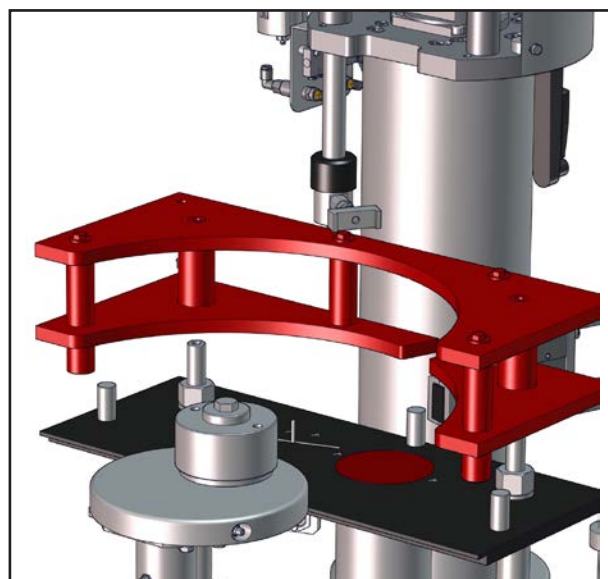


Fig. 64

10.1.2.2 INSTALLATION

For installation proceed as follows:

- Position the conveyor on the support columns.

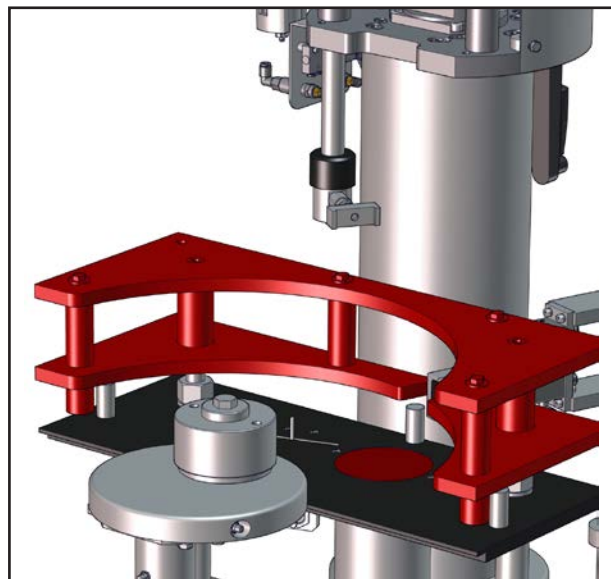


Fig. 65

- Fasten the conveyor by knobs.

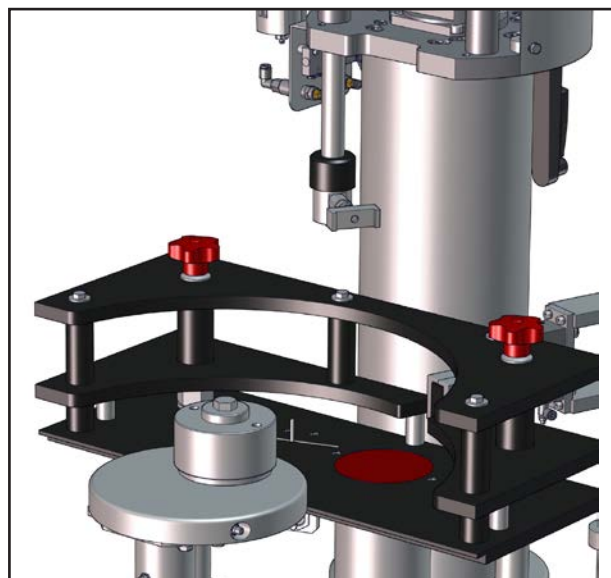


Fig. 66



10.1.3 CONTAINER ANTI-ROTATION DEVICE



10.1.3.1 ACTIVATION OF THE CONTAINER ANTI-ROTATION DEVICE

It may be necessary to use the anti-rotation device depending on the format of container to be processed.

To enable the antirotation device it is necessary to activate the functioning of the container antirotation cylinder acting on the relative commands.

If necessary, replace the piston position spacer with the one suitable for the format of container to be processed:

- loosen the fastening screws of the cylinder-holding brackets.

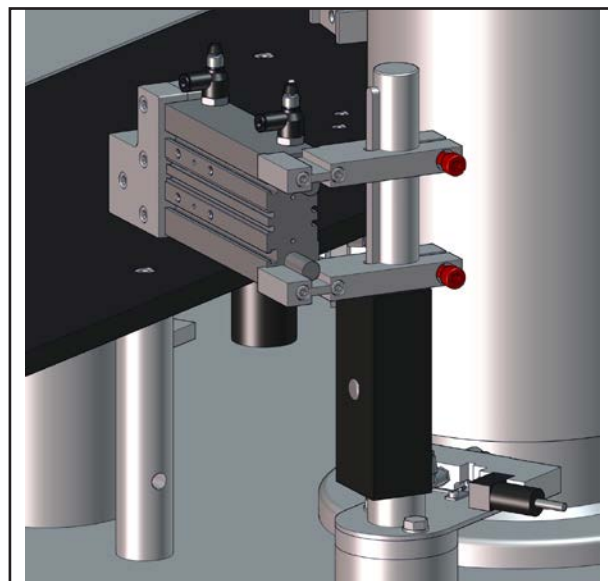


Fig. 67

- Lift the container anti-rotation cylinder unit.

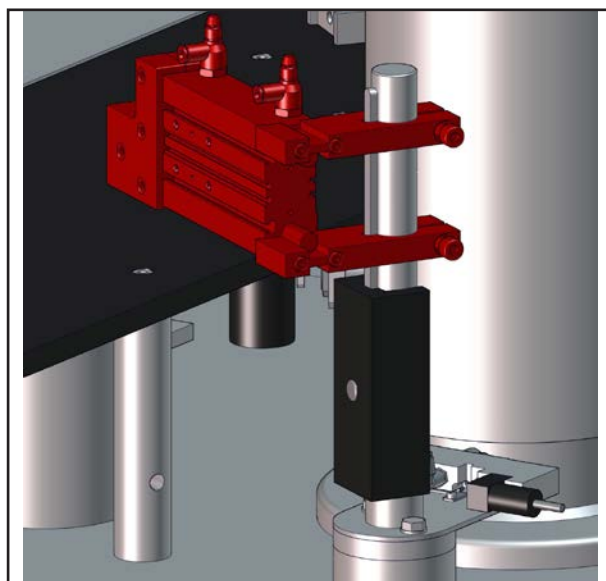


Fig. 68



- Remove and replace the piston supporting spacer.



To identify the correspondence between piston supporting spacer and the relevant container, see the TOOLING COLOUR TABLE attached to this manual.

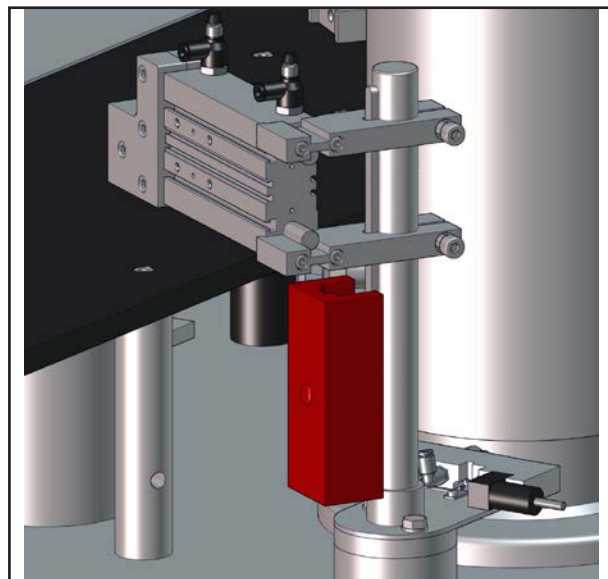


Fig. 69

- Lay the anti-rotation cylinder unit on the new piston supporting spacer.
- Tighten the fastening screws of the previously loosened cylinder-holding brackets.

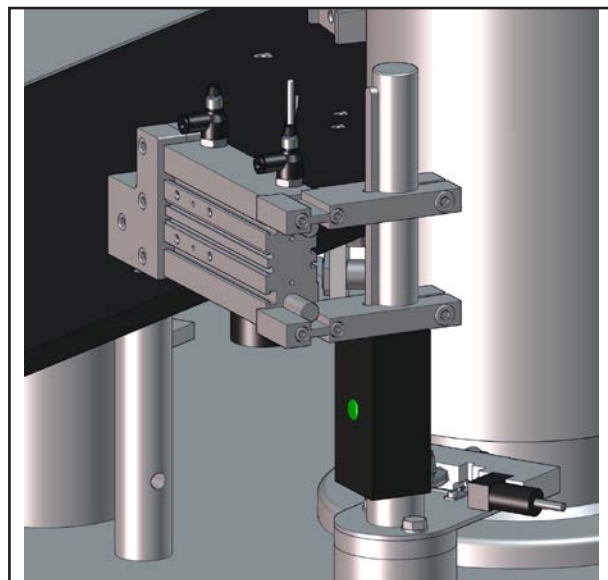


Fig. 70

10.1.3.2 DE-ACTIVATION OF THE CONTAINER ANTI-ROTATION DEVICE

To activate the operation of the anti-rotation device, it is necessary to cut off the operating of the bottles anti-rotation cylinder, acting on the specific controls.



10.1.4 REPLACEMENT OF THE BOTTLE SUPPORTING PLATE

To replace the bottle supporting plate proceed as follows:



Before working on the closing machine, isolate and segregate the sources of energy and dissipate the remaining energy.

- Lift the closing carousel in order to allow the removal of the bottle guide equipment as described in paragraph 9.3.1 CLOSING MACHINE HEIGHT ADJUSTMENT page 34.
- Remove the central star-wheel. (See sub-chapter 10.1.1 CENTRAL STAR-WHEEL page 50).
- Remove the conveyor. (See sub-chapter 10.1.2 CONVEYOR page 54).

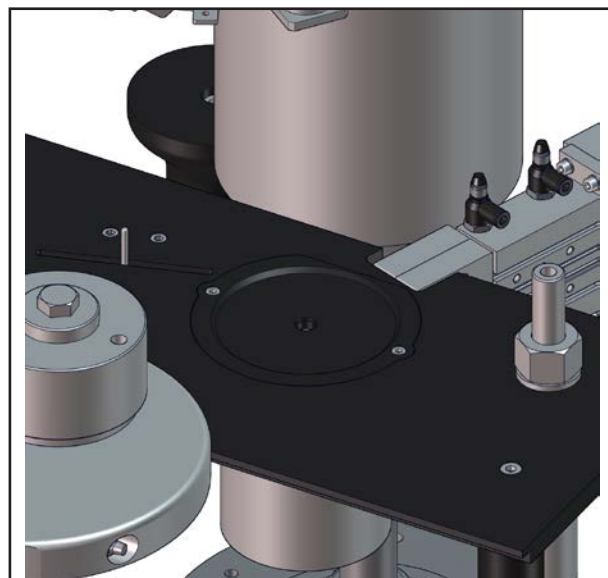


Fig. 71

- Unscrew the Allen screws using a 5 mm Allen wrench.

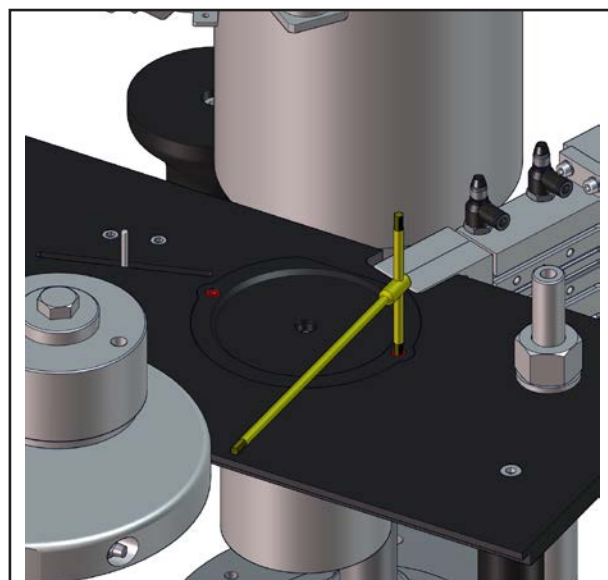


Fig. 72



- Remove the fixing Allen screws.

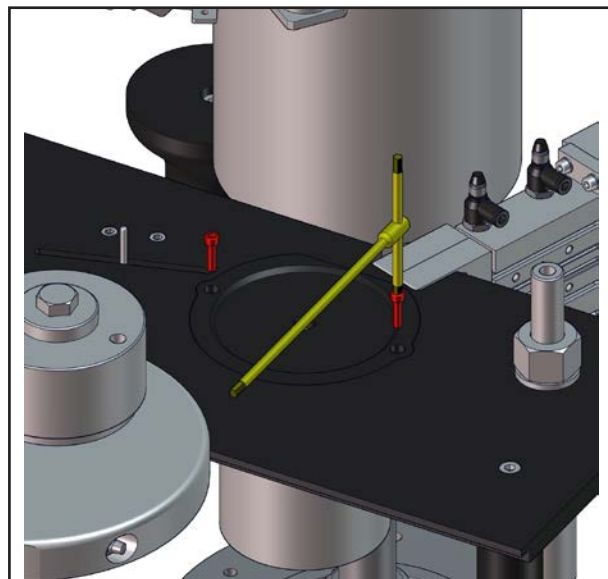


Fig. 73

- Lift and remove the bottle supporting plate.
- Replace the bottle supporting plate and fasten again proceeding in the reverse order of the above.

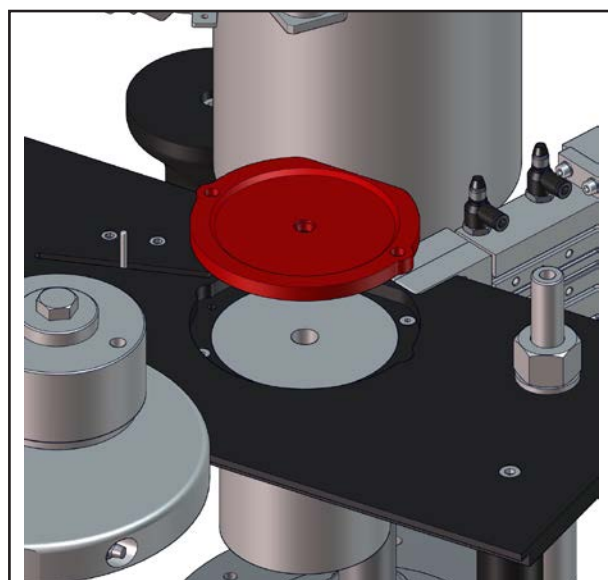


Fig. 74

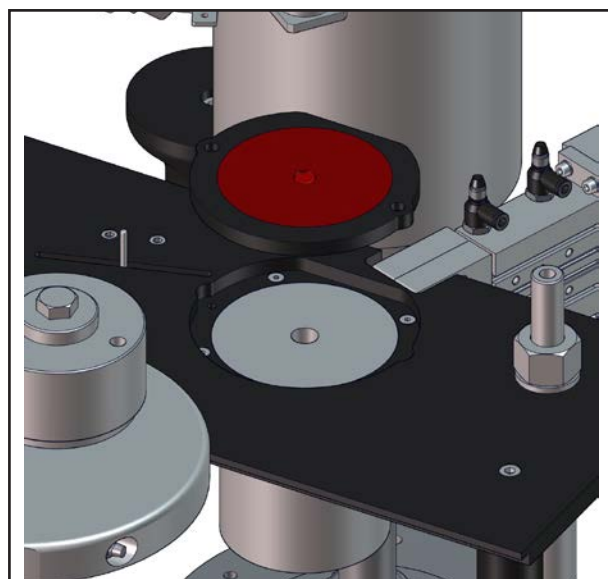


Fig. 75



10.2 CAP SIZE CHANGE



Having to change the format of the caps to be treated, you must:

- replace the closure chuck. (See sub-chapter 10.2.5 REMOVAL AND INSTALLATION OF THE CLOSURE CONE page 62).
- Replace the caps chute, if provided. (See sub-chapter 10.2.7 CAPS CHUTE page 69).
- If required, replace the caps transport pin. (See sub-chapter 10.2.8 CAPS TRANSFER DEVICE page 75).



Fig. 76

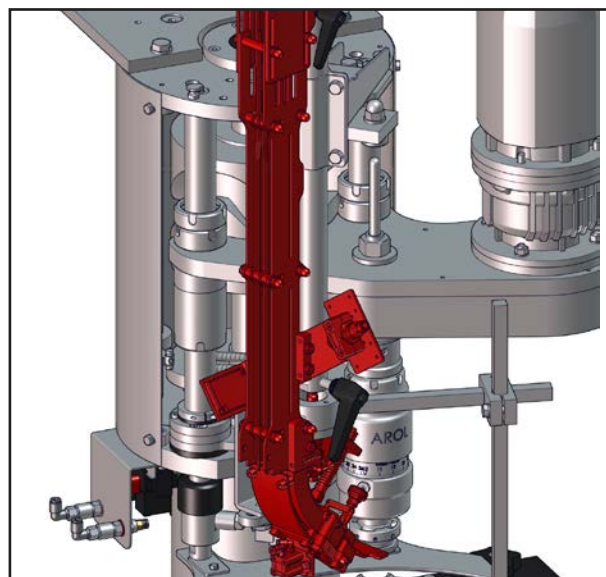


Fig. 77

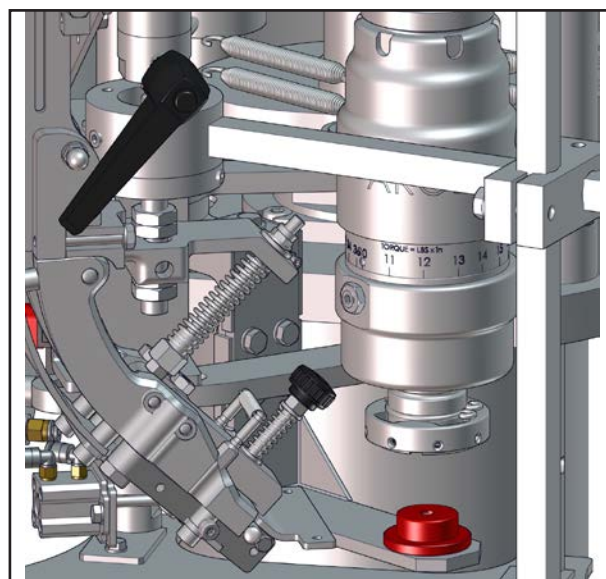


Fig. 78



- If provided, replace the tank of the caps sorter. (See sub-chapter 10.2.9 CAP SELECTION EQUIPMENT FOR VIBRATION ORIENTATOR page 77).



Fig. 79



10.2.5 REMOVAL AND INSTALLATION OF THE CLOSURE CONE

10.2.5.1 REMOVAL OF THE CLOSURE CHUCK "PK314 L"

The table below shows the dedicated tools for disassembly and assembly of the closure chuck.

TOOLS	
	Hexagonal wrench 34 mm

To remove the closure chuck, operate as follows:

- fit the slim hook wrench in the hole of the ejector guide bush.

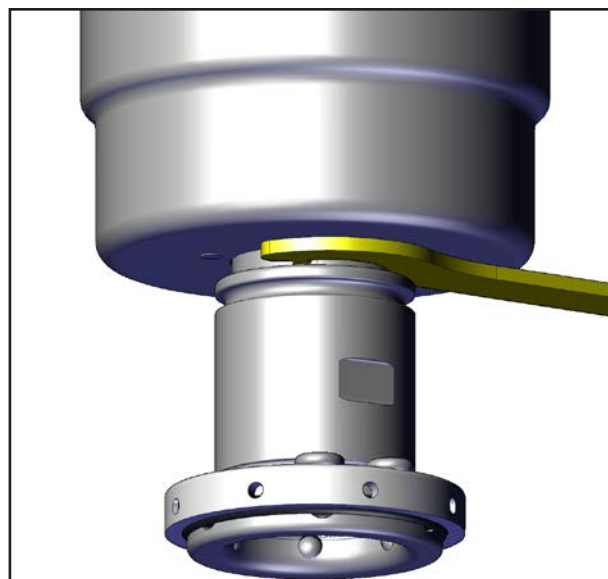


Fig. 80

- Fit the 34 mm hexagonal wrench in the seats of the closure chuck.
- By means of the two wrenches, unscrew the closure chuck; if the closure chuck is locked, it is sufficient to give a light stroke on the handle of the hexagonal wrench.

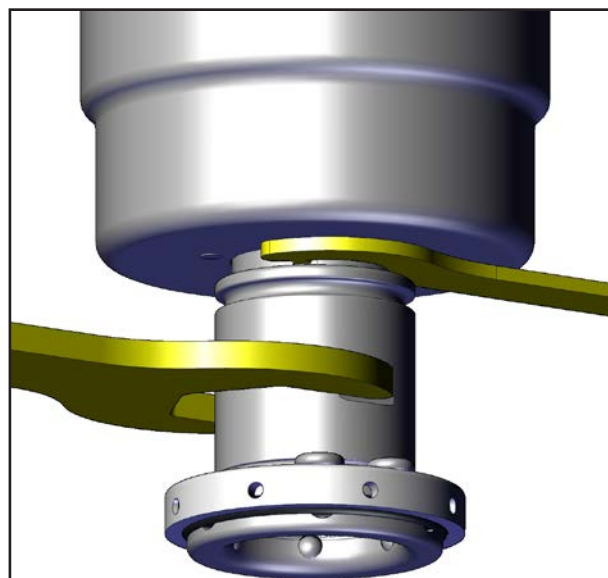


Fig. 81



- Remove the closure chuck unscrewing by hand.

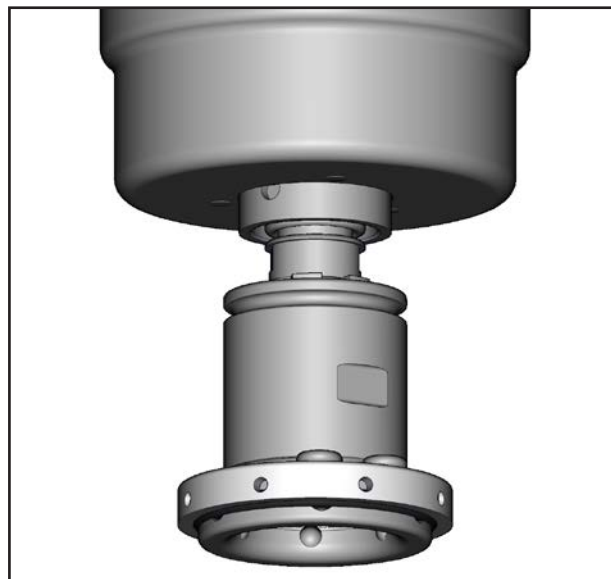


Fig. 82



10.2.5.2 INSTALLATION OF THE CLOSURE CHUCK "PK314 L"

To install the closure chuck, proceed as follows:

- Screw by hand the chuck on ejector guiding bush.

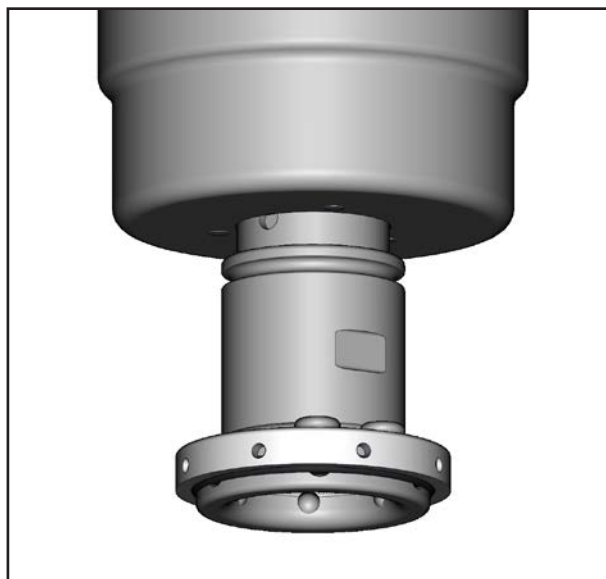


Fig. 83

- fit the slim hook wrench in the hole of the ejector guide bush.

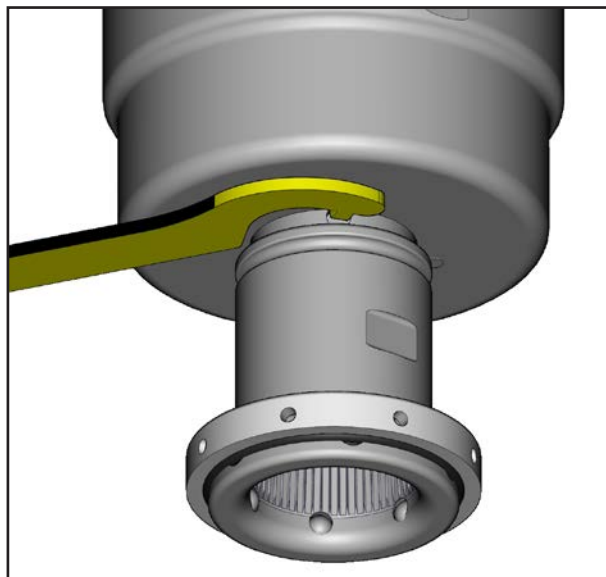


Fig. 84

- Fit the 34 mm hexagonal wrench in the seats of the closure chuck.
- Using the two wrenches, lock the closure chuck.

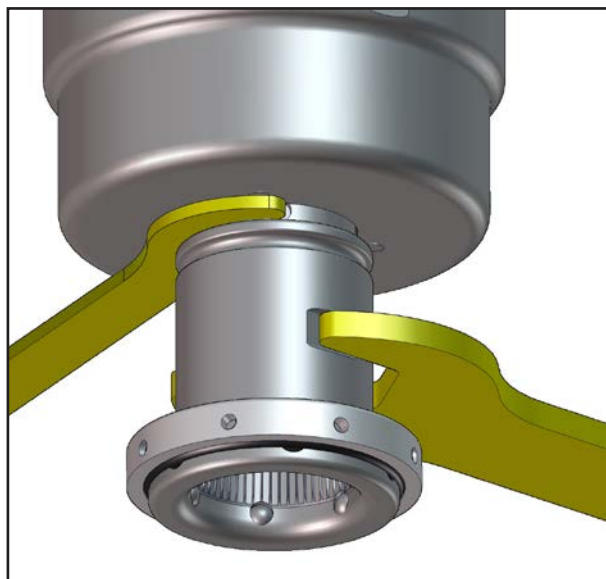


Fig. 85



10.2.6 REMOVAL AND INSTALLATION OF THE CLOSURE HEAD

The table below shows the dedicated tools for disassembly and assembly of the closure head.

TOOLS	
	Hook wrench 58-62
	Hook wrench 45-50

10.2.6.1 "PK311" CLOSURE HEAD REMOVAL

To remove the closure head from the spindle body, proceed as follows:

- Insert a 45-50 hook wrench in the hole of the head extension.

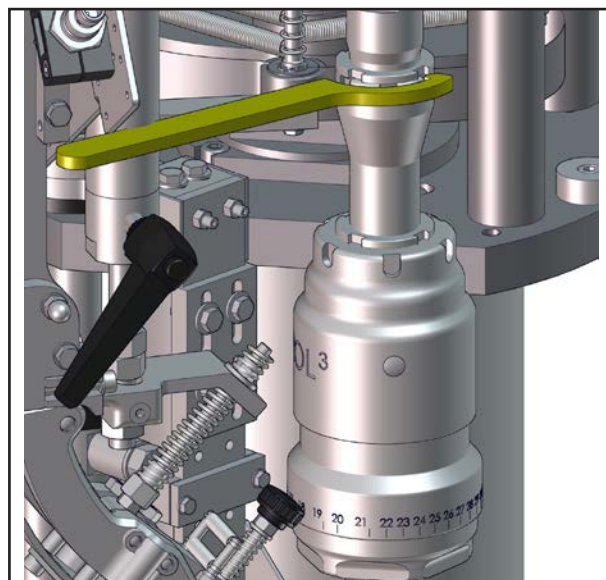


Fig. 86

- Insert 45-50 hook wrench in a seat of the head fixing ring-nut.
- Hold the 45-50 hook wrench into the seat of head extension and loosen the fastening ring nut using a 45-50 hook wrench.

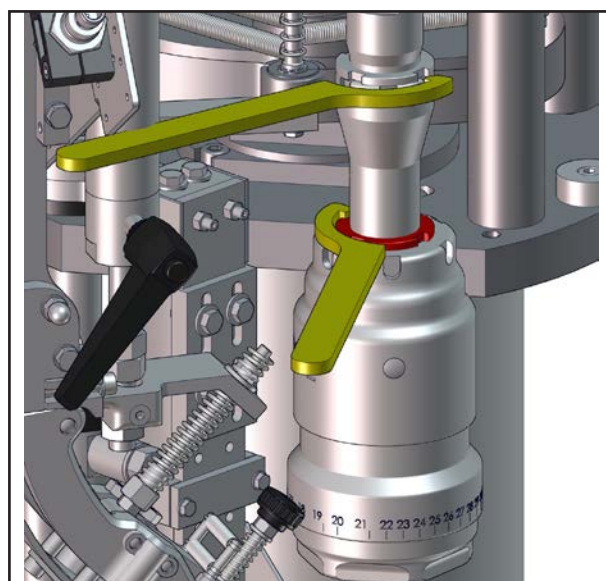


Fig. 87



- Insert a 45-50 hook wrench in the hole of the head extension.
- Insert the 58-62 hook wrench in the seats of the closure head.
- Hold the 45-50 hook wrench into the seat of head extension and loosen the closure head with the 58-62 hook wrench.

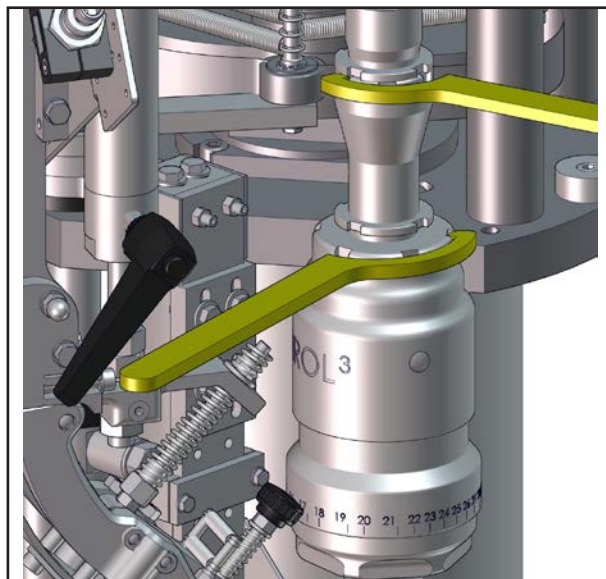


Fig. 88

- Pull the closing head down so as to remove it from the spindle.

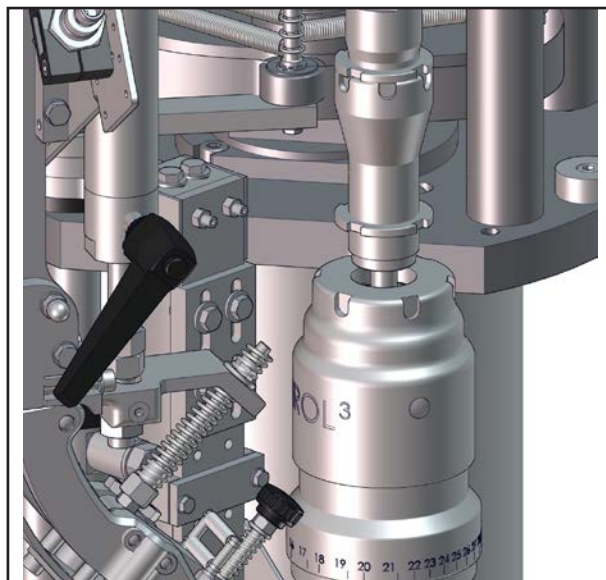


Fig. 89



When the closure head is removed, the ejector rod tends to remove as well: support it during the removal.

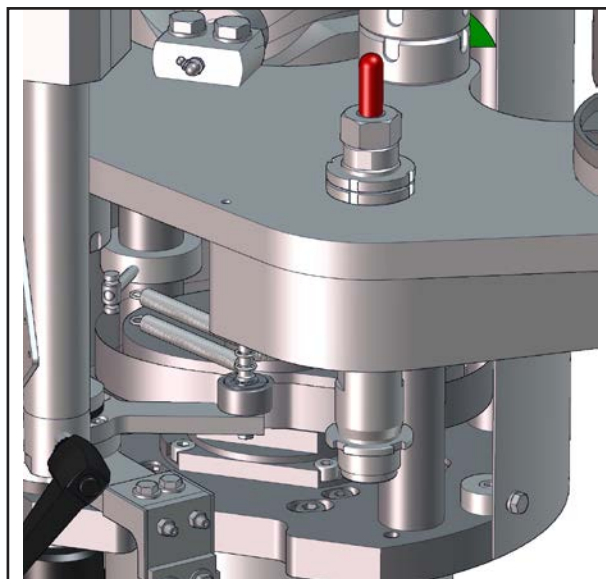


Fig. 90

10.2.6.2 "PK311" CLOSURE HEAD INSTALLATION

To install the closure head on the spindle body, proceed as follows:

- Insert the ejector rod inside the spindle body and support it until starting to screw the closure head

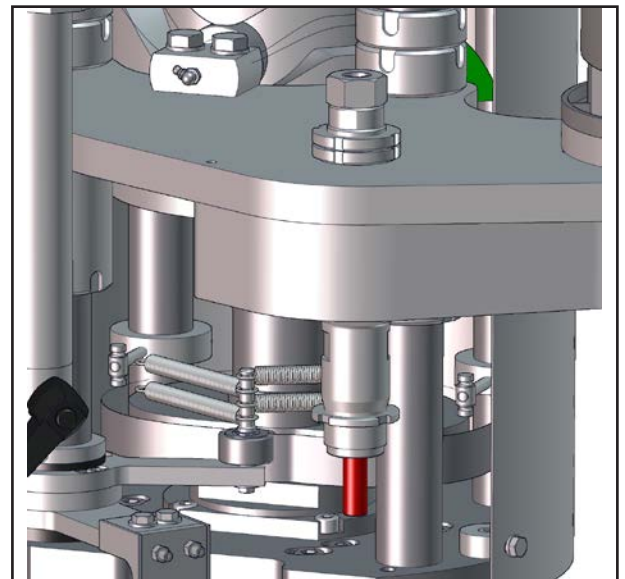


Fig. 91

- Insert the 58-62 hook wrench in the seats of the closure head.

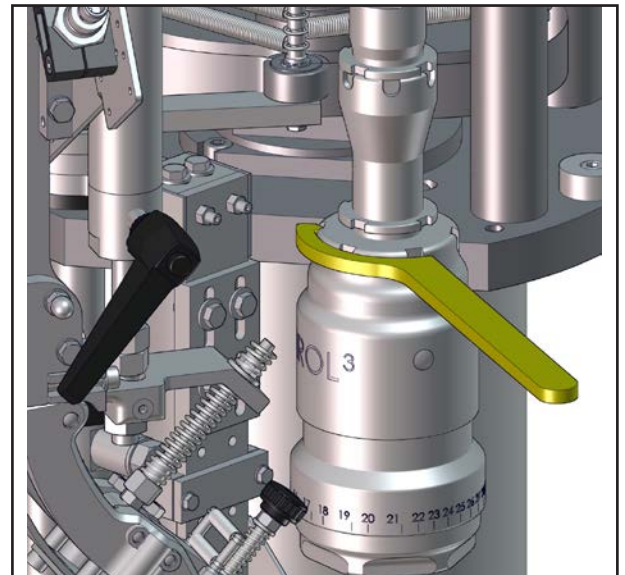


Fig. 92

- Manually fit the thread of the closing head on the body of the spindle.

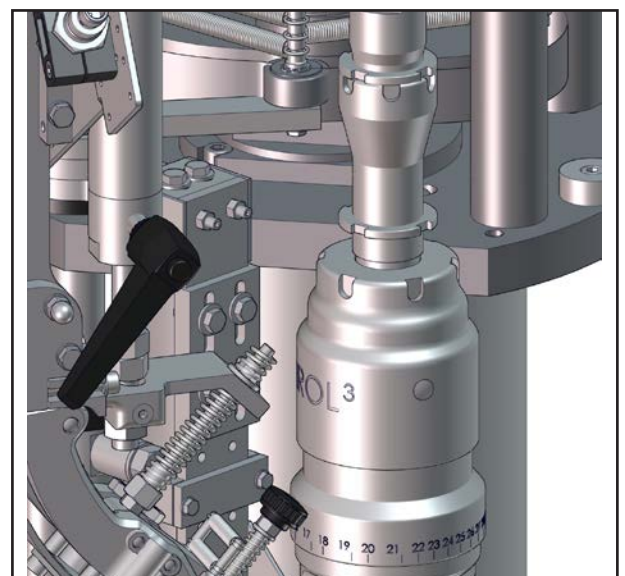


Fig. 93



- Insert a 45-50 hook wrench in the hole of the head extension.
- **non tradotto**

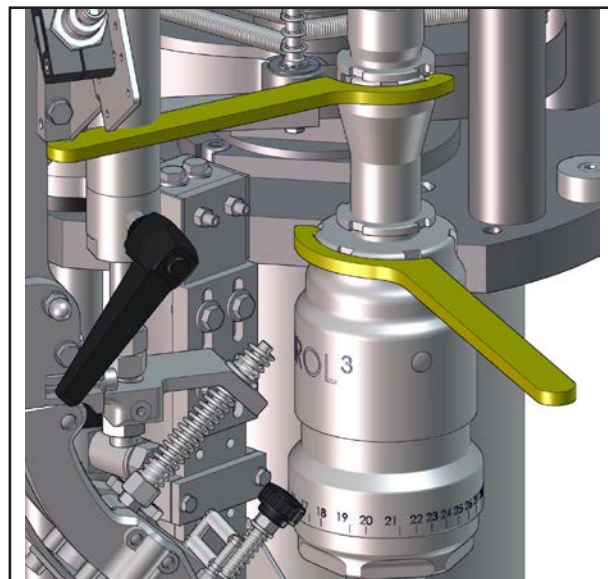


Fig. 94

- Insert a 45-50 hook wrench in the hole of the head extension.
- Insert 45-50 hook wrench in a seat of the head fixing ring-nut.
- Hold the 45-50 hook wrench into the seat of head extension and tighten the fastening ring nut using a 45-50 hook wrench.

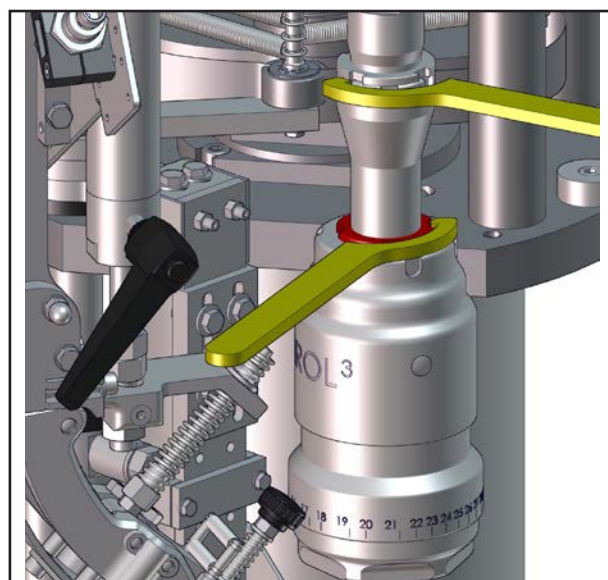


Fig. 95



10.2.7 CAPS CHUTE

10.2.7.1 CAPS CHUTE REMOVAL

Remove the caps chute in the following way:

- Disconnect the pneumatic connections of the caps stop cylinder from their fittings.

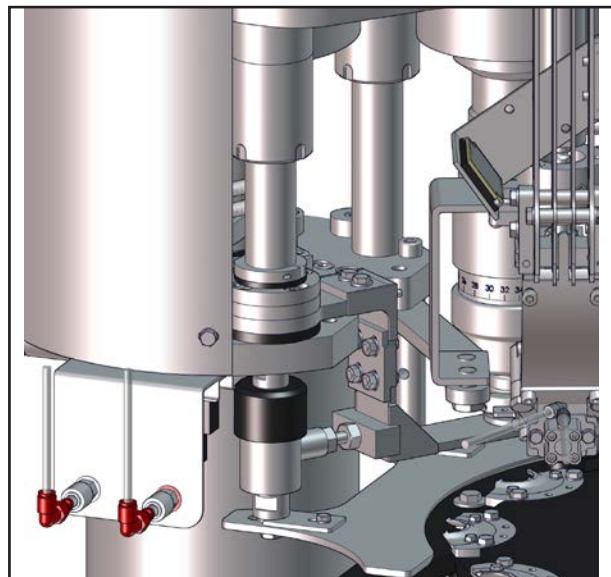


Fig. 96

- Disconnect the electrical connection from the caps presence photocell (one or more).

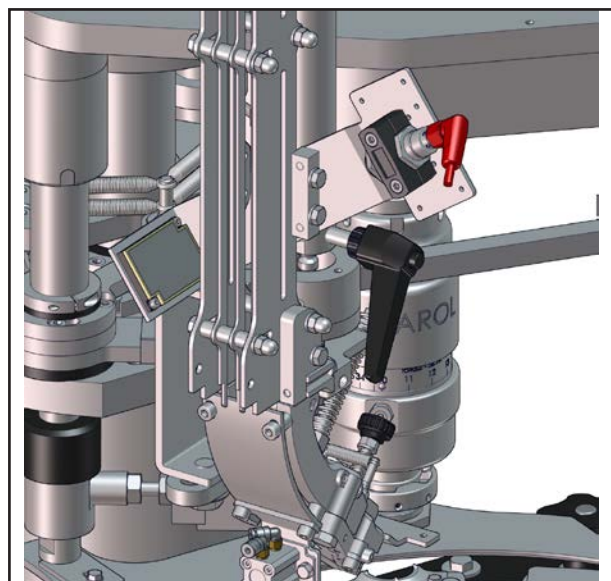


Fig. 97



- Loosen the return handles located on the support chute.

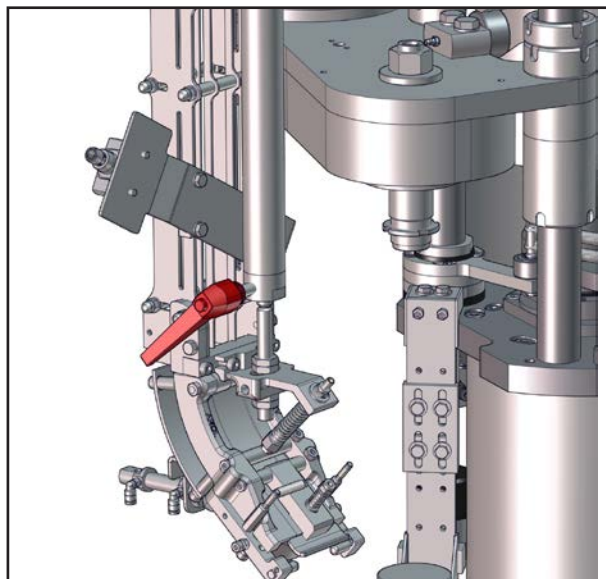


Fig. 98

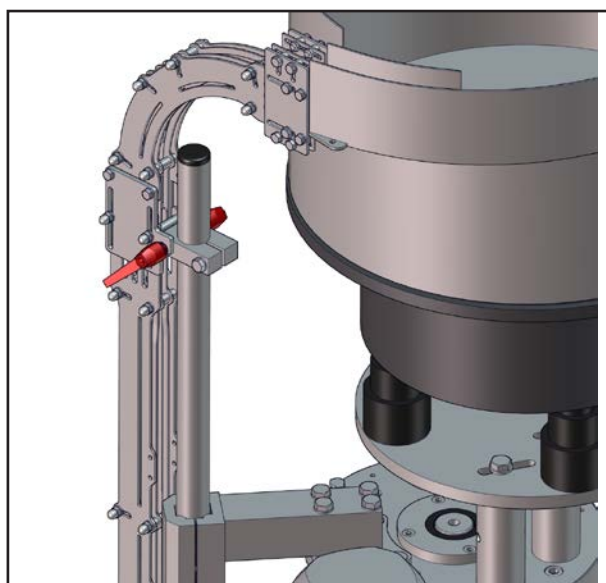


Fig. 99

- Lift the complete caps chute, in order to extract it from the pins of the return handles.

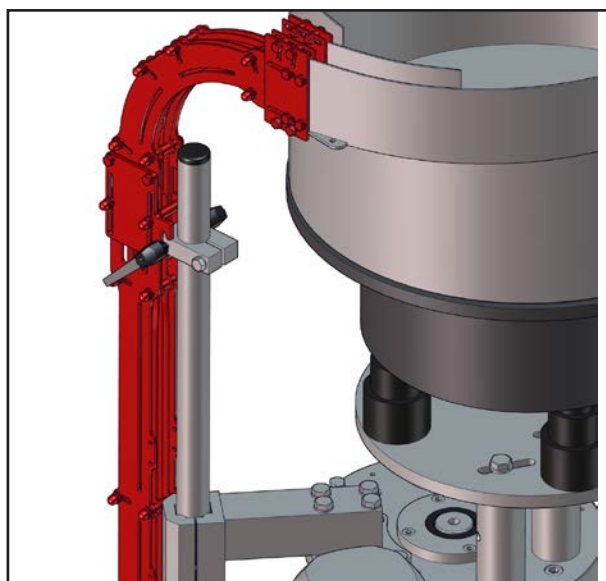


Fig. 100



- Successively lower the caps chute so that the head height adjustment pin come out from the hole of the chute support, then remove the chute.

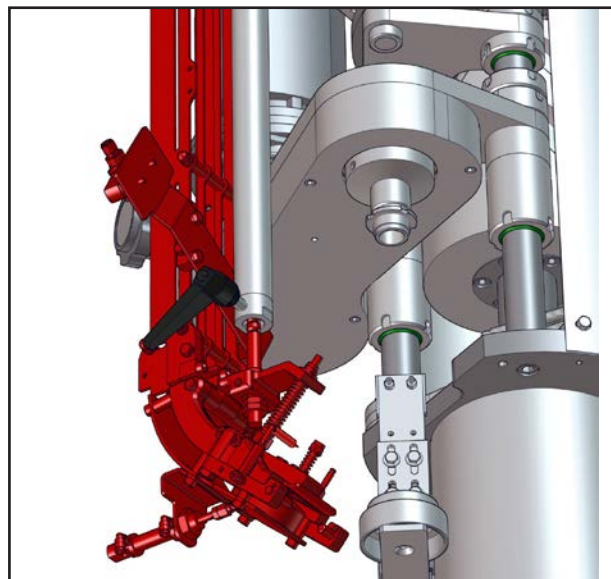


Fig. 101



10.2.7.2 CAPS CHUTE INSTALLATION

To install the caps chute on the machine it is necessary:

- Loosen the return handles located on the support chute.

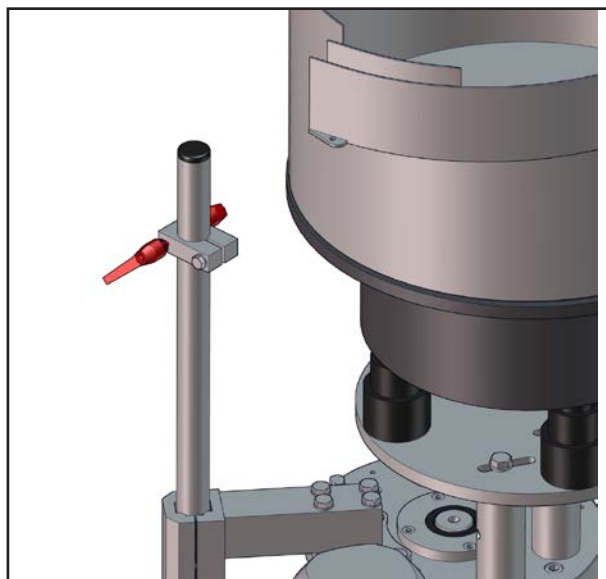


Fig. 102

- Insert the head height adjustment pin into the hole of the chute support and lift the complete caps chute.

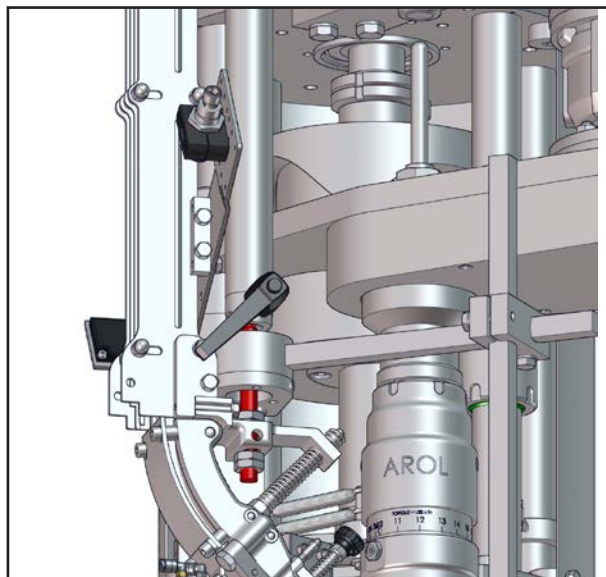


Fig. 103

- Place the caps chute so that the fixing brackets are positioned on the pins of the return handles.

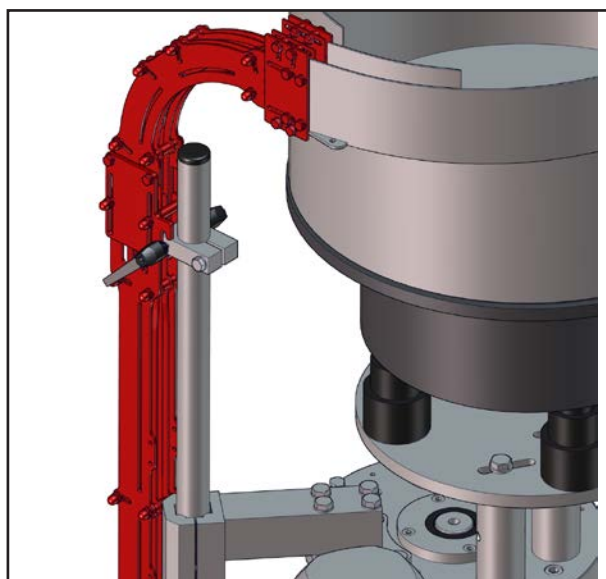


Fig. 104



- Lock the return handles located on the caps chute support.

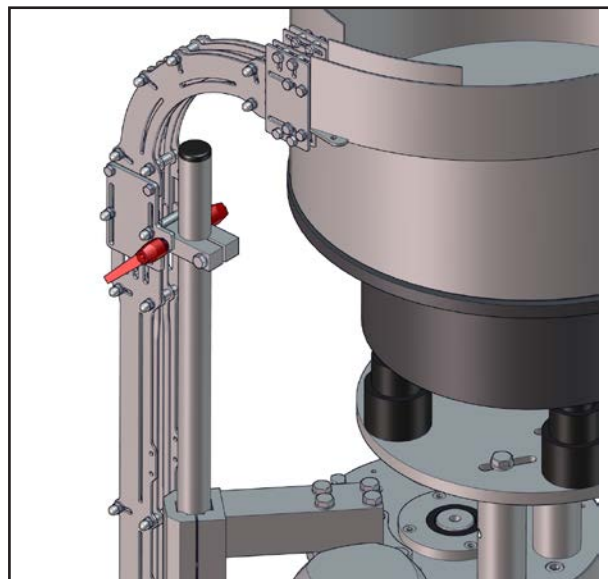


Fig. 105

- Lock the return handles of the caps distribution head.

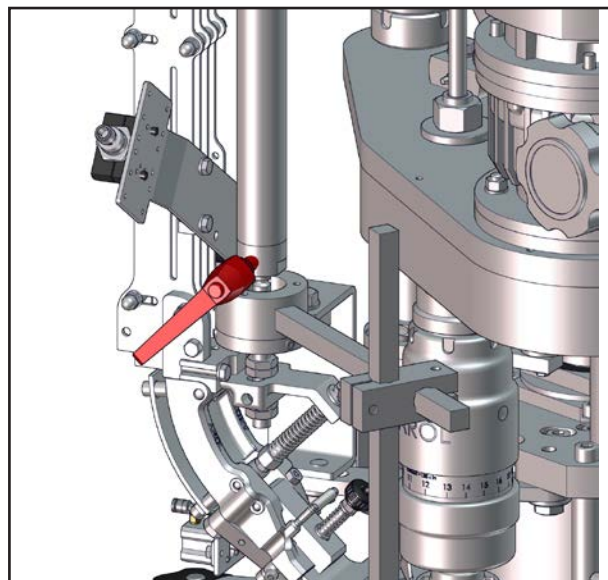


Fig. 106



- Connect the electrical connection of the caps presence photocell (one or more).

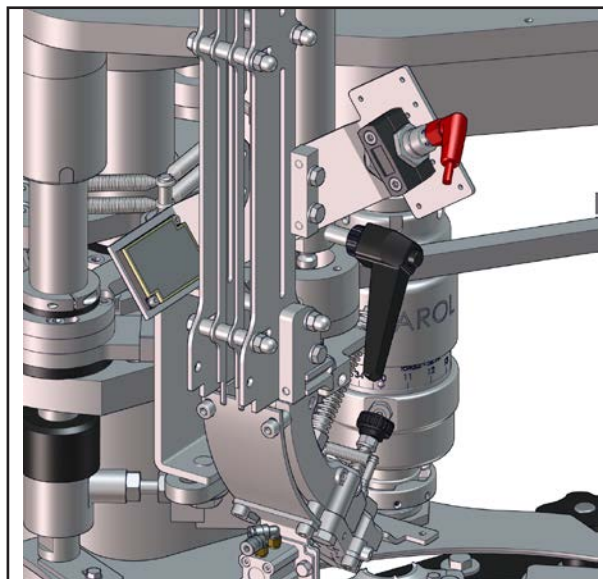


Fig. 107

- Connect the pneumatic connections of the caps stop cylinder to their fittings.

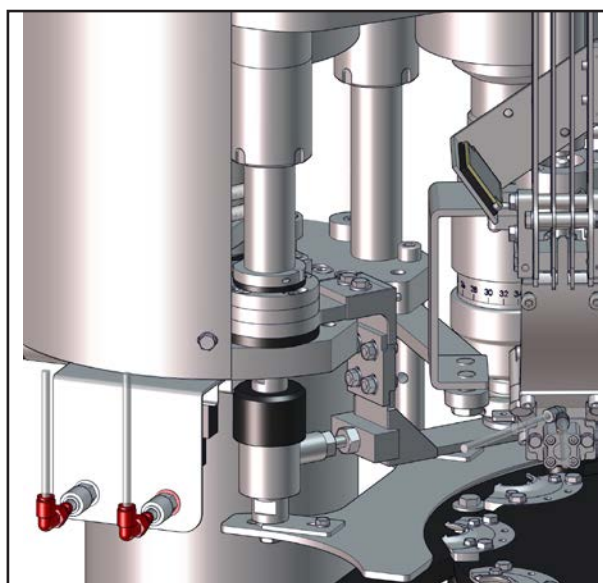


Fig. 108

10.2.8 CAPS TRANSFER DEVICE

10.2.8.1 REMOVAL

Remove the pin of the caps transport device as follows:

- Loosen the retaining screw using a 5 mm Allen wrench and remove it.

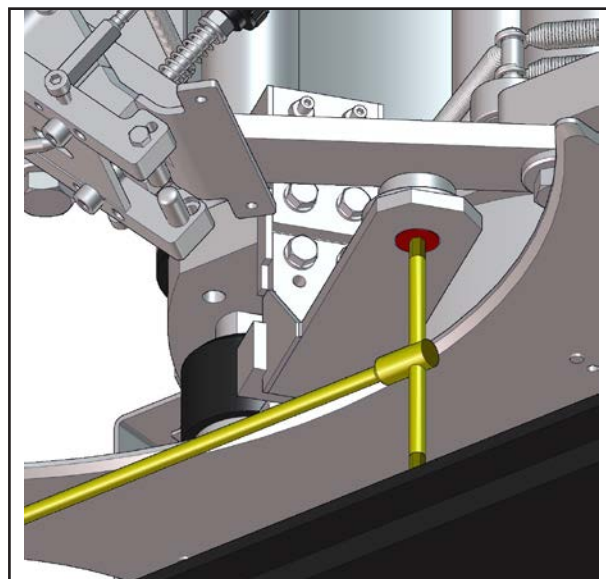


Fig. 109

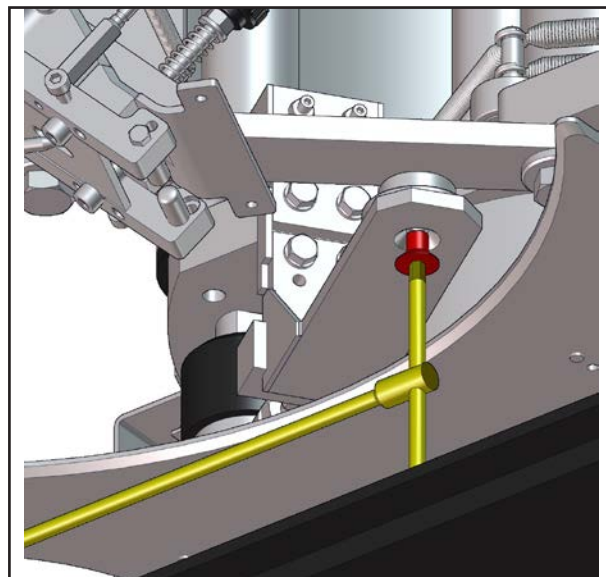


Fig. 110

- Remove the caps transport pin.

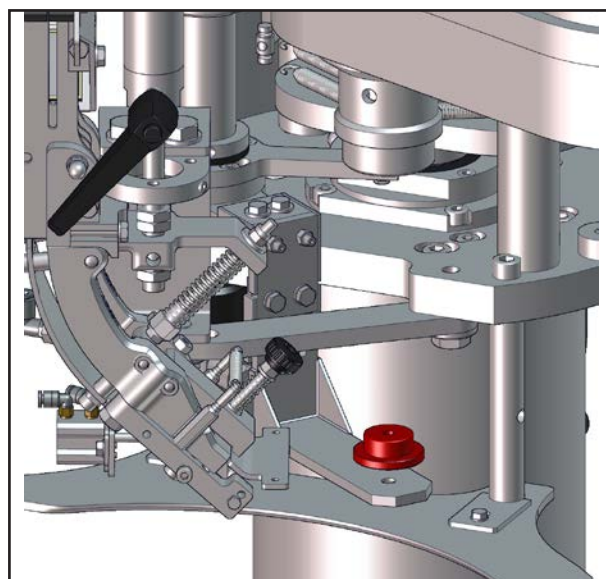


Fig. 111



In case the entire caps transport device is to be removed, you need to:

- Unscrew the fixing bolts with a 10 mm hexagonal wrench and remove the bolts.

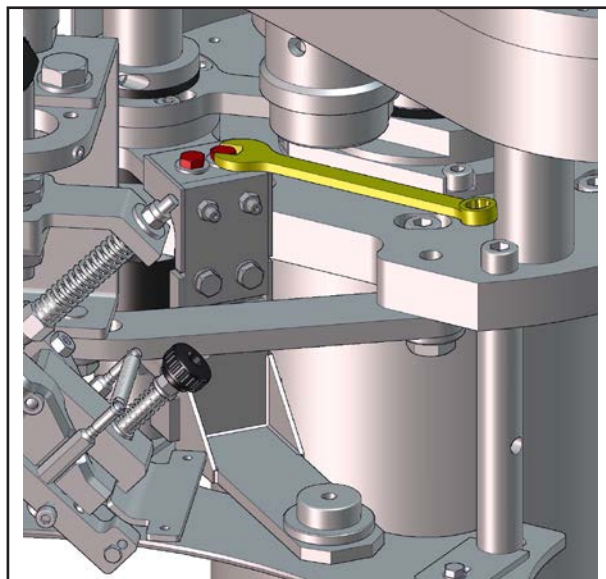


Fig. 112

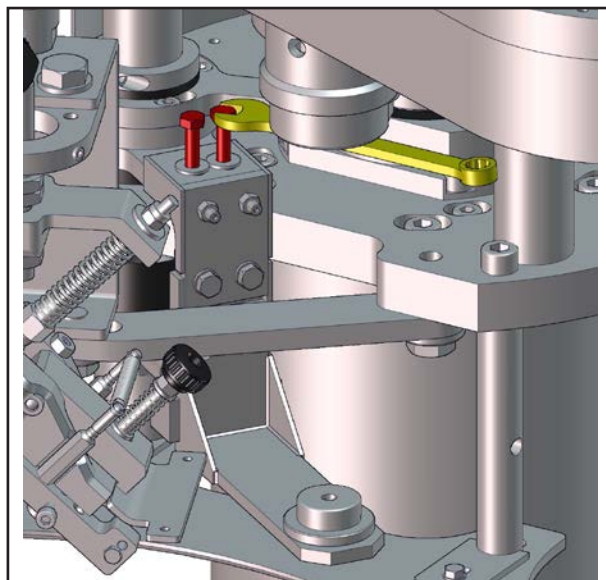


Fig. 113

- Remove the entire caps transport device.

10.2.8.2 INSTALLATION



To install the caps transport device, follow the instructions for its removal in reverse order.

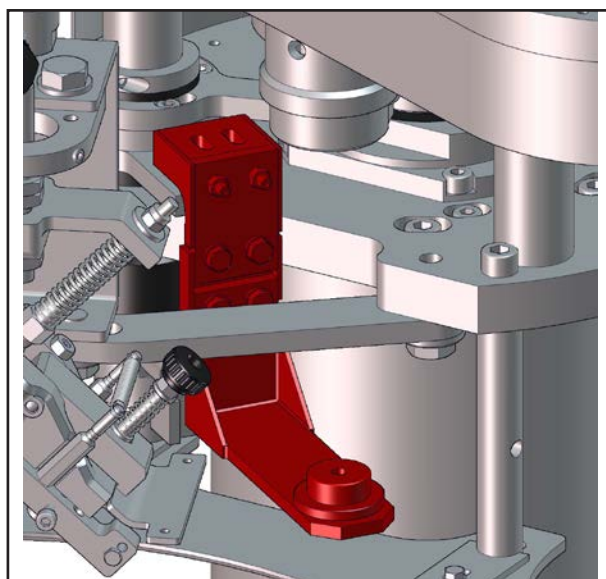


Fig. 114



10.2.9 CAP SELECTION EQUIPMENT FOR VIBRATION ORIENTATOR



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

10.2.9.1 SELECTION TANK REMOVAL



To remove the tank, proceed as follows:

- unscrew and remove the central screw that fix the tank to the vibrating body.

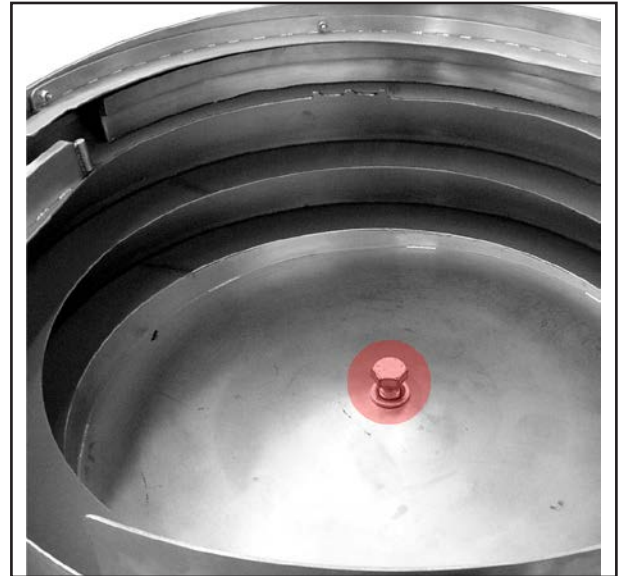


Fig. 115

- If provided, disconnect the pneumatic connections from the supply tank.



Fig. 116



- Lift and remove the orientation tank.



Fig. 117



10.2.9.2 SELECTION TANK INSTALLATION

To install the tank, proceed as follows:

- position the orientation tank on the vibrating body.



Fig. 118

- Fasten the tank to the vibrating body with the central screw.



Fig. 119

- If provided, connect the pneumatic connections on the orientation tank.



Fig. 120



10.3 ADJUSTMENTS OF THE CLOSURE HEAD

10.3.1 CLOSURE HEAD "PK311"

The table below shows the necessary tools for adjustment of the closure head.

TOOLS	
	Threaded ring adjusting wrench

10.3.1.1 TIGHTENING TORQUE ADJUSTMENT

This step can be carried out with the head installed on the closing machine.

The adjustment is made by a graduated scale engraved on the slotted ring-nut (in Pound x Inch) and a notch engraved on the magnet adjustment ring nut:



Fig. 121

- Lift the slotted threaded ring.



Fig. 122



- Turn the magnets adjustment threaded ring with the threaded ring adjustment wrench, to obtain the alignment between the engrave and the value of the desired torque.

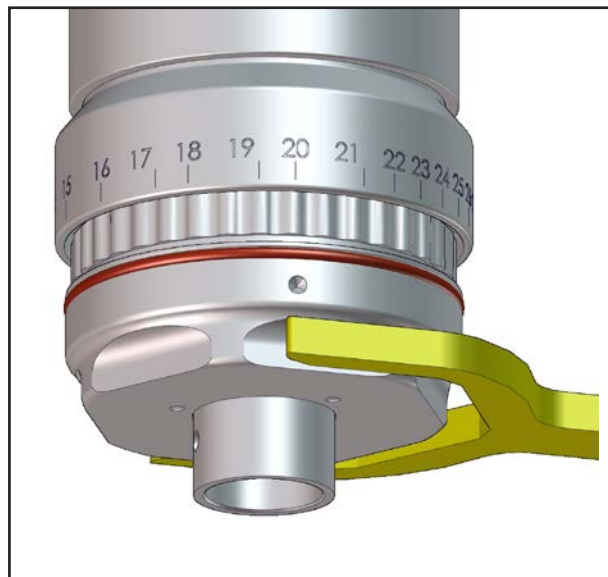


Fig. 123

- After adjustment, lower the slotted threaded ring checking the correct coupling between the inner teeth of the slotted threaded ring and the outer teeth of the magnet adjustment threaded ring.



The adjustment must be made on all closure heads with the same tightening torque.



Fig. 124



10.3.1.2 VARIATION IN THE VERTICAL LOAD

To change the vertical load, remove the head from the closing machine and replace the compensating spring:

- To remove the closure head, operate as indicated in sub-chapter 10.2.6 REMOVAL AND INSTALLATION OF THE CLOSURE HEAD page 65.
- Push the head upper bush downwards.



Fig. 125



Fig. 126

- Turn the head upper bush counterclockwise, while maintaining the lower part, and pull upward.



Fig. 127



- Remove the compensating spring.
- Replace the spring with a fit to the vertical load to be achieved.
- Remount proceeding reversely than the removal, taking care to grease the spring.

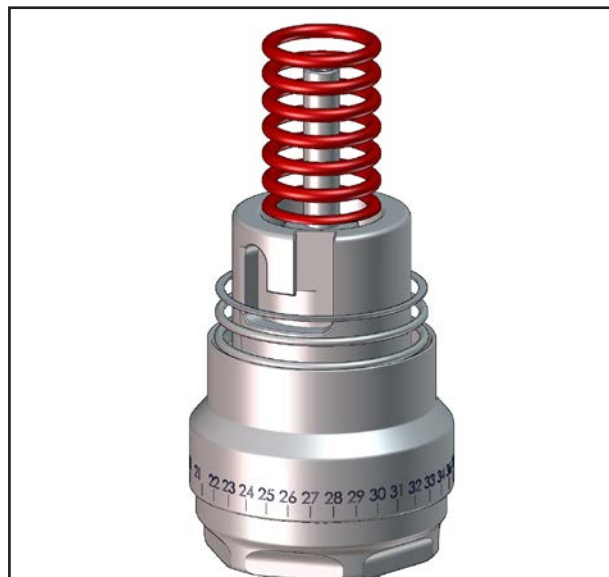


Fig. 128



Are available as standard, the springs shown in the table indicated below ($1 \text{ kg}_f = 9.81 \text{ N}$).

Load Kg_f	Colour	Code
4	None	P07D00621300
8	YELLOW	P07D00621700
12	BROWN	P56D02555900
16	WHITE	P07D00621900
20	RED	P03D00363500
25	BLUE	P52D02240100
32	GREEN	P52D02240300
61	VIOLET	P52D02240400
100	BLACK	P52D02240700



11 MAINTENANCE OF THE CLOSING MACHINE

11.1 MAINTENANCE GENERAL INFORMATION



11.1.1 MAINTENANCE SCHEDULE

From the constructive point of view, interventions concern mechanical, electrical and pneumatic parts.

The envisaged operations are grouped according to work time intervals. Each intervention or group of interventions can concern about mechanical, electrical and pneumatic aspects.

The ordinary maintenance is divided in two categories:

- scheduled ordinary maintenance (or preventive).
- Maintenance according to the conditions (caused by wear and tear).

The scheduled ordinary maintenance (so called periodical or preventive) includes inspections, checks and interventions that, to prevent stops and faults, keep under control:

- the lubrication state of the closing machine.
- The state of parts that can be affected by wear and tear.

The maintenance according to special conditions recommends a regular inspection of components, to check their conditions and perform the maintenance only if it is required.

11.1.2 MAINTENANCE INTERVALS



The following tables show the maintenance deadlines and the operations needed for a correct operation and long life of the closing machine.

The interventions described below have been arranged at intervals of time expressed in hours.

Interval (hours)	Working in 1 shift	Working in 2 shifts	Working in 3 shifts
8	-	-	-
40	1 week	-	-
120	3 weeks	-	1 week
250	6 weeks	3 weeks	2 weeks
500	3 months	6 weeks	1 month
1000	6 months	3 months	2 months
3000	12 months	9 months	6 months
6000*	*	*	12 months

(*) every 6000 hours or anyway once a year.



11.1.3 GENERAL INDICATIONS ON THE SCHEDULED MAINTENANCE PLAN

A logbook of the scheduled maintenance and lubrication interventions must be arranged according to the maintenance indications in the sub-chapters below and to the closing machine working conditions.

Example:

MAINTENANCE SCHEDULED AT 1000 HOURS					
INTERVENTION	DATE	MAINTENANCE MAN	SIGNATURE	INTERVENTION RESULT	NOTES
Check of the height sliding of the closure carousel.	XX/XX/XX	Mario Rossi		positive	
Control gear lubrication	XX/XX/XX	Mario Rossi		lubricated	

11.2 MAINTENANCE AND LUBRICATION PLANNING



Interval	Intervention type
40 hours	Ordinary maintenance
250 hours	lubrification
250 hours	lubrification
	Ordinary maintenance
500 hours	lubrification
	Ordinary maintenance
1000 hours	Ordinary maintenance
3000 hours	Ordinary maintenance
5000 hours	Ordinary maintenance
6000 hours	Ordinary maintenance
12000 hours	Ordinary maintenance
18000 hours	Ordinary maintenance



11.3 CLOSURE HEAD WEARING PARTS PK311



It is recommended to stock a spare part kit.

SPARE PART KIT	Working hours			
	3000	6000	12000	18000
Compensating spring	x			
Ejector rod supporting spring	x			
Central oblique bearing		x		
Thrust bearing on the ejector rod		x		
Bush for dowel		x		
Magnetic rings		x	x	
O-ring		x		
Ejector guide bush			x	
Rod guide bush				x
Upper bush group.				x



Service the closing head following the instructions in sub-chapter 11.5.2
OF THE CLOSURE HEAD page 92.

MAINTENANCE

11.4 PK314 L CLOSURE CHUCK WEARING PARTS



It is recommended to stock a spare part kit.

SPARE PART KIT	Working hours
	3000
Chuck model	x
Spheres	x
O-ring	x
Lid sealing ring	x



Service the chuck following the instructions in sub-chapter 11.5.3
THE CLOSURE CHUCK page 115.

MAINTENANCE

OF



11.5 SCHEDULED MAINTENANCE



11.5.1 ROUTINE MAINTENANCE

11.5.1.1 EVERY 40 WORKING HOURS

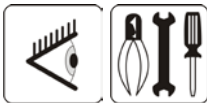
OPERATION	AIM	NOTES
Pneumatic system check.  	Check the air supply pressure to the various utilization points. Check by the gauges, that the pressure reducers are adjusted at the work pressure (the pressure is indicated on the pressure gauges) Drain the possible condensate collected under the discharge filter. Checking the filter clogging. If necessary, replace it.	

11.5.1.2 EVERY 500 WORKING HOURS

OPERATION	AIM	NOTES
Lubrication of the closure chuck. 	Lubricate the chuck.	See sub-chapter 11.6.7 LUBRICATING THE CLOSURE CHUCK page 138.
Pneumatic system check.  	Check the correct operation and possible leaks (from fittings, lines, etc.): ▪ verify the air pressure at the closing machine entry. ▪ Supplying the pneumatic system with compressed air, check there is no leak, broken or disconnected pipe. ▪ Check that the closing machine pressure gauges indicate the applied pressure on the suitable data plate.	
Verify the conditions of the bottle centering devices.  	Check if the teeth of the centering devices are worn and, if necessary, replace them.	See sub-chapter 11.5.4 REPLACEMENT OF TOOTHED CENTERING DEVICE page 118.



11.5.1.3 EVERY 1000 WORKING HOURS

OPERATION	AIM	NOTES
Check the conditions of the closure chuck. 	Check the wear status of the components of the closure chuck. Replace the sphere retaining O-rings.	See sub-chapter 11.5.3 MAINTENANCE OF THE CLOSURE CHUCK page 115.
Check of the height sliding of the closure carousel. 	Check the correct vertical sliding, making the closure carousel slide from the min. to the max. value, so as to ensure a correct component sliding in time.	
Equipment timing check. 	Check the timing of the whole equipment and, if necessary, re-time it.	See sub-chapter 11.7.1 TRANSPORT EQUIPMENT TIMING page 139.
Electric connection check. 	Carefully check all the connectors and connection cables and if necessary replace all the damaged parts.	
Sight and sound check of possible noises, clearances and unbalancing. 	Check, starting and stopping the closing machine several times without bottles, for possible noises, faults, clearances and vibrations in the motor-drive components (motors, reducers, gears, bearings). Replace possible worn components. Lubricate the new installed components. Clean and lubricate the disassembled components. Report any anomalies you need to contact <i>Technical Support</i>.	



11.5.1.4 EVERY 3000 WORKING HOURS

OPERATION	AIM	NOTES
Overhaul of the closure head. 	Overhaul the closure head, replacing the wear components.  The component being replaced are indicated in sub-chapter 11.3 CLOSURE HEAD WEARING PARTS PK311 page 86. Check the head correct tightening torque.	See sub-chapter 11.5.2 MAINTENANCE OF THE CLOSURE HEAD page 92.



To avoid production line long stops at the complete head overhaul, AROL S.p.A. offers its customers the “HEADS EXCHANGE PROGRAM” service.

The service includes:

- Temporary supply of substitute heads for the time strictly necessary for the overhaul.
- Complete head overhaul carried out at AROL S.p.A. by its technicians.
- Additional 1-year guarantee on the heads.

Contact the *TECHNICAL SERVICE* for info.

Lubrication of the closure head movable parts. 	Lubricate the springs. Lubricate the upper bush and the seat of the head central body. Lubricate the bush for dowel.	See sub-chapter 11.6.6 LUBRICATION OF THE CLOSURE HEAD page 133.
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11.5.1.5 EVERY 5000 WORKING HOURS

OPERATION	AIM	NOTES
Container anti-rotation device check. 	Check for the wear of the rubber sector of the container anti-rotation device.	See sub-chapter 11.5.6 CONTAINERANTI-ROTATION SECTOR page 121.
Check the wear of the rubber sectors of the central star-wheel. 	Check the wear of the rubber sectors of the central star-wheel and change it, if necessary.	See sub-chapter 11.5.5 S T A R - W H E E L ANTIROTATION SECTORS page 119.

11.5.1.6 EVERY 6000 WORKING HOURS

OPERATION	AIM	NOTES
Overhaul of the closure head. 	Overhaul the closure head, replacing the wear components.  The component being replaced are indicated in sub-chapter 11.3 CLOSURE HEAD WEARING PARTS PK311 page 86. Check the head correct tightening torque.	See sub-chapter 11.5.2 MAINTENANCE OF THE CLOSURE HEAD page 92.
Lubrication of the closure head movable parts. 	Lubricate the springs. Lubricate the upper bush and the seat of the head central body. Lubricate the bush for dowel.	See sub-chapter 11.6.6 LUBRICATION OF THE CLOSURE HEAD page 133.
Replacement of the bottles centering devices. 	Replace the bottles centering devices.	See sub-chapter 11.5.4 REPLACEMENT OF TOOTHED CENTERING DEVICE page 118.



11.5.1.7 EVERY 12000 WORKING HOURS

OPERATION	AIM	NOTES
Overhaul of the closure head. 	Overhaul the closure head, replacing the wear components.  The component being replaced are indicated in sub-chapter 11.3 CLOSURE HEAD WEARING PARTS PK311 page 86. Check the head correct tightening torque.	See sub-chapter 11.5.2 MAINTENANCE OF THE CLOSURE HEAD page 92.
Lubrication of the closure head movable parts. 	Lubricate the springs. Lubricate the upper bush and the seat of the head central body. Lubricate the bush for dowel.	See sub-chapter 11.6.6 LUBRICATION OF THE CLOSURE HEAD page 133.
Closing machine general check. 	Check the correct operation of the machine; if faults are found, contact the <i>Technical Service</i> .	

11.5.1.8 EVERY 18000 WORKING HOURS

OPERATION	AIM	NOTES
Overhaul of the closure head. 	Overhaul the closure head, replacing the wear components.  The component being replaced are indicated in sub-chapter 11.3 CLOSURE HEAD WEARING PARTS PK311 page 86. Check the head correct tightening torque.	See sub-chapter 11.5.2 MAINTENANCE OF THE CLOSURE HEAD page 92.
Lubrication of the closure head movable parts. 	Lubricate the springs. Lubricate the upper bush and the seat of the head central body. Lubricate the bush for dowel.	See sub-chapter 11.6.6 LUBRICATION OF THE CLOSURE HEAD page 133.
Overhaul of the star-wheel shaft unit. 	Overhaul the star-wheel unit by replacing the components below: ▪ Bearings. ▪ Snap rings. ▪ MIM seal rings.	For the overhaul of the star-wheel shaft, contact the <i>Technical Service</i> .



11.5.2 MAINTENANCE OF THE CLOSURE HEAD

11.5.2.1 CLOSURE HEAD "PK311"

11.5.2.1.1 CLOSURE HEAD OPENING

To open the closure head it is necessary to:

- remove the closure head from the closing machine as described in sub-chapter 10.2.6 REMOVAL AND INSTALLATION OF THE CLOSURE HEAD page 65.
- Push the head upper bush downwards.



Fig. 129



Fig. 130



- Turn the head upper bush counterclockwise, while maintaining the lower part, and pull upward.



Fig. 131



Fig. 132



11.5.2.1.2 REPLACING THE SPRINGS

11.5.2.1.2.1 COMPENSATING SPRING REPLACEMENT

To remove the compensating spring proceed as described below:

- Open the closure head as described in sub-chapter 11.5.2.1.1 CLOSURE HEAD OPENING page 92.



Fig. 133

- Remove the compensating spring.



Fig. 134



11.5.2.1.2.2 THREADED RING PRESSURE SPRING REPLACEMENT

To remove the threaded ring pressure spring proceed as described below:

- Open the closure head as described in sub-chapter 11.5.2.1.1 CLOSURE HEAD OPENING page 92.



Fig. 135

- Pull out the threaded ring pressure spring.



Fig. 136



The threaded ring pressure spring is tapered. Therefore must be mounted in such a way that the black painted spire is at the bottom.

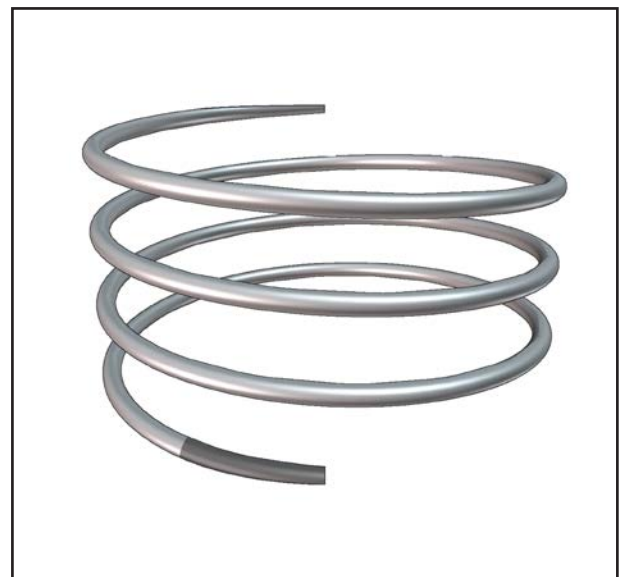


Fig. 137



11.5.2.1.2.3 REPLACEMENT EJECTOR ROD SUPPORTING SPRING

To remove the threaded ring pressure spring proceed as described below:

- Open the closure head as described in sub-chapter 11.5.2.1.1 CLOSURE HEAD OPENING page 92.



Fig. 138

- Remove the compensating spring.

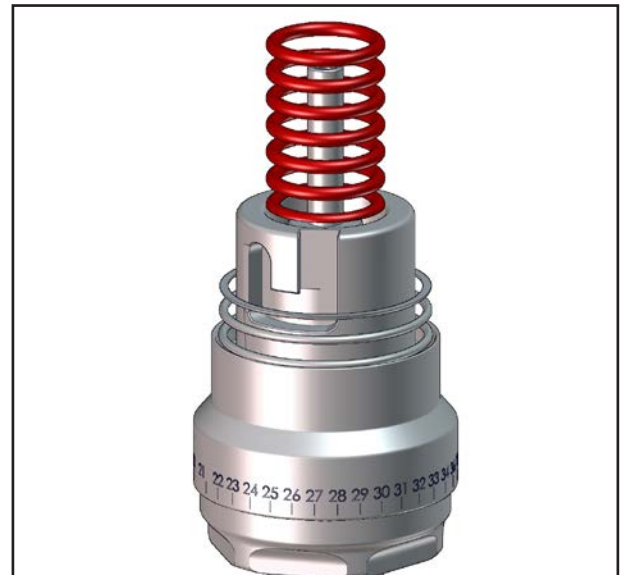


Fig. 139

- Pull out the ejector driving rod.



Fig. 140



- Pull out the ejector rod holder spring.



Fig. 141



11.5.2.1.3 MAGNETIC RING OVERHAUL

11.5.2.1.3.1 MAGNETIC RING CHECKING

Magnetic rings are components that tend to lose their original features in time; thus, they must be replaced one or more times during the head life.

Check using a torquemeter that the torque corresponds to the static torque value set on the slotted ring nut.



Tolerance $\pm 5\%$ of the stroke end value engraved on the slotted ring nut. (Example: if the graduated scale has values between 10 and 36 Lbs x inch and want to get a screwing torque equal to 30, the admitted tolerance value is between 28,2 - 31,8).

If the obtained value is higher than the tolerated value, proceed to replace of the magnetic rings.

11.5.2.1.3.2 REMOVING OF LOWER MAGNETIC RING



Note down the value of the screwing torque set on the graduated scale etched on the slotted ring nut.



Fig. 142

When it is necessary to remove the lower magnetic ring, proceeding as described below:

- Lift the slotted threaded ring.



Fig. 143



- Loosen and remove the magnet adjustment ring-nut from the head central body.



Fig. 144

- Loosen the Allen screws fastening the lower magnetic ring on the magnet adjustment ring nut with a 3 mm Allen wrench.

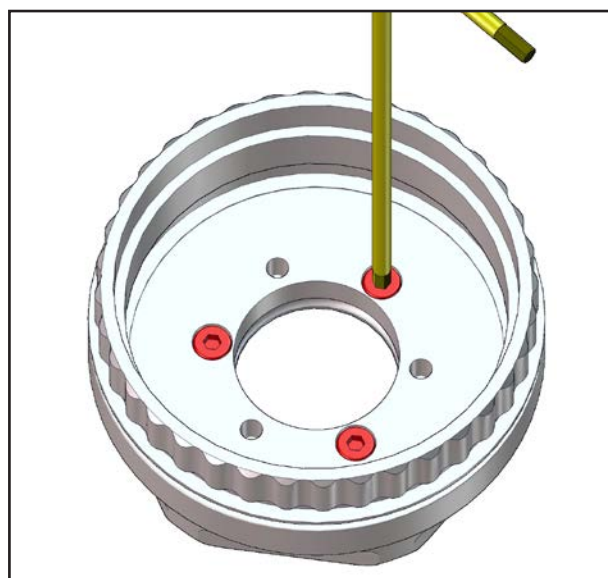


Fig. 145

- Remove the lower magnetic ring.

Remove the shim, if installed.



Use a medium resistance thread locking agent when tightening the Allen screws during the installation of the magnetic rings.



Fig. 146



11.5.2.1.3.3 REMOVING OF UPPER MAGNETIC RING

The table below shows the necessary tools for maintenance of the closure head.

TOOLS	
	Supporting tool

When it is necessary to remove the upper magnetic ring, proceed as follows:

- Loosen and remove the magnet adjustment ring-nut from the head central body as described in sub-chapter 11.5.2.1.3.2 REMOVING OF LOWER MAGNETIC RING page 98.



Fig. 147

- Fit the supporting tool fitting the pin in one of the holes of the upper magnetic ring.

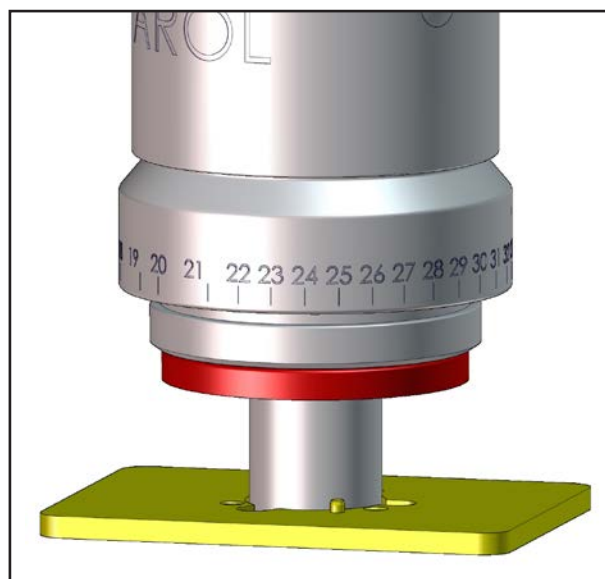


Fig. 148

- Position the supporting tool so that the Allen screws of the upper magnetic ring can be seen from the suitable openings.

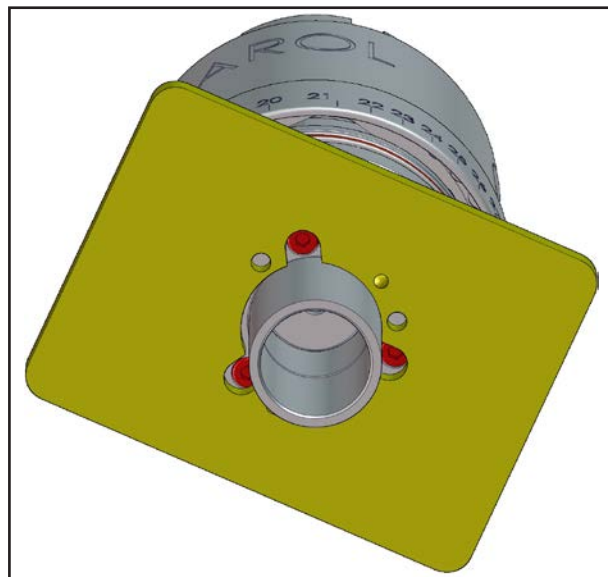


Fig. 149

- Holding in position the supporting tool and loosen the Allen screws that secure the magnetic ring with a 3 mm Allen wrench.

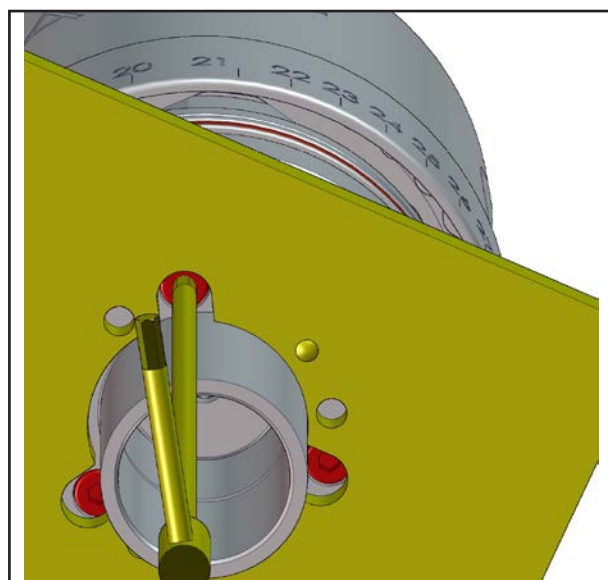


Fig. 150

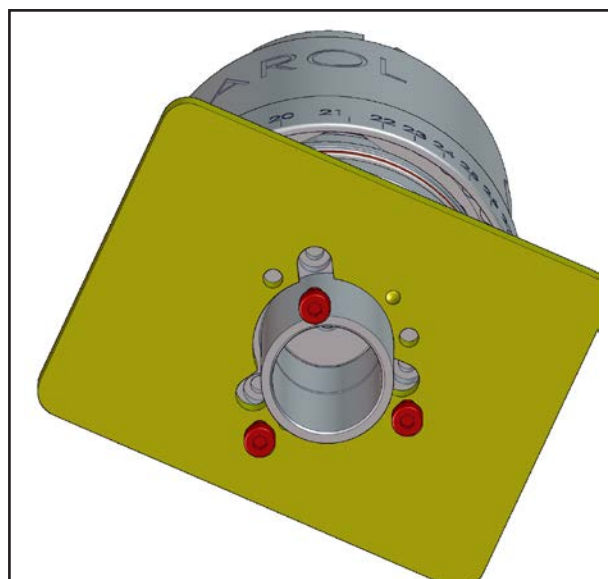


Fig. 151



- Remove the magnetic ring from the head central body.



Use a medium resistance thread locking agent when tightening the Allen screws during the installation of the magnetic rings.



Fig. 152



11.5.2.1.4 BEARINGS OVERHAUL

The table below shows the necessary tools for maintenance of the closure head.

TOOLS	
	Hook wrench 45-50 with ball
	Supporting tool

11.5.2.1.4.1 REMOVING OF CENTRAL BEARING

When it is necessary to remove the central bearing, proceeding as described below:

- Remove the upper magnetic ring as described in sub-chapter 11.5.2.1.3.3 REMOVING OF UPPER MAGNETIC RING page 100.



Fig. 153

- Remove the snap ring with a snap ring pliers.

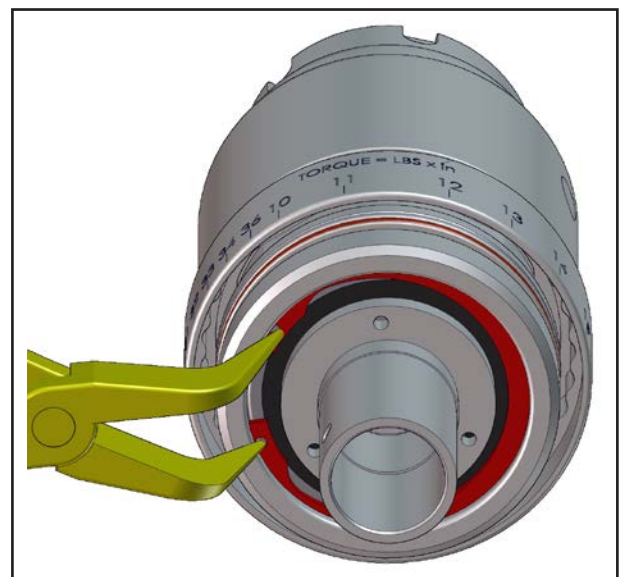


Fig. 154



- Remove the ejector guide bush complete of bearing from the head central body.



Fig. 155

- Fit the supporting tool on the ejector guide bush and tighten three Allen screws in the hole with a 3 mm Allen wrench.

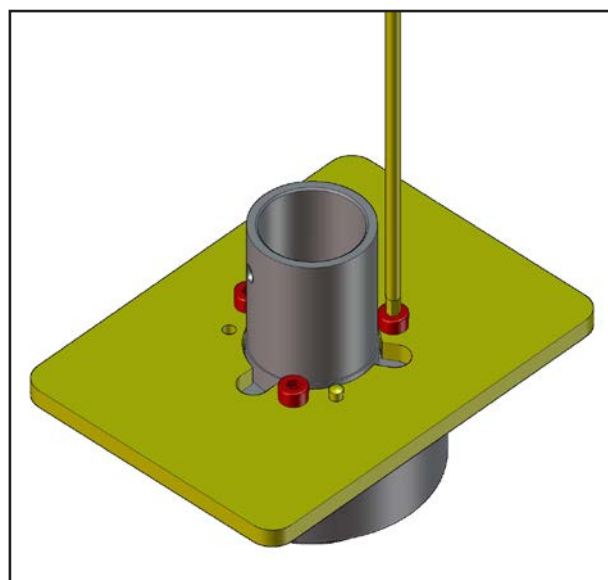


Fig. 156

- Fit the supporting tool locked in this way on the bearing by a vice.

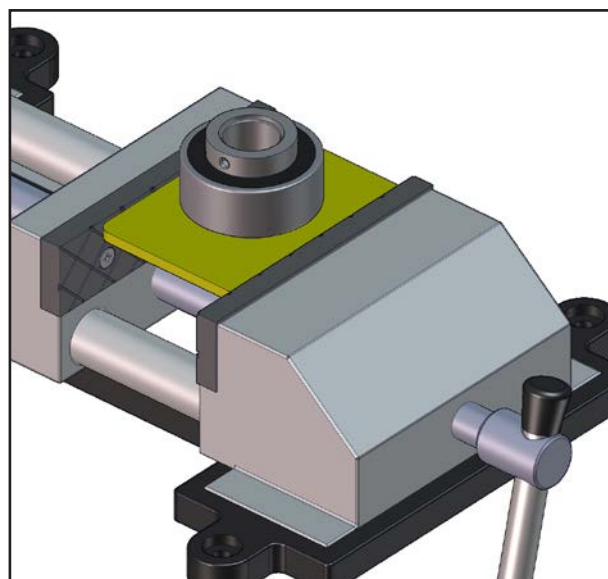


Fig. 157



- Unscrew the bearing locking ring-nut with the 45-50 hook wrench with ball, and remove the bearing locking ring-nut.

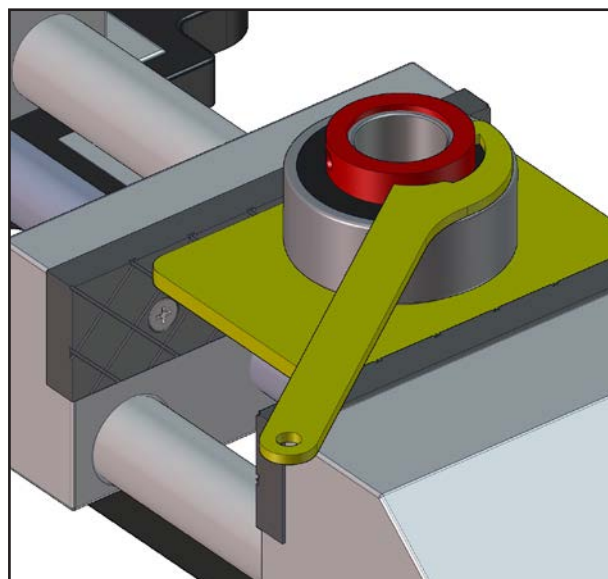


Fig. 158

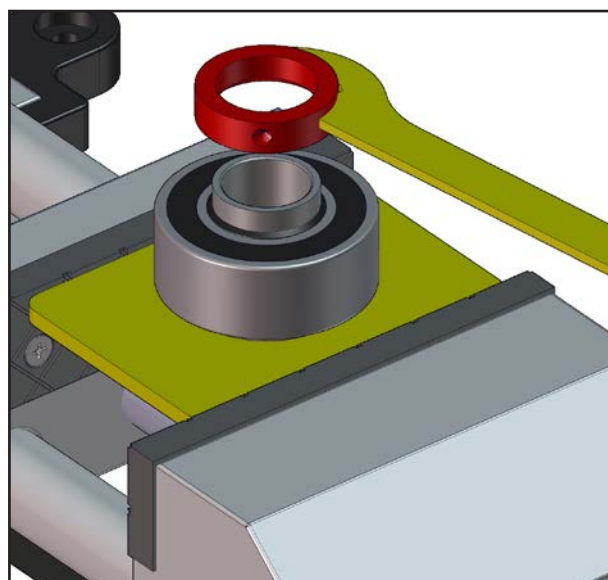


Fig. 159

- Remove the O-ring.

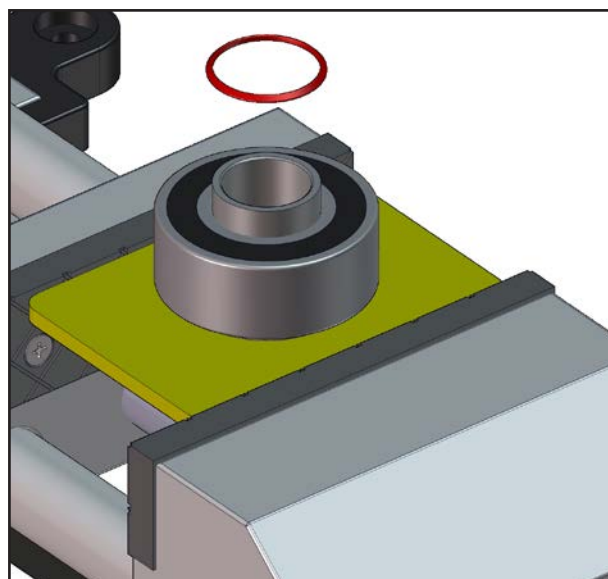


Fig. 160



- Extract the central bearing from the ejector guide bush.

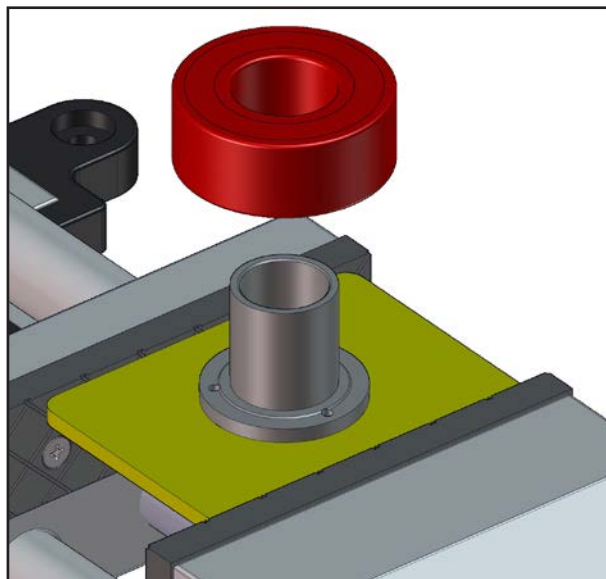


Fig. 161



11.5.2.1.5 GASKET OVERHAUL

The table below shows the necessary tools for maintenance of the closure head.

TOOLS	
	Hook wrench 45-50
	Hook wrench 45-50 with ball
	Supporting tool

11.5.2.1.5.1 REMOVING OF THE O-RINGS

To remove the O-rings, proceed as follows:

- Open the closure head as described in sub-chapter 11.5.2.1.1 CLOSURE HEAD OPENING page 92.
- Pull out the upper bush from the lower part of the head.



Fig. 162

- Remove the slotted threaded ring.



Fig. 163



- Loosen and remove the magnet adjustment ring-nut from the head central body.



Fig. 164

- Remove the O-ring from the head central body with a screwdriver or suitable tool.

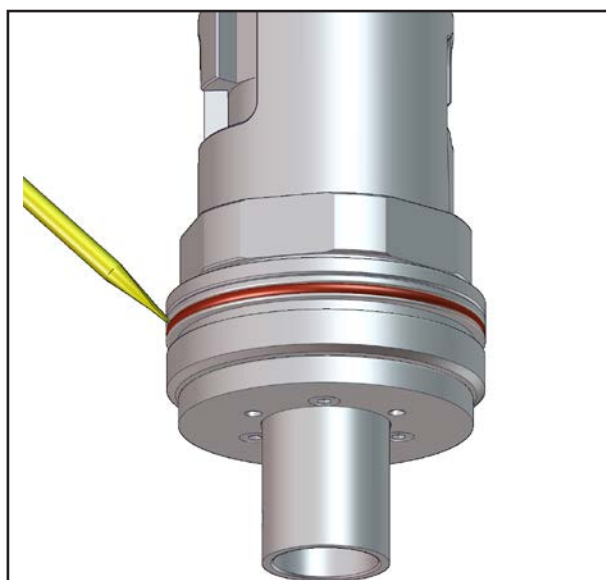


Fig. 165



- Remove the O-ring from the magnets adjustment threaded ring with a screwdriver or suitable tool

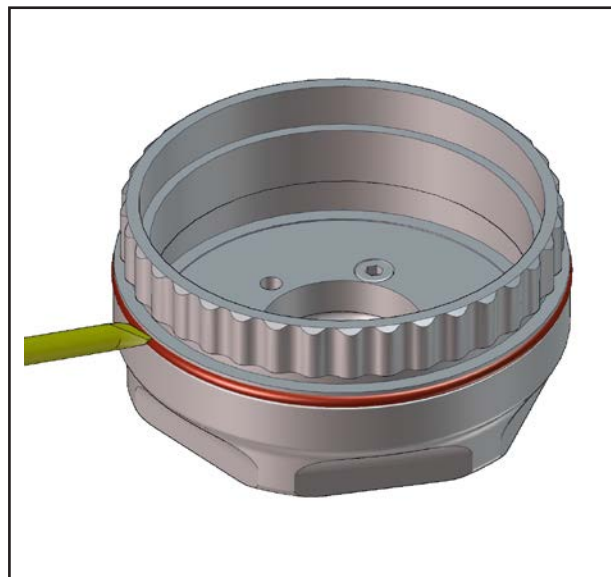


Fig. 166

- Remove the ejector guide bush complete of bearing from the head central body. (See sub-chapter 11.5.2.1.4.1 REMOVING OF CENTRAL BEARING page 103).



Fig. 167

- Fit the supporting tool on the ejector guide bush and tighten three Allen screws in the hole with a 3 mm Allen wrench.

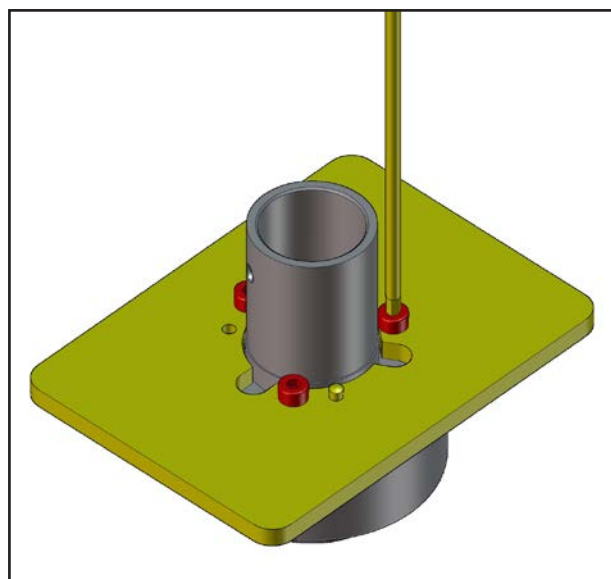


Fig. 168



- Fit the supporting tool locked in this way on the bearing by a vice.

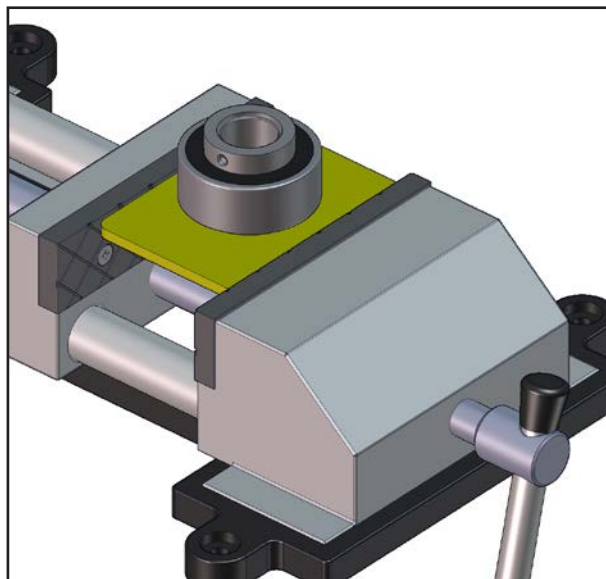


Fig. 169

- Unscrew the bearing locking ring-nut with the 45-50 hook wrench with ball, and remove the bearing locking ring-nut.

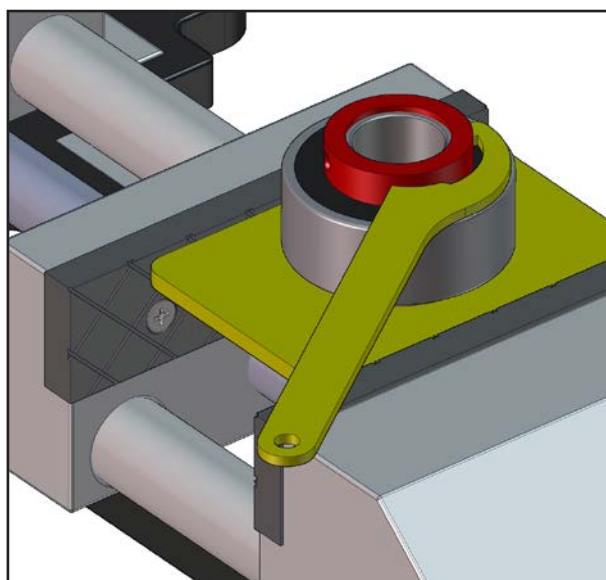


Fig. 170

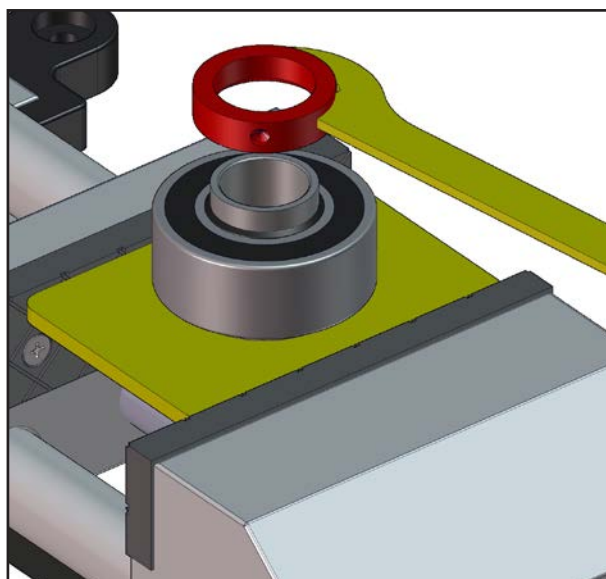


Fig. 171



- Remove the O-ring.

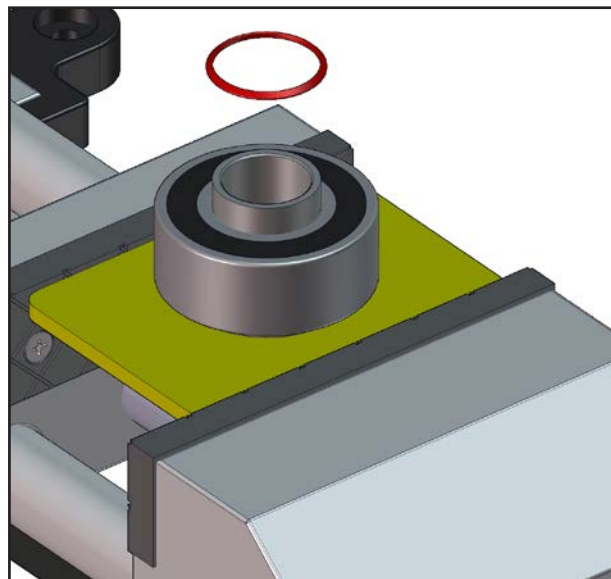


Fig. 172



11.5.2.1.6 REMOVING OF BUSH FOR DOWEL

When it is necessary remove the bushes, proceed as follows:

- Open the closure head as described in sub-chapter 11.5.2.1.1 CLOSURE HEAD OPENING page 92.
- Pull out the upper bush from the lower part of the head.



Fig. 173

- Remove the dowel bush pulling it out from the dowel.

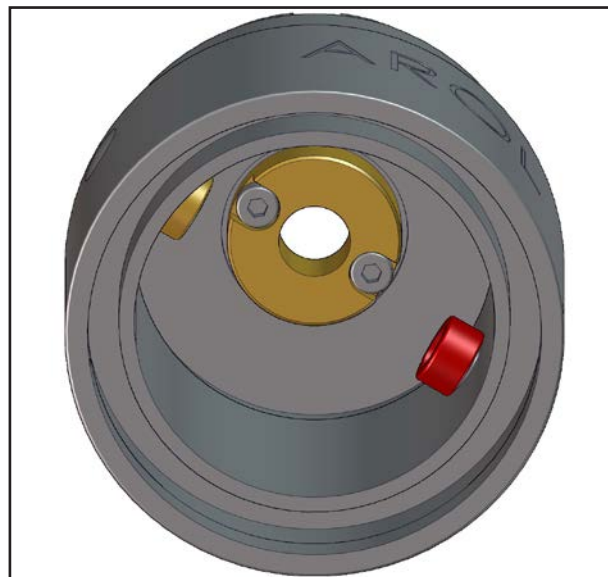


Fig. 174

- Re-install the bush caring to position it correctly, with the most curved side facing the head inner side.



The dowel bushes must be both replaced in the same intervention period.

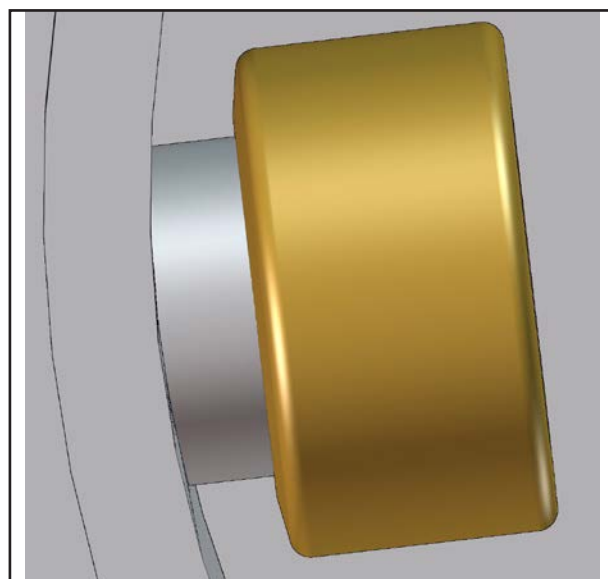


Fig. 175



11.5.2.1.7 CLOSURE HEAD INSTALLATION

Remount, proceeding inversely to that described for disassembly.



Before reassemble the closure head, clean and grease all components.

11.5.2.1.7.1 MOUNTING OF THE ADJUSTMENT THREADED RING



In order to get the same torque as the one set before the maintenance operation when the head is restored, the head must be installed aligning correctly the references etched on the parts to be matched. Only after the components have been correctly matched the torque value can be adjusted by setting the one noted at the operation start.

For the correct mounting, operate as follows:

- screw the magnet adjustment threaded ring till the upper edge coincides with the horizontal reference engrave made on the head central body.
- Align the vertical engraving of the head central body with the engraving of the magnet adjustment threaded ring.

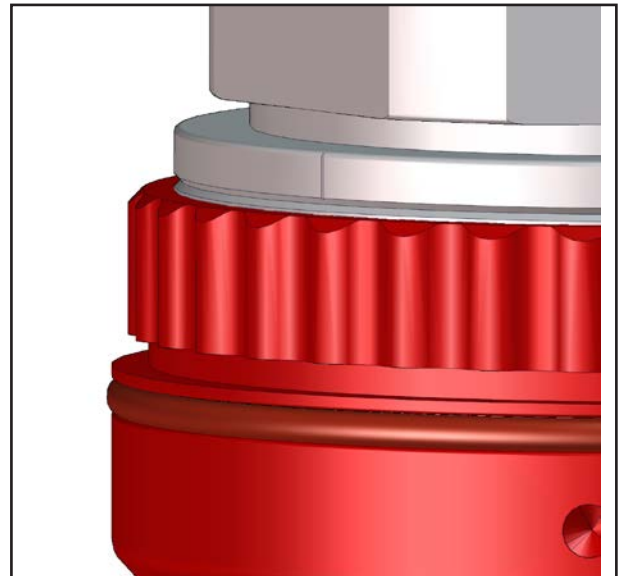


Fig. 176

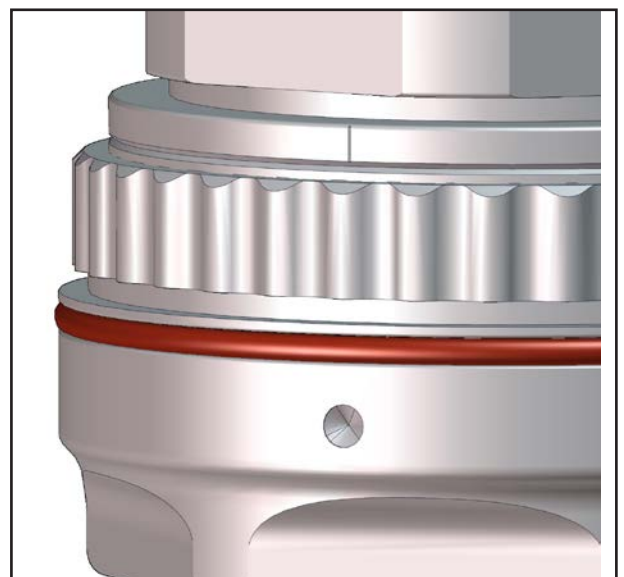


Fig. 177



- Fit the slotted threaded ring on the head central body so that the torque indication of **16 LBSxin** engraved on the threaded ring is aligned with the engraving of the magnet adjustment threaded ring.



Fig. 178

- Lower the slotted threaded ring till its inner teeth totally mesh with the outer teeth of the magnet adjustment threaded ring.



Fig. 179



Having to set a value of the tightening torque different from **16 LBSxin**, as indicated in sub-chapter 10.3.1.1 TIGHTENING TORQUE ADJUSTMENT page 80.



11.5.3 MAINTENANCE OF THE CLOSURE CHUCK

11.5.3.1 "PK314 L" CLOSURE CHUCK

11.5.3.1.1 REMOVING OF THE O-RINGS AND SPHERES

To remove the O-rings and the spheres , proceed as follows:

- Remove the cone lid blocking ring with a pliers for elastic ring.

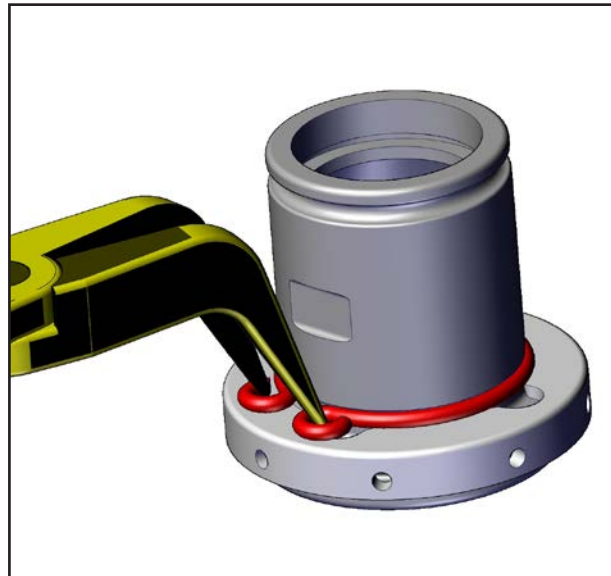


Fig. 180



Fig. 181



- Slide out the cone protecting lid.

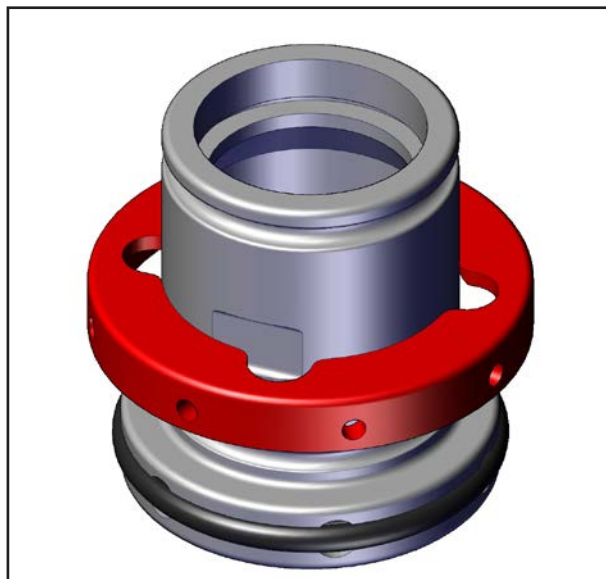


Fig. 182

- Remove the O-Rings (one or more depending on the type cap treated) from their seat with a screwdriver or a suitable tool.



If are damaged, replace the O-Rings.



If the O-rings are not damaged, replacing them has to happen from time to time as described in sub-section 11.4 PK314 L CLOSURE CHUCK WEARING PARTS page 86.

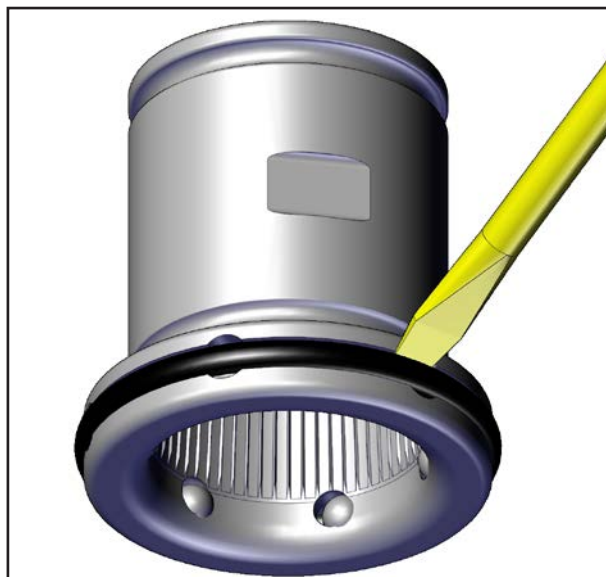


Fig. 183

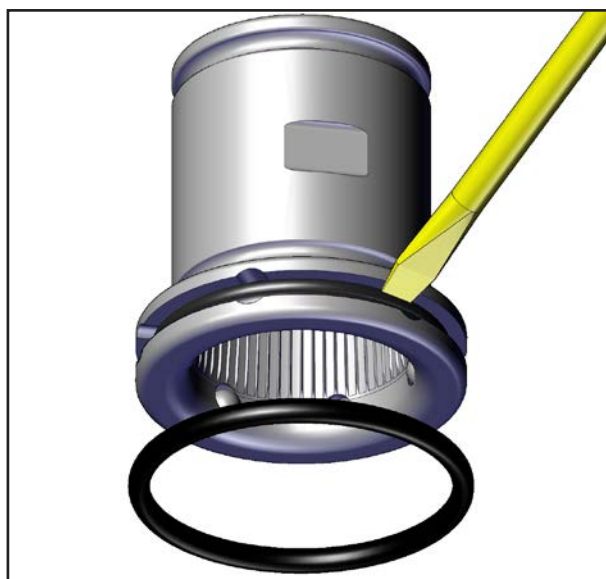


Fig. 184



Removing the O-rings, the chuck spheres are free and can scatter.

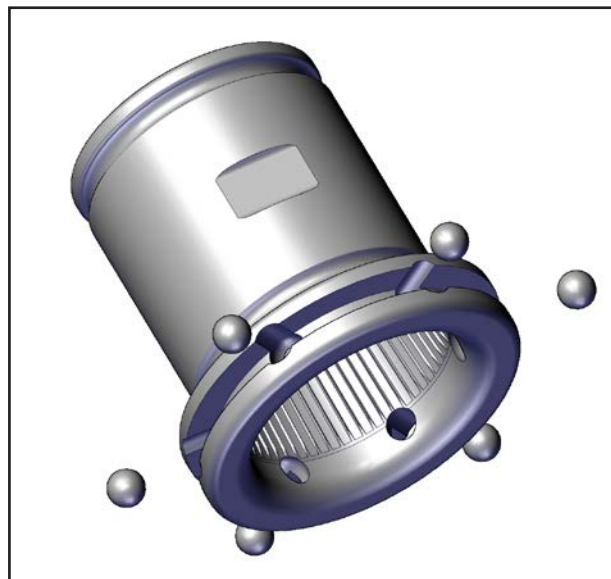


Fig. 185



11.5.4 REPLACEMENT OF TOOTHED CENTERING DEVICE



The teeth of the centering devices to carry out their bottle anti rotation task have a triangular shape. If the teeth are no longer sharp, it is necessary to replace them.

To replace the worn centering devices, operate as follows:

- Unscrew the fixing bolts using a 7 mm hexagonal wrench and remove them.



Fig. 186

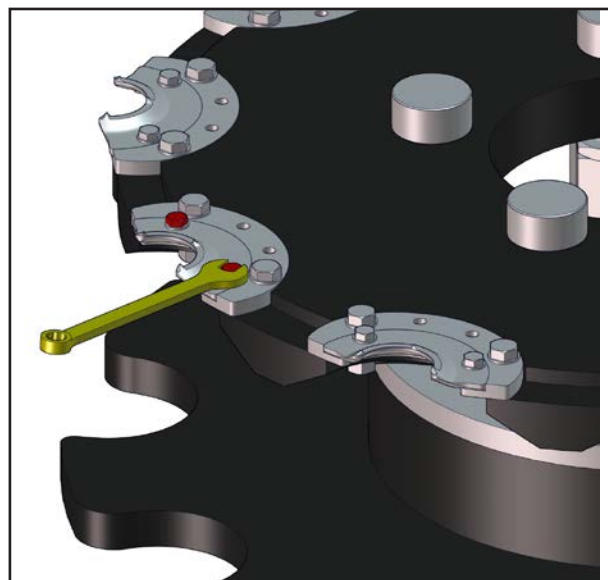


Fig. 187

- Remove the bottle centering devices.

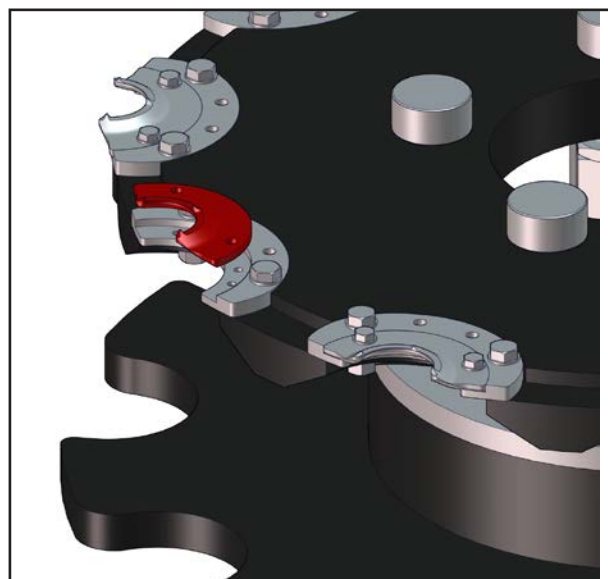


Fig. 188

11.5.5 STAR-WHEEL ANTIROTATION SECTORS



To remove proceed in the following way:

- Remove the central star-wheel (see the sub-chapter 10.1.1 CENTRAL STAR-WHEEL page 50).
- Unscrew and remove the fastening screws to free the star-wheel.

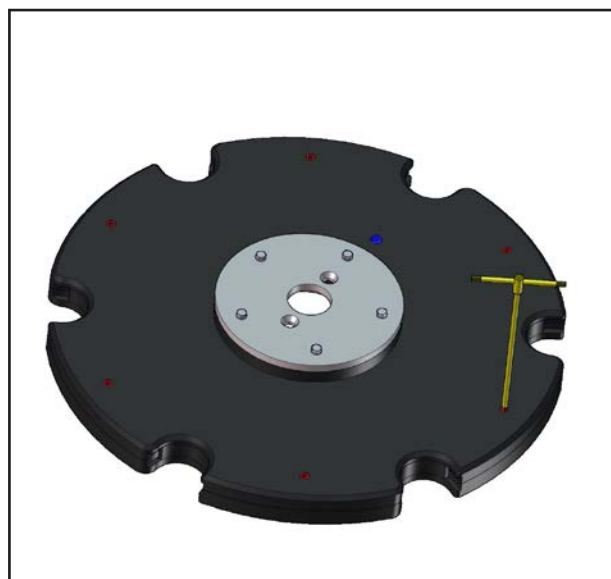


Fig. 189

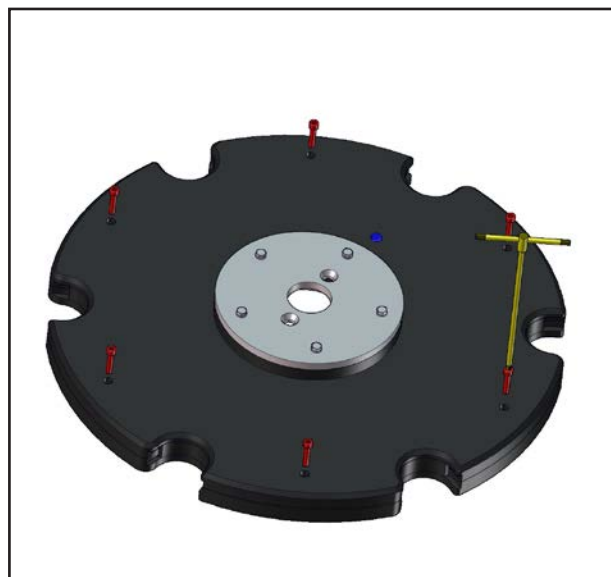


Fig. 190

- Separate the upper part of the star-wheel from the lower one.



Fig. 191



- Remove the rubber sectors and replace them.
- Refit, in reverse order to that described above.



Fig. 192

11.5.6 CONTAINER ANTI-ROTATION SECTOR



To remove proceed in the following way:

- Loosen and remove the fastening screws to separate the sector fastening supports.

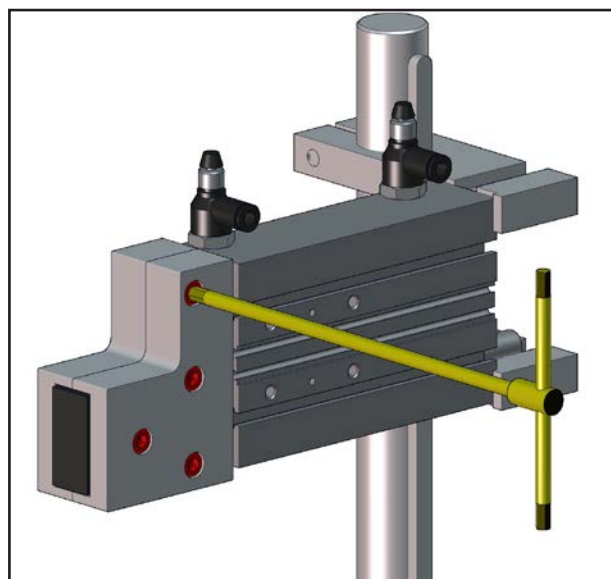


Fig. 193

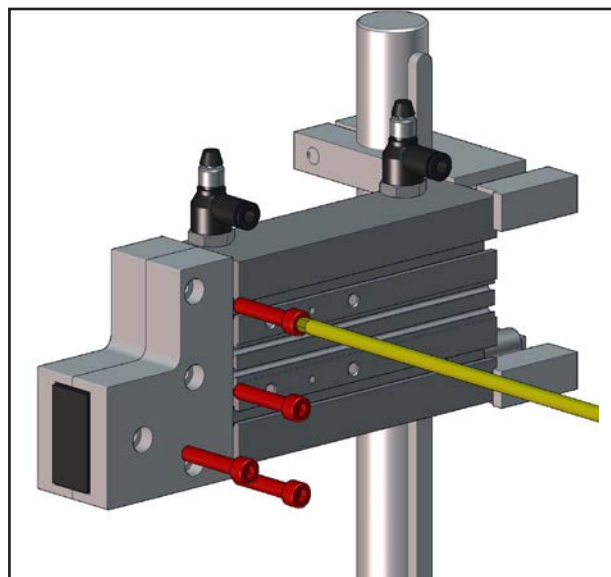


Fig. 194

- Separate the sector fastening supports so as to release the anti-rotation sector.

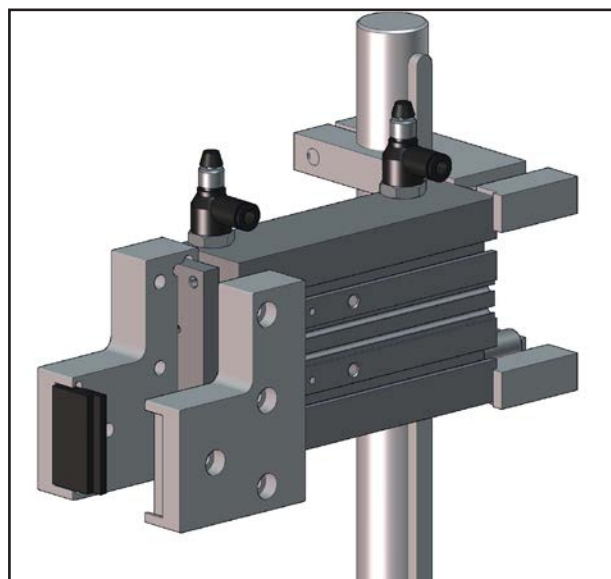


Fig. 195



- Remove the rubber sector and replace it.

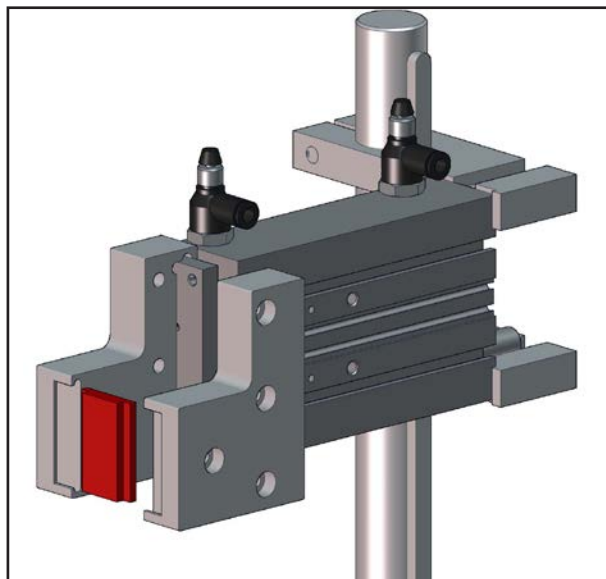


Fig. 196

- Refit, in reverse order to that described above.



11.5.7 ADJUSTING THE HEAD ROTATION MOTOR BELT TENSION



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

If the tension of the head rotation motor belt needs to be adjusted, proceed as follows:

- Using a 5 mm Allen wrench, loosen the fixing grub screws of the slide casing and remove the grub screws.

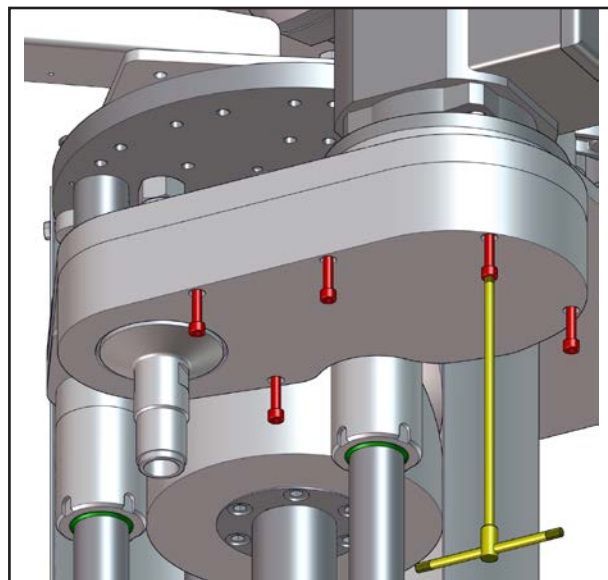


Fig. 197

- Remove the slide casing.

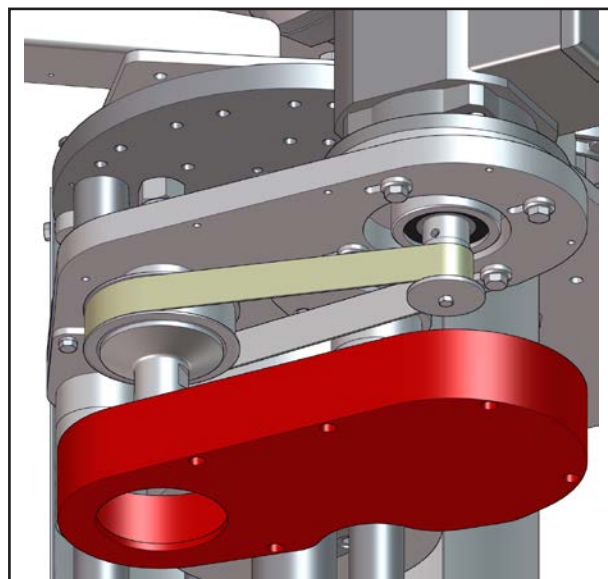


Fig. 198



- Loosen the plate fixing bolts with a 13 mm hexagonal wrench.

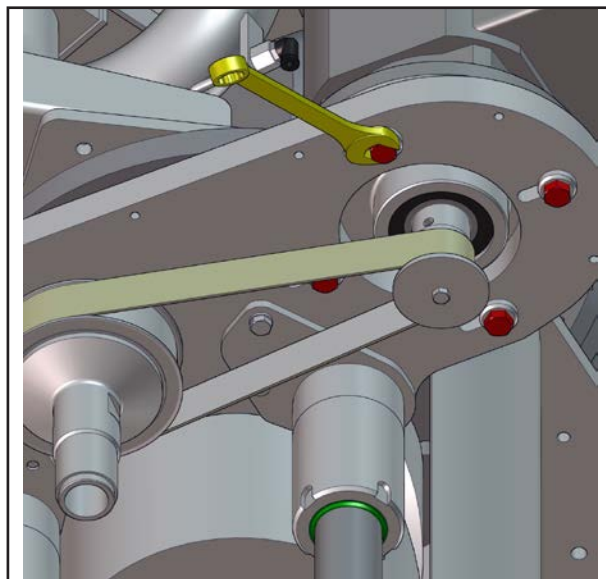


Fig. 199

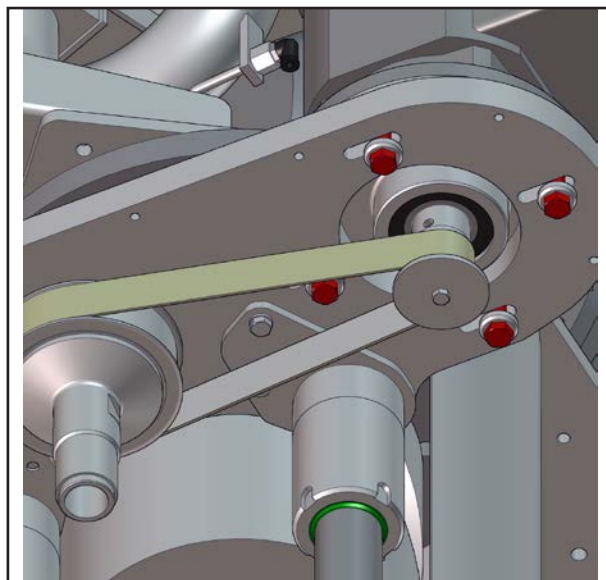


Fig. 200

- Make the head rotation motor with its supporting flange slide on the slide plate to adjust the tension of the motor rotation belt.

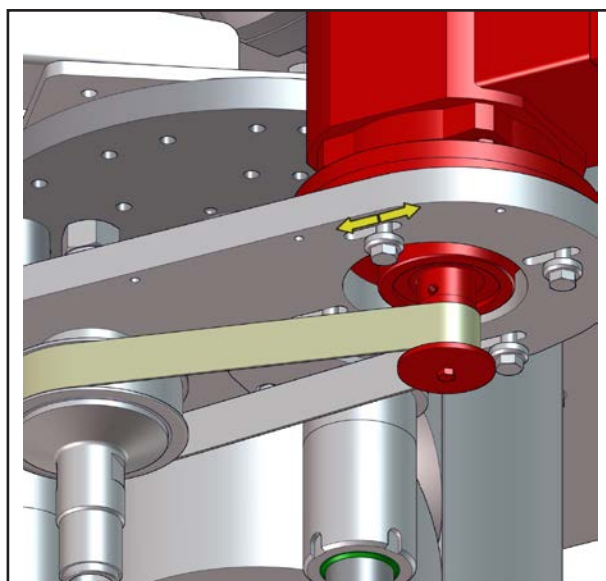


Fig. 201



- After the adjustment, tighten the plate fixing bolts.
- Install the slide casing and fix it with the fixing grub screws previously removed.

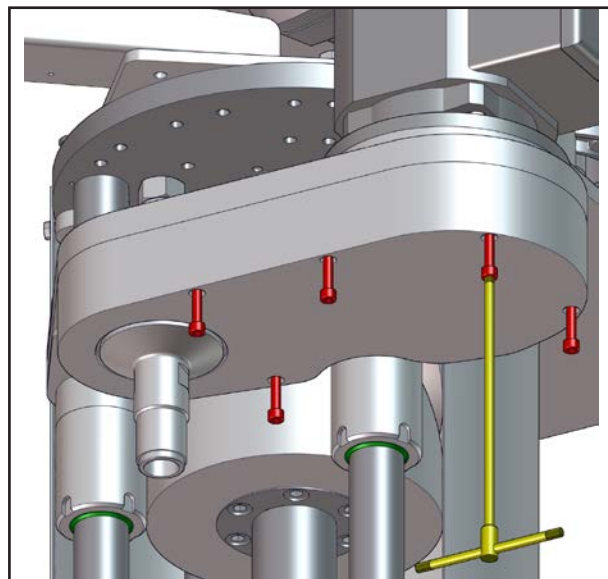


Fig. 202



11.6 LUBRICATION



11.6.1 GENERAL INFORMATION



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.



Before starting the closing machine, check the lubrication of all components described in the following subchapters.

If the closing machine is not used for a long time, lubricate it again.

Mineral oils, when not used for over six months, lose their properties and must be changed.

11.6.2 LUBRICANT TYPE-APPROVAL ACCORDING TO NSF

Due to the fact that the closing machine is used in food industry, AROL S.p.A. dedicated and special lubricants should be used on the machines, to prevent possible contamination problems of the food/drink in case of occasional or accidental contact between lubricant and food and drink. To this purpose, the selection of the lubricant was made according to the lubricant approval standard, according to NSF, a standard applied all over the world for food industry.

The type-approval according to NSF includes two categories: NSF - H1 and NSF - H2.

- **H1** indicates *Food grade lubricants*, that can thus be used in all friction points of machines and systems for the food and pharmaceutical industries, where occasional, and technically unavoidable, contacts can occur between lubricant and food.
- **H2** indicates lubricants that are recommended for the general use in the food and pharmaceutical industries, supposing that a contact with the food is totally to be excluded.

Due to the fact an accidental contact with the food/drink cannot be totally avoided, AROL S.p.A. decided to use always lubricants of category **H1** (*Food grade lubricants*).

The user company must use category H1 lubricants as described in the following tables.



11.6.3 TYPES OF LUBRICANTS

AROL S.p.A., after careful studies and functional tests, recommends to use only the lubricants indicated in the table below to lubricate its machines:



The lubricants indicated in the table are classified **NSF-H1**.

SIGNS		PRODUCT	NLGI	Base oil	Thickener [Viscosity Index]	Base oil viscosity [cSt]	Worked penetration [0.1mm]	T. Min [°C]	T. Max [°C]
A		Klubersynth UH1 64-62	2	Synth-PAO	Gel silica	60	265-295	-40	150
		G4501	1	Synth-PAO	AIK	100	310-340	-40	150
B		Kluberlub NH1-11-222	2	MO	AIK	220	265-295	-15	110
		Klubersynth UH1-14-222	2	Synth-PAO	AIK	220	265-295	-2	120
		Kluberfood NH1 94-402	1/2	Synth	Complex calcium	400	280-300	-30	160
C		Paraliq P150		Synth+MO		150		-15	100
D		Klubersummit FG46		Synth-PAO	[130]	46		-45	135
		L0346FG		MO	[100]	46		-21	
		Purity AW 46		HT/MO	[108]	46		-15	
E		Purity FG Synth	2	Synth-PAO	Complex sulfonate calcium	46	294	-45	200
F		Kluberpaste UH1 84-201	1	Synth-PAO	PTFE	200			
		P1900	1	MO	AIK	85	290-340	-30	300
G		Barrierta L55 / 1	1	Synth-PFPE	PTFE	400		-40	260
		Krytox GPL 226	2	Synth-PFPE	PTFE [155]	240		-30	288
H		Klubersynth UH1 64-62	2	Synth-PAO	Gel silica	60	265-295		
		G4501	1	Synth-PAO	AIK	100	310-340	-40	150
I		Purity MF Spray		HT/MO	[150]	151		-9	150



SIGNS		PRODUCT	NLGI	Base oil	Thickener [Viscosity Index]	Base oil viscosity [cSt]	Worked penetration [0.1mm]	T. Min [°C]	T. Max [°C]
I		Purity MF Spray		HT/MO	[150]	151		-9	150
J		Klubersynth UH1 64-62	2	Synth - PAO	Gel silica	60	265-295	-40	150
		G4501	1	Synth - PAO	AIK	100	310-340	-40	150
K		Paraliq GB 363	3	Synth+MO	Bentonite	2400	215-245	-30	140
L		Paraliq GB 363	3	Synth+MO	Bentonite	2400	215-245	-30	140
M		Kluberlub NH1-11-222	2	MO	AIK	220	265-295	-15	110
		Klubersynth UH1-14-222	2	Synth - PAO	AIK	220	265-295	-25	120
		Kluberfood NH1 94-402	1/2	Synth	Complex calcium	400	280-300	-30	160
N		Kluberoil 4 UH1-220N		Synth - PAO	[150]	220		-25	120
		L1122FG		Synth - PAO / MO	[142]	197		-33	
		Purity FG Gear Oil 220		HT/MO	[92]	206		-18	
O		G4501	1	Synth - PAO	AIK	100	310-340	-40	150
P		Klubersynth UH1 64 1302	2	Synth	Silicate	1300	265-295	-10	150
Q		Purity FG EP 150		MO	[95]	144		-18	247
R		Kluberfood NH1 4-220N Spray		Synth - PAO	[150]	220		-30	120
S		Klubersynth UH1 64 1302	2	Synth	Silicate	1300	265-295	-10	150
		Kluberfood NH1 94-402	1/2	Synth	Complex calcium	400	280-300	-30	160
T		Cassida Grease GTS 2	2	Synth - PAO	Complex sulfonate calcium	46	265-295	-45	170
U		Kluberfood NH1 1-17		MO					



11.6.3.1 LUBRICANT RE-ORDERING

The lubricants used for the machine can be directly ordered to *AROL S.p.A.* or to the local distributors. See the table here below for the international addresses.

	http://www.klueber.com
	http://www.molykote.com
	http://www.petro-canada.ca
	http://www.krytox.com
	http://www.fuchs-oil.com

11.6.4 LEGEND FOR LUBRICATION SCHEMES

SYMBOL	DESCRIPTION
	Lubricate with manual pump for grease
	Lubricate with brush for grease
	Lubrication glass with disposable cartridge
	Lubricate with spray
	Lubricate with manual pump for oil
	Lubrication glass



11.6.5 POINTS TO BE LUBRICATED



To protect the closing machine from wear, seizure or other severe damage to the mechanisms, it is necessary to lubricate and grease periodically all indicated points.



AROL S.p.A. guarantees the maintenance intervals indicated hereby and the correct operation of the mechanisms exclusively with the recommended lubricants.

11.6.5.1 INSTRUCTIONS FOR READING THE "LUBRICATION POINT LIST" TABLE

The following pages show:

- the description table (*Lubrication point table*) relevant to time intervals, methods and used lubricants for the maintenance interventions on the closing machine herein.
- the position of the lubricating points (*Lubrication point scheme*) described herein.

The table "*Lubrication point list*" is designed to find immediately the operations to be carried out for the closing machine lubrication.

The "*Lubrication point scheme*" indicates the position of the points described in the table on the closing machine.

The points marked by a double frame indicate that the lubrication is carried out in a centralized way. Thus, the quantity of lubricant indicated in the table must be filled; then, the system will distribute the correct quantities to the different components.

POINT CODE	The column indicates the reference of the point to be lubricated, indicated on the closing machine next to the greaser and in the " <i>Lubrication point scheme</i> "
POINT DESCRIPTION	This column briefly describes the greasing point.
LUBRICANT TYPE	The reference letter for the lubricant to be used is indicated in this column. See the table for the lubricant type to be used.
METHOD	This column shows the tool by which the relevant greasing point must be greased.
INLET NUMBER [NO.]	It indicates the number of greasing points on the closing machine.
QUANTITY/POINT [gr.]	It indicates the lubricant quantity to be fed in each greasing point.
TOTAL QUANTITY [gr.]	It indicates the total lubricant quantity to be used.
INTERVAL [h]	It indicates every how many hours the different lubrication points must be greased.



After the lubrication of each points by grease, check that there are not leaks from seal gaskets or from the ditribution pipes.



11.6.5.2 LUBRICATION POINT LIST AND SCHEMES

LUBRICATION POINTS LIST

Order Ack.	S/N	Version	Reference	Typ	Model	P.D.	H.N.
	A4344	0		T	EAGLE PK	410	1
Point	Description	Lubr. Type	Lubr. Method	Number of points	Qty per point [g]	Total qty [g]	Lubr. Int. [h]
65	Saddle columns	C		2	5	10	250
4	Pistons controlling cam	M					500
6	Central bearings or central slewing bearing	M		1	5	5	500
8A	Piston upper roller	A		1	3	3	500
56	Transmission gears	B					500
60	Star-wheel shaft gears	M		1	10	10	500
61	Star-wheel rotation cam	M					500
62	Front bearing support	M		1	5	5	500
63	Rear bearing support	M		1	5	5	500
64	Pick and place cam	M					500
67	Star-wheel shaft bearings	M		1	10	10	500

Rotating point

Fig. 203 Lubrication point table

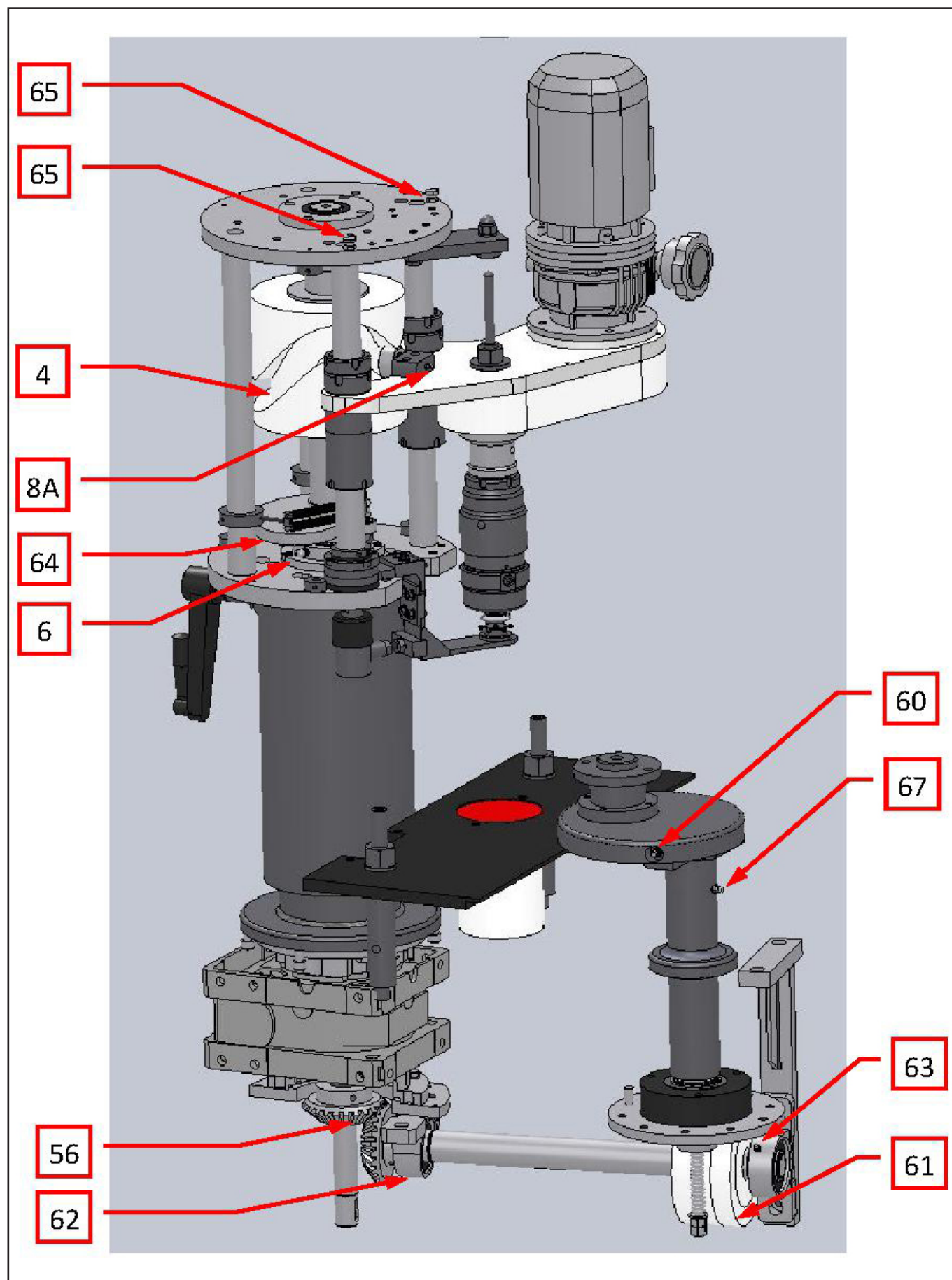


Fig. 204 Lubrication point scheme



11.6.6 LUBRICATION OF THE CLOSURE HEAD



Since it is objectively difficult to express an amount of lubricant to be laid with a brush, with "stroke" refers to the amount of grease that is retained, plunging up to half the bristles of a brush number 8 off about 10 mm high 18 mm bristle.

Use lubricant "**K**" type.

11.6.6.1 CLOSURE HEAD "PK311"

- Clean the springs and apply a layer of lubricant on all the coils. Use a brush.



Fig. 205

- Apply a layer of lubricant on the inside of the head upper bush. Use a brush.
- Apply a layer of lubricant on the dowel bushes. Use a brush.

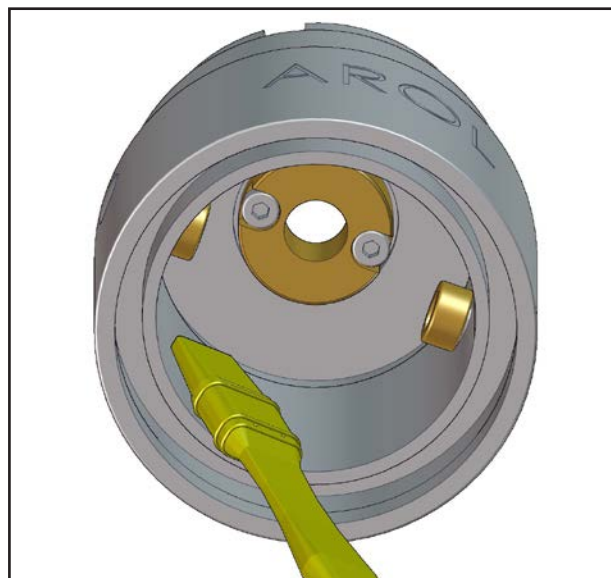


Fig. 206



- Apply a layer of lubricant on the seat of the head upper bush. Use a brush.



Fig. 207

- Apply a layer of lubricant on the inner rod. Use a brush.

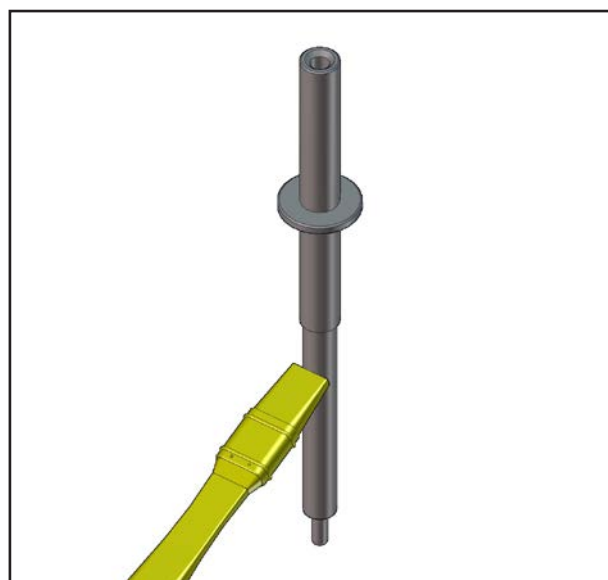


Fig. 208

- Clean the bearings and apply a layer of lubricant on all surfaces. Use a brush.

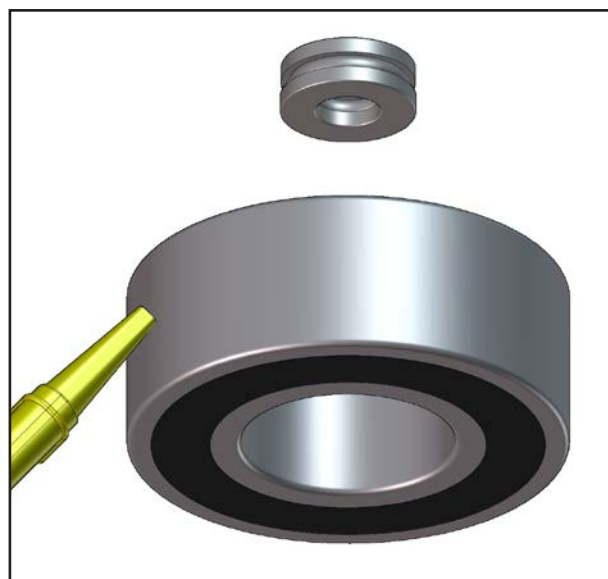


Fig. 209



- Clean the magnetic rings and apply a layer of lubricant on all surfaces. Use a brush.

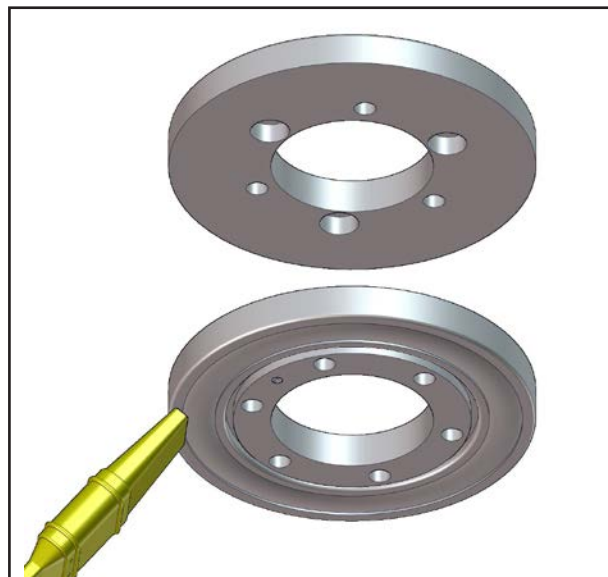


Fig. 210

- Clean inside and outside the head central body; apply a layer of lubricant inside and outside. Use a brush.

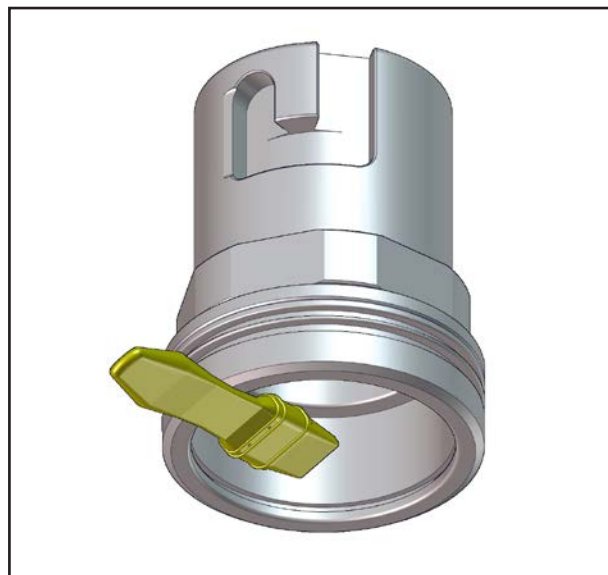


Fig. 211

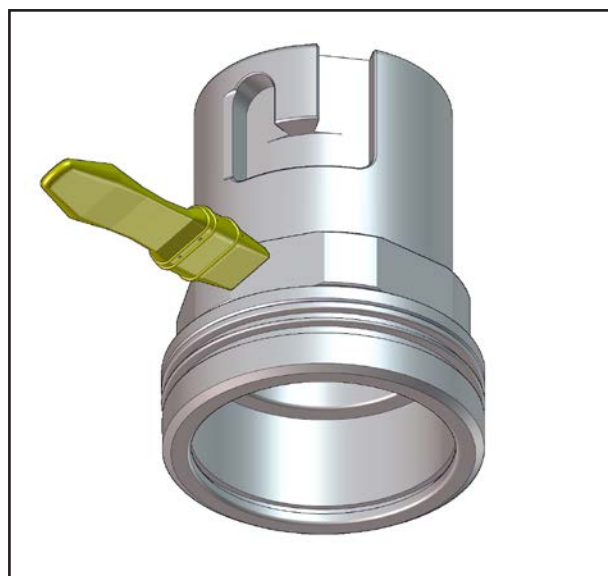


Fig. 212



- Clean the release seats of the head central body; apply a layer of lubricant inside the seats. Use a brush.

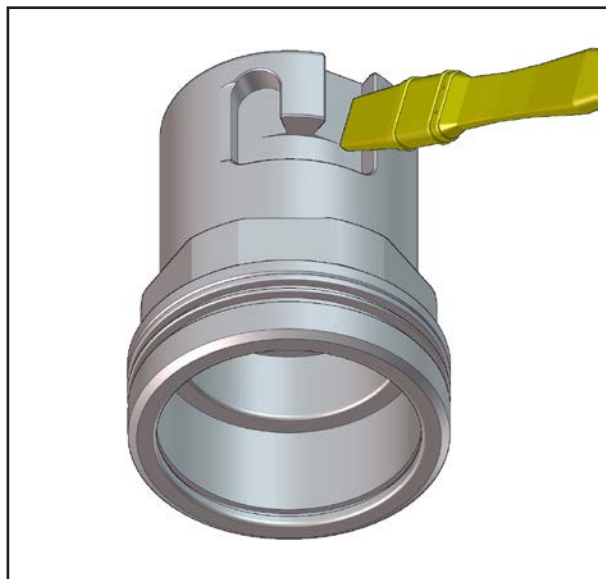


Fig. 213

- Clean the magnets holder ring-nut; apply a layer of lubricant inside the ring-nut. Use a brush.

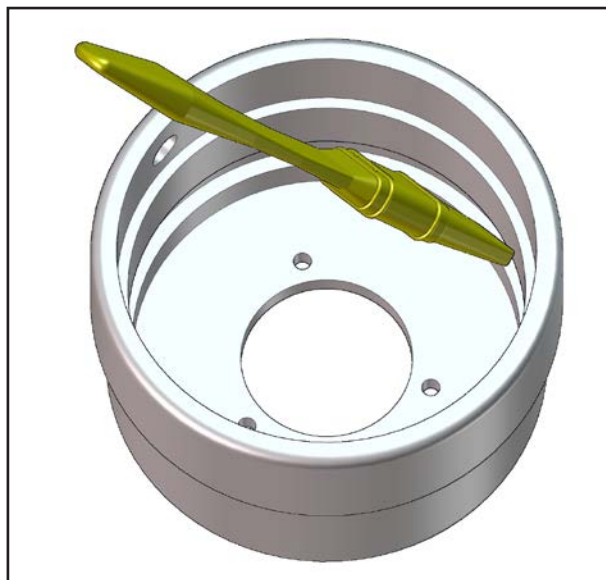


Fig. 214

- Clean the slotted threaded ring; apply a layer of lubricant inside the ring-nut. Use a brush.

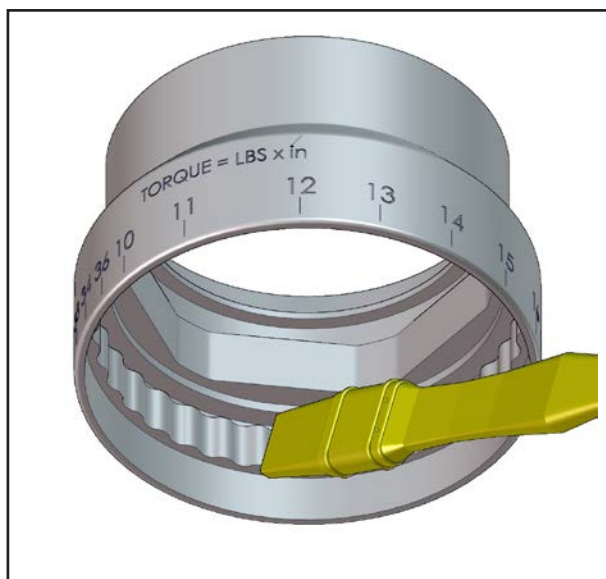


Fig. 215



- Apply a layer of lubricant on the inside of the ejector guide bush. Use a brush.

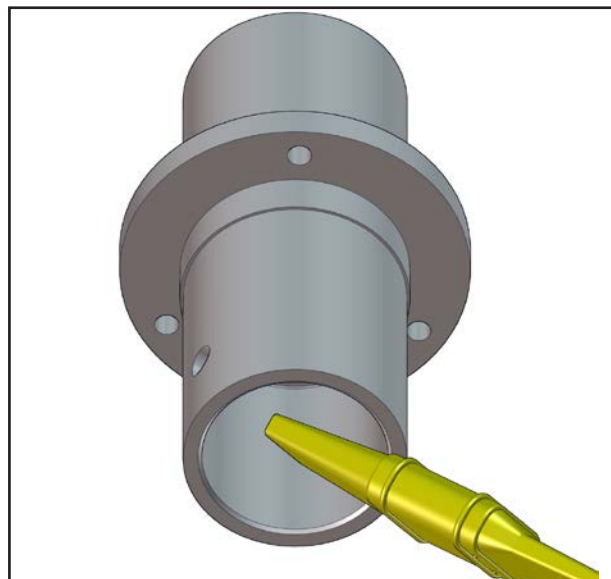


Fig. 216



11.6.7 LUBRICATING THE CLOSURE CHUCK



Since it is objectively difficult to express an amount of lubricant to be laid with a brush, with "stroke" refers to the amount of grease that is retained, plunging up to half the bristles of a brush number 8 off about 10 mm high 18 mm bristle.

Use lubricant "K" type.

11.6.7.1 "PK314 L" CLOSURE CHUCK

- Apply a layer of lubricant on the inside thread of the closure chuck. Use a brush.

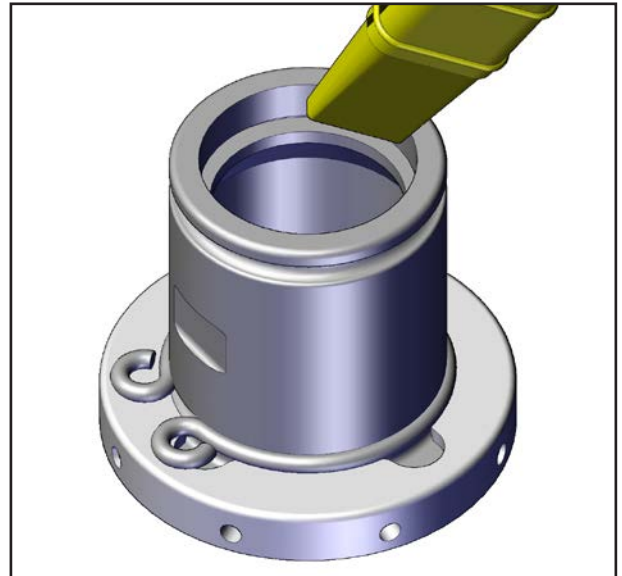


Fig. 217



11.7 EXTRAORDINARY MAINTENANCE

11.7.1 TRANSPORT EQUIPMENT TIMING

11.7.1.1 CENTRAL STAR-WHEEL TIMING

The star-wheel holder shaft is equipped with a clutch device in its lower part.

If, due to a collision, the clutch device causes the phase shift of the central star-wheel with respect to the closing carousel, it is necessary to restore it operating as follows:



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

- Position again the star-wheel, rotating it manually around itself till the clutch device returns to its original seat (a click is heard).

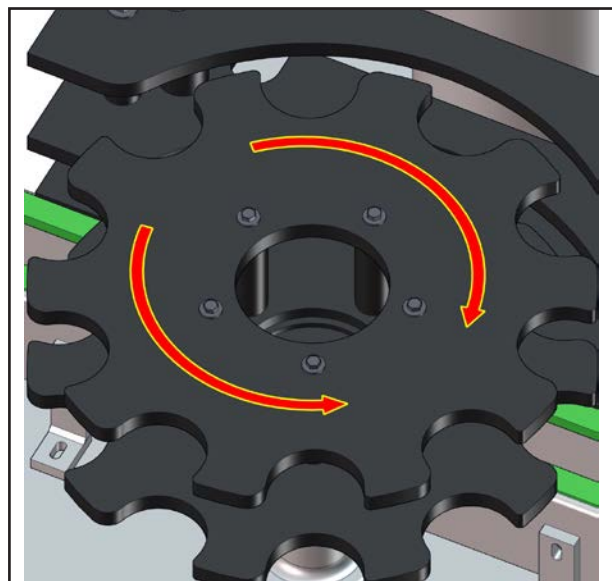


Fig. 218



11.7.2 REPLACEMENT OF THE HEAD ROTATION MOTOR BELT



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

If the head rotation motor belt needs to be replaced, proceed as follows:

- Using a 5 mm Allen wrench, loosen the fixing grub screws of the slide casing and remove the grub screws.

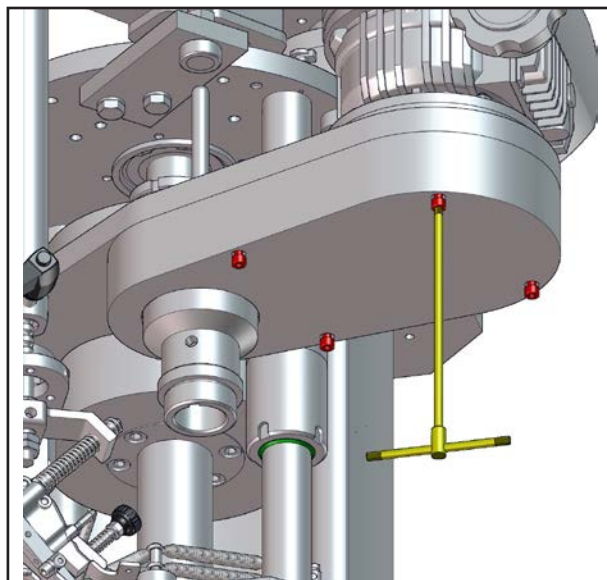


Fig. 219

- Remove the slide casing.

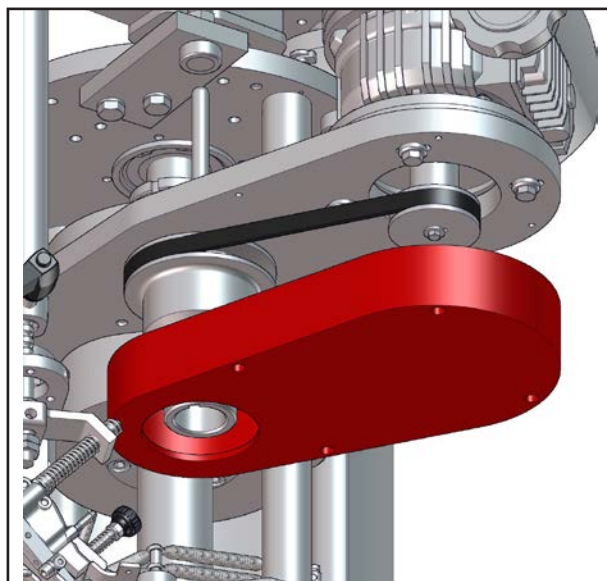


Fig. 220

- Loosen the plate fixing bolts with a 13 mm hexagonal wrench.

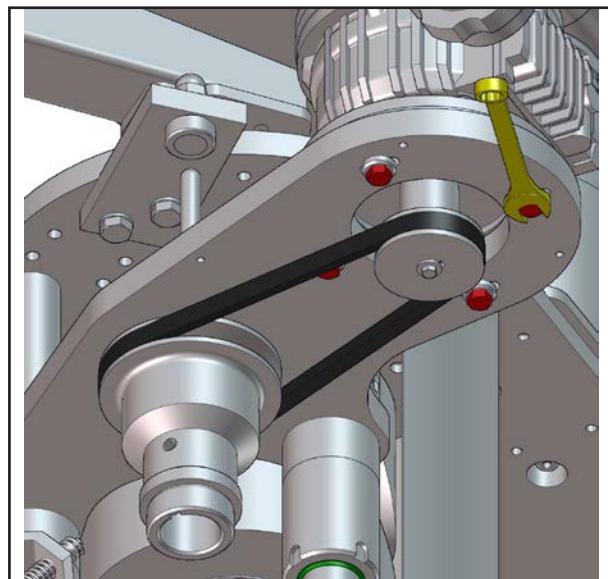


Fig. 221

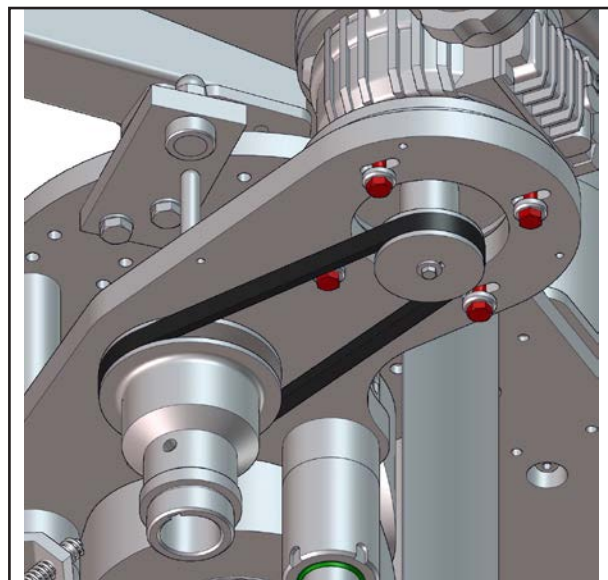


Fig. 222

- Make the head rotation motor with its supporting flange slide on the slide plate toward the spindle body to reduce the tension of the motor rotation belt.

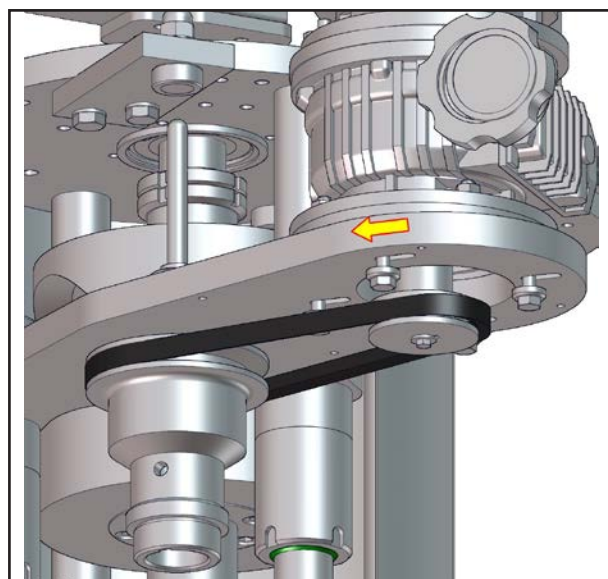


Fig. 223



- Pull the belt out of the motor pulley and of the spindle body pulley and replace it with a new one with the same characteristics.

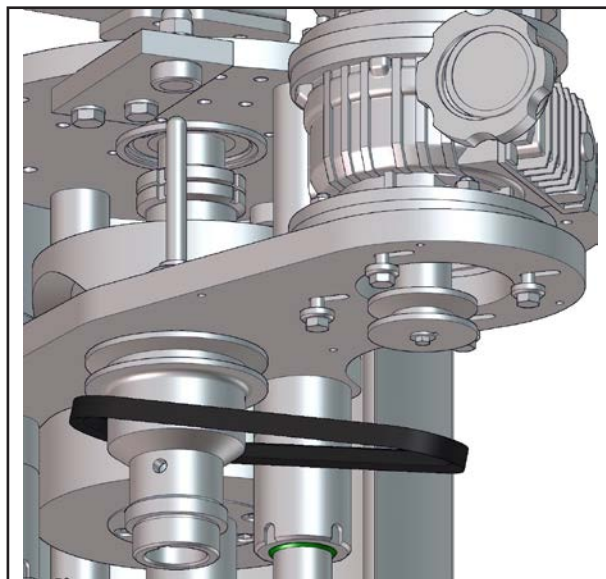


Fig. 224

- After the replacement, adjust the tension of the belt as described in sub-chapter 11.5.7 ADJUSTING THE HEAD ROTATION MOTOR BELT TENSION page 123.
- Install the slide casing and fix it with the fixing grub screws previously removed.

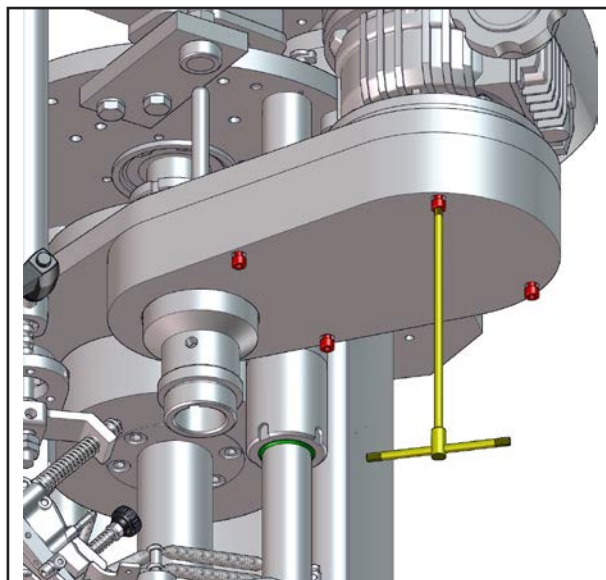


Fig. 225



11.7.3 MAINTENANCE OF THE VIBRATIONS CAPS SORTER



Rarely it happens that after the firsts two or three month of continuous functioning there is a "settling" of the caps sorter that involves a vibration decrement.

The same phenomenon can happen after years of work, for yield at "exertion" of the leaf spring springs.

In both the described cases and in the remote possibility that the caps sorter is subjected to a too low or too high voltage, it is necessary to calibrate the vibrating base of the caps sorter.

Proceed as follows:



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

- remove the fixing bolts of the vibrating mass covering guard.
- Remove the vibrating mass covering guard.

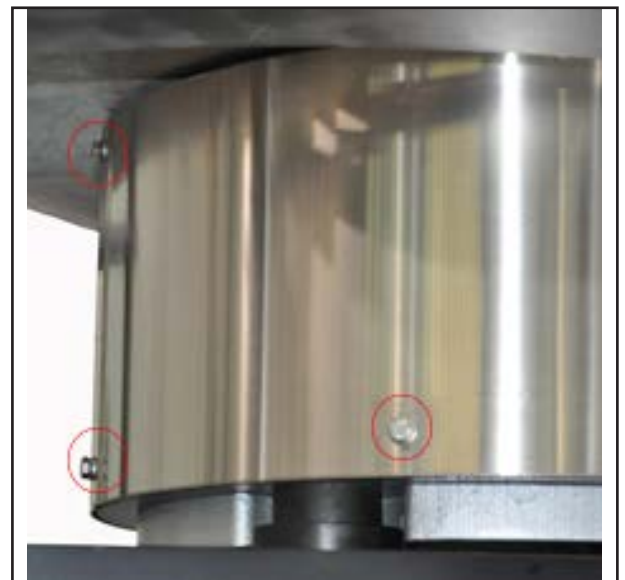


Fig. 226

- Screw to the end all the fixing screws of the leaf springs.

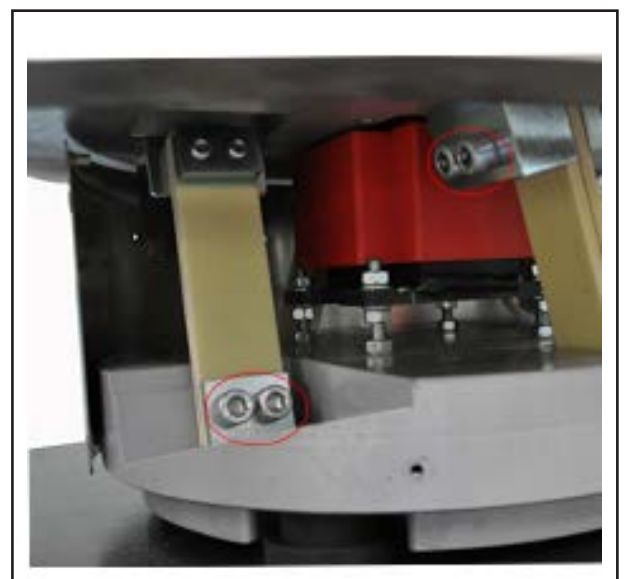


Fig. 227



- Start up the caps sorter and adjust the speed at the maximum value by the appropriate command.
- Check during the operating, the distance between the magnet and the fixed plate.

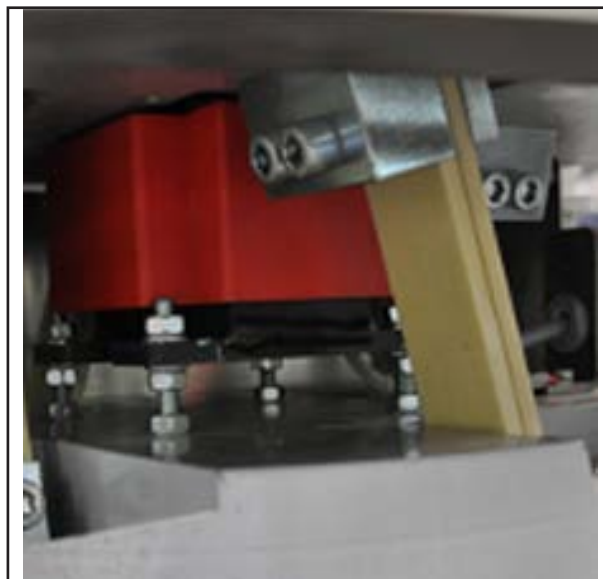


Fig. 228

- Loosen the fixing threaded rings of the threaded registers.
- Adjust the distance between the magnet and the fixed plate, in order to have a value of 1,2 mm without causing a mechanical beat.
- After adjustment, tighten the fastening nuts of the threaded registers.



Fig. 229



- Slowly unscrew the lower fixing screws on a pack of leaf springs.
- One can notice a caps speed change inside the tank of the vibrator:
 - The speed decreases: mount other leaf spring. First a leaf spring in a pack. If this is not sufficient, progressively mount a leaf spring (with relevant spacers) for each pack.
 - The speed increases: dismount one or more leaf springs, in the same previous order.



In the calibration of the vibrating base it is necessary to pay attention in order to don't create an excessive air gap. An excessive air gap, in fact, cause a power absorption such as overheats the coil and/or make it burn.

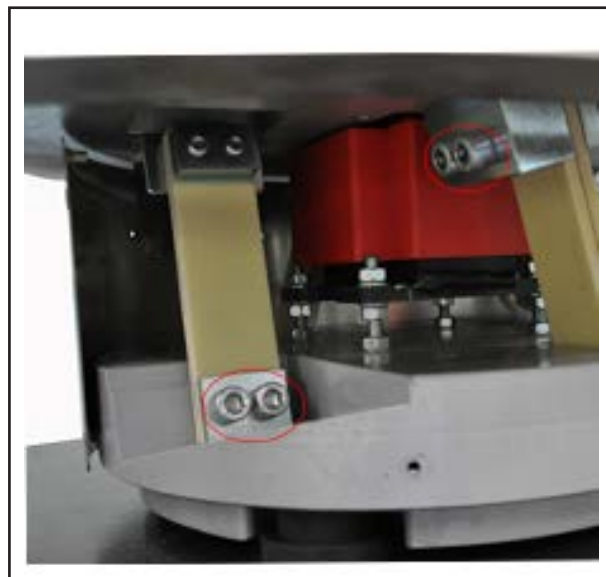


Fig. 230



12 FAULTS AND REMEDIES

The list of the faults in the following pages is useful to detect as fast as possible their reasons and possible solutions.

The research and the interventions must be carried out methodically (from the most probable to the less probable reason) and in logical sequence.

Proceed by steps: a first intervention, a test on the obtained result, then passage to the next intervention.

The main closing machine units are:

- Closure unit
- Caps/corks distribution.
- Caps/corks feeding.



Before operating on the closing machine, insulate all the energy sources and dissipate the residual energies.

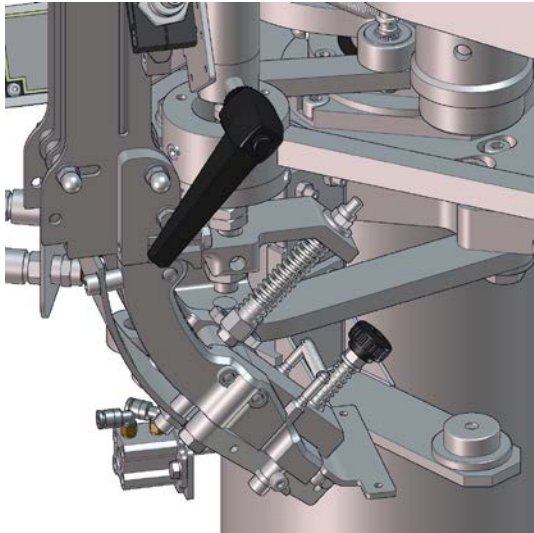


FAULT	CAUSE	REMEDY
Lifting or lowering the closure carousel, the same does not move vertically.	Interference with the container conveying equipment, foreign matters in the lifting mechanism, other.	Identify the reason for the overload (interference with the container conveying equipment, foreign matters in the lifting mechanism, other).
	The fastening screw of the sliding bush has not been loosened.	Loosen the fastening screw.
	Min. or max. limit of closing carousel vertical range reached.	Make sure the closing machine has been preset for the container size to be processed. If so contact <i>Technical Assistance Service</i> .
The containers break or are badly guided in the loading step.	Container transfer equipment not suitable.	Replace the equipment with the one suitable for the size to be processed.
The containers break in the closure step.	The closure carousel is too low.	Position the closure carousel at the correct height for the container to be processed.
	Container transfer equipment not suitable.	Replace the equipment with the one suitable for the size to be processed.
Caps are not descending from the sorter.	The sorter exit-mouth is clogged.	Remove possible caps, cap parts or foreign matters. Check on the pressure gauges if the pressure of the air jets is correct.
	Caps are not correctly sized or are strained.	Use caps with the suitable size and even features.
The cap screws badly on the bottle.	Closure chuck not exact.	Use the closure chuck suitable for the cap to be processed.
	Cap fragments in the closure chuck.	Remove the fragments from the closure chuck inside.
	The rubber spheres seal ring on the closure chuck is worn.	Replace it with one of the correct size.
	Bottle-neck guides not suitable for the bottle size to be processed.	Replace the guides in the next with those suitable for the bottle size to be processed.
	Broken compensating spring.	Replace the spring with a spare one.
	Container threads/cap not compatible.	Check that the bottle to be processed and the used cap correspond exactly.



FAULT	CAUSE	REMEDY
The cap is not totally screwed.	Insufficient magnetic torque.	Increase the head tightening torque. For the adjustment see sub-chapter 10.3 ADJUSTMENTS OF THE CLOSURE HEAD page 80.
	The cap slips in the closure chuck.	Check the wear status of the closure chuck and replace it with a spare one if necessary. Check whether the closure chuck is suitable for the relevant cap and replace it with a suitable one if necessary.
	The bottles rotate around themselves.	Container anti-rotation sector worn. Check the wear of the container anti-rotation sector and if necessary replace it as described in paragraph 11.5.6 CONTAINER ANTI-ROTATION SECTOR page 121.
		Container anti-rotation sector of the star-wheel worn. Check the wear of the container anti-rotation sectors and if necessary replace them as described in paragraph 11.5.5 STAR-WHEEL ANTIROTATION SECTORS page 119.
		The bottles rotate around themselves.
	Closure head springs not working.	Push the closure head body upwards. The elastic reaction of the compensating spring must be felt. For the replacement see sub-chapter 11.5.2 MAINTENANCE OF THE CLOSURE HEAD page 92.
The cap is too much screwed.	Excessive magnetic torque.	Decrease the head tightening torque. For the adjustment see sub-chapter 10.3 ADJUSTMENTS OF THE CLOSURE HEAD page 80.



FAULT	CAUSE	REMEDY
Problems in the cap distribution.	The caps fall from the closure head.	The rubber spheres seal ring is worn or the ball seats are dirty. Replace worn-out parts and clean carefully.
	Incorrect cap gripping.	Check the height of the closing carousel. Check the operating of springs of the caps positioning device.
		
		Check that the caps positioning device is properly operating. If it does not contact the <i>Technical Service</i> . Check that the transport pin is suitable to the relevant cap. Otherwise, replace the caps transport pin.



13 SCRAPPING AND DISPOSING



As there are in force different norms of observation, it is necessary to follow the prescriptions fixed by laws and authorities of the states where the closing machine will be installed.

Proper precautions must be taken for out-of-order and scrapping of the closing machine in order to avoid possible dangerous situations.

In particular, follow these general instructions:

- disconnect the closing machine from the electric network and check that no residual voltage exists.
- Empty all the lubrication points of oil paying attention in not dispersing and dispose it according the laws in forces in the country where the closing machine has been used.
- Lock all the moving parts of the closing machine, making sure that there are no possibilities of movement, also due to accidental collision during the transfer operation.
- With the help of qualified and authorized personnel proceed to the dismantling of components and the differentiated collection of materials (steel, plastic, electrical components, wires, etc.) in order to dispose everything according to the indications foreseen by the laws in force.



Always use security clothes during the operations that require direct intervention on the closing machine.